

BUY^{THE} FEAR SELL^{THE} GREED



7 BEHAVIORAL QUANT
STRATEGIES FOR TRADERS

LARRY CONNORS
CONNORS RESEARCH, LLC

BUY THE FEAR, SELL THE GREED:

**7 Behavioral Quant Strategies
for Traders**

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Larry Connors
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TRADINGMARKETS
Publishing

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The Connors Group, Inc.
185 Hudson St.
Suite 2500
Jersey City, NJ 07311

Connors Research, LLC
185 Hudson St.
Suite 2500
Jersey City, NJ 07311

CONTENTS

ACKNOWLEDGMENTS

CHAPTER **1** FEAR, GREED, AND MARKETS

CHAPTER **2** RSI POWERZONES

CHAPTER **3** CRASH

CHAPTER **4** VOLATILITY TRADING

CHAPTER **5** VOL PANICS

CHAPTER **6** VXX TREND STRATEGY

CHAPTER **7** TRADING NEW HIGHS

CHAPTER **8** TPS; *FEAR AND GREED RISING*

CHAPTER **9** TERROR GAPS

CHAPTER **10** BUYING FEAR, SELLING GREED

APPENDIX

- i. Connors Research Traders Journal (Volume 7):
Superior Trade Construction for Asymmetrical Gains
- ii. Connors Research Traders Journal (Volume 3):
Stops (Still) Hurt
- iii. AmiBroker Add-on Code for the Trading
Strategies in *Buy the Fear, Sell the Greed*
- iv. More Research from Larry Connors
- v. Private Mentoring
- vi. Private Consulting for Asset Managers and Family Offices
- vii. Introduction to AmiBroker Programming
- viii. Correlation Risk
- ix. Trading Resources
- x. Relative Strength Indicator Formula
and ConnorsRSI Calculation

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CHAPTER 1

Fear, Greed, and Markets

"In war tomorrow, we shall be dealing with men subject to the same emotions as were the soldiers of Alexander."

George S. Patton
From George Patton's Personal Papers
The Secrets to Victory
March 26, 1926

Let's imagine you're asking for professional advice on an important financial matter from someone who is experiencing the following symptoms:

- Nausea
- Dizziness
- Chest pain
- Headaches
- Neck aches
- Stomach upset
- Pulsing in the ear
- Burning skin
- Shortness of breath
- Electric shock feeling
- Shooting pains in the face
- Heart palpitations
- Weakness in legs
- Feelings of going crazy
- Fear of impending doom

On a scale of 1-10, how confident are you of this person's judgment?

8-10 - Highly confident. This guy definitely has his act together.

5-7 - Somewhat confident. Maybe the shooting pains in his face and his heart palpitations are a bit bothersome, but you're willing to overlook everything else, including his feelings of burning skin, nausea, dizziness, and the weakness in his legs.

2-4 - Little confidence. Feelings of going crazy and fear of impending doom are not quite the qualities that inspire strength and confidence in someone's judgment in making financial decisions for you.

1 - None. Nada. Nul. Nicht. ZERO! The guy's likely a nut. Who in their right mind would rely on the judgment of a person in this condition???

The Correct Answer

You likely answered either 2-4 or 1 (OK, I know you answered 1).

THE OFFER

Now let's assume I make the following offer to you: I offer you the chance to trade directly against traders and investors who are having the exact symptoms listed above, including the feelings of going crazy and impending doom. And as a bonus...

1. You can trade one-on-one against them;
2. You get to determine the times you'll trade against them;
3. You get to construct the trade however you feel is best for you;
4. And you get to do this multiple times every month for the rest of your life. On top of this you know that had you done this over the past quarter century, you would have won anywhere from over 70% of the time to up to 97% of the times you did this.

How does this make you feel? Would you accept this offer?

Most people certainly would and I'm sure you would too. This book is going to show you when, where, and how to do this.

So now I'm sure you're excited. You're also saying to yourself, "What's the catch? This sounds too good to be true."

Well, it's true. You'll see that quantitatively for years and even decades it's been true. The catch is that the times you will be buying and selling these historically large-edge high-

probability times are psychologically the hardest times to take these trades. This is why the edges you are about to learn have existed for years and, in many cases, decades.

Unfortunately, many traders and investors are psychologically unable to fully take advantage of these types of opportunities. (Warren Buffett can and does on the investment side, but this is a big part of why Buffett is Buffett.)

The outside fear when these trading opportunities occur is often too extreme and when fear is too extreme, the crippling symptoms mentioned earlier kick in and one normally goes into protection mode. Traders and investors are usually not willing or able to shut off the noise to go into offensive mode. You'll see this throughout this book — there are many behavioral components going on at the same time and none add up to one rationally saying to themselves, "It's time to buy." In fact, the majority of rational people are in panic mode pushing prices even lower and making edges even greater for those who step in at the right time.

The goal of this book is to make you aware of when and why short-term market edges exist in stocks and in ETFs, and then give you the quantified strategies to trade them.

I wrote my first book on trading more than two decades ago and have since authored a half dozen more books. Each book contained either strategies and/or years of price data that showed where trading edges exist.

Over the past five years, it has become more and more apparent to me that the edges I saw were not being driven by technical analysis, nor fundamental analysis. The majority of the time they were being driven by behavioral factors of which fear and to a lesser extent greed were the catalysts. In a few cases, for example the RSI PowerZones Strategy in Chapter Two, we've seen these edges as measured by the times the signals were correct actually increase over recent years. Part of this is due to the 2009-2017 bull market. Another part of it is behavioral.

Thirty years ago, when a news event would occur, it could take days to assimilate it. The majority of the assimilation was done by professionals, not individual traders, who over time moved prices to where they should be. Fear and greed were certainly always prevalent, but outside of large events (for example, 1987), the response was smoothed by time and contained within a small universe, mostly made up of investment professionals.

Today it's far different. News now drives prices every day, and at times every minute. The number of news sources today is vastly larger compared to 30 years ago. Thirty years ago, there was Dow Jones Newswires and maybe a few other financial news outlets. Bloomberg

had only begun gaining some traction (I believe Michael Bloomberg had just made the decision to go into the news business) and CNBC didn't exist, nor did the internet.

Today there are literally dozens of potential news and information sources for traders and investors to get their information from, ranging from Bloomberg, which is primarily for professional traders and investors, to many sites I won't mention because there are so many, which get millions of pageviews a month because they put their own editorial spin to things. Add in CNBC, which runs before and during market hours and is followed by (as of May 2018) *Fast Money*, which moves stocks after hours, along with the ubiquitous Jim Cramer coming on the air right after with his hour-long show, plus dozens of message boards including the large user-generated content site StockTwits, and the news and chatter never shut off.

If fear is a contagion, then when fear hits, it quickly goes viral. Nearly two decades before D-Day, then-Major George Patton so astutely pointed out that human nature in soldiers (and I believe he likely meant all humans) never changes. The same can be said for traders and investors. The only thing that's changed is the timing of their emotion; today it occurs faster and at times is more extreme primarily due to the role the media (and especially social media) plays in disseminating the news that triggers this behavior. We'll see this played out and quantified in each of the strategies in the book.

We'll start the book off with two chapters, with one that shows when fear is at its greatest and has correctly predicted the short-term direction of liquid ETFs over 80% of the time since 2006. For the S&P 500 ETF (SPY), the strategy has correctly predicted its direction over 91% of the time since 1993.

We'll then look at a strategy that takes advantage of greed; in this case it's extreme greed in stocks where the stocks reach parabolic levels, often because of news and crazy rumors, and then they reverse (they CRASH) the majority of the time.

From there we'll go to my favorite topic: trading volatility via VXX. No market gets inside fear better than the volatility market, and we'll walk through the history and, more importantly, the structure of VXX, where we'll see why and how this product was built to go to zero.

We'll also look at the five types of buyers in VXX, a group that has historically lost substantial money year after year since 2009. They range from well-meaning but uninformed investment advisors all the way to outright gamblers. With this knowledge, we will then learn two strategies to trade this structurally inefficient product, one that applies a very short-term time frame (on average approximately one week) to a longer-term trend-following strategy that climbs aboard being short VXX on average for three

months while VXX trends lower. There are strong historical test results in these two strategies and these edges have been in existence since VXX was created and brought to the market in early 2009.

We'll then look at the Trading New Highs Strategy, a strategy we created based on a seminal academic study that was originally published in 2004 and is still often referenced today. This study shows how traders and investors use New 52-Week Highs as an anchor point. We added a fear component to this anchor point and you'll see that when combining their findings with fear, those stocks have risen over 77% of the time within days.

Chapter Eight looks at the times fear and greed build up in equity indexes, not only in the United States, but around the world. Applying a scaling-in approach, this strategy has correctly identified prices moving higher well over 80% of the time. This strategy is robust, meaning there are literally thousands of permutations you can apply to buy fear and sell greed in ETFs.

Chapter Nine looks at overnight fear. This strategy combines taking an ETF that has been pounded down by incessant selling and then gaps lower. Then a further intraday sell-off creates panic and terror to the owners of these ETFs. The majority of the time, many of these owners can't take the pain any longer (they exhibit a number of the symptoms we talked about earlier) and they sell out of their position in order to avoid any further pain. You'll learn to step in and take the position from them, oftentimes with large edges in place.

Then we'll tie everything together, including looking at how closely this type of trading resembles Warren Buffett's type of investing. Buffett has made a fortune from "buying fear." He has had a repeatable process in place — often buying when everyone else is selling. On a longer-term basis you'll see how Buffett has done this for decades with these investments. We'll tie together these same behavioral biases and doing so on a short-term trading basis. We'll also discuss the many ways to take these strategies and apply them to your trading.

In the Appendix, you'll gain further knowledge with additional resources I recommend that are available to you, along with additional ways to structure your trades to buy fear and sell greed. By the time you're done with this book, you'll be able to move ahead, quantitatively applying behavioral edges that will likely be in place for many years to come. Markets may change but, as you will repeatedly see, human behavior does not.

16 THINGS YOU NEED TO KNOW TO GET THE MOST OUT OF THE STRATEGIES IN THIS BOOK

Before we get to the strategies, let's cover the following additional notes:

1. Every Strategy in This Book Is Quantified

This means every strategy has a fixed set of rules (fixed set of inputs) and is backed by anywhere from as short as nine years of historical test results to as long as 25 years of test results.

Buy signals occur when fear is high, and short selling signals on stocks and ETFs occur when greed is at its greatest. Throughout this book, we're going to do exactly as the title states: We're going to buy the fear and sell the greed. And the main reason we're going to do this is because there are quantified behavioral edges that occur over and over again for you to take advantage of.

We systematize and quantify fear and greed in many different ways. This includes doing so in both stocks and ETFs, trading volatility via VXX, trading both at the end of the day and also intraday, and trading both on the long side and the short side, along with every strategy trading with rule-based exits.

Every trade will have a reason to be entered, applying simple indicators to profit from fear and greed. You'll understand the behavioral reason for the trade ahead of time. Most importantly, because everything is structured and systematized, there will be no guessing when to enter or exit a position.

2. Here Is How We Ran the Tests

- a) Each test has a specific start date and the end date was the last day of trading in 2017.
- b) Norgate Data survivorship-free data was used for all tests. The data was adjusted for dividends and splits, and the universe of stocks or ETFs are predefined for you in each strategy.
- c) Unless limit orders were used, all tests were done on end-of-day prices. Commissions and slippage were not included.

d) Each test was verified a second time with the same data. This means at least two separate researchers, located in different locations, came up with the exact same test results from the written rules.

3. Indicators Used

a) Price - The strategies in this book are highly reliant on price behavior. Price behavior often tells you the main story as to how much fear and greed exist at any given time.

b) Trend - In many cases we rely upon the longer-term trend to buy in the direction of the trend. This is primarily done with the 200-day simple moving average (SMA). Hedge fund legend Paul Tudor Jones said it best in Tony Robbins' *MONEY Master the Game* book: "My metric for everything I look at is the 200-day moving average (MA) of closing prices."

We agree. As a whole, equity index prices that trade above their 200-day moving average tend to be more predictable to the upside than when they are trading below their 200-day moving average. There's usually a reason an index or stock is trading below its 200-day MA, whether it's for market reasons, or sector reasons, or company-specific reasons. These reasons are often valid and trading in the direction of the trend (for example, only trading an equity on the long side when it's above its 200-day MA) has, on a quantified basis, proven to increase short-term edges.

c) Relative Strength Index (RSI) to accurately measure how overbought and oversold a security is.

You'll see throughout this book that the more fear there is in the marketplace, the more oversold a security becomes. The same is true for greed, especially excessive greed. The more overbought a security becomes, the greater the greed there is in the marketplace. This is where the largest reversals take place and where the greatest quantified edges have existed.

Over the past 30+ years of researching equity prices, no indicator has proven better on a quantified basis than RSI. (You can find the formulas for RSI and our enhanced version of RSI, ConnorsRSI, in the Appendix.)

Throughout this book, you will see the use of the Relative Strength Index. RSI was originally created by Welles Wilder in the 1970s in his seminal book, *New Concepts in Technical Trading Systems*. For the next three decades after he published it, traders relied

upon his default setting of a 14-period RSI. In fact, even today, most software packages use this as the default setting.

In 2002 I began overlaying stock prices with a much shorter reading: a 2-period RSI. Up until then, I had yet to see anyone use RSI in this manner. What struck me was just how in sync it was with short-term price movement, especially in equity indexes and in the S&P 500 futures markets.

I had my research team begin running back tests and we saw test results higher than anything we had ever tested. One year later, we published a strategy around our findings, and as far as we know we were the first to publish these findings applying a shorter-term RSI.

It's now a decade and a half later and Wilder's RSI, shortened to a shorter time frame, still tests better than any indicator available. We certainly have looked for something even better and we're always open and excited to test anything and everything. We've yet to see anything surpass Wilder's RSI.

You will see three different RSI time frames used in the strategies: the 2-period RSI, the 4-period RSI, and ConnorsRSI (I'll explain this one in a minute). For each strategy in the book, we stayed with the original way it was applied. This is our way of saying that RSI is robust and, as a whole, remains today a key to measuring short-term market sentiment (fear and greed) in equity prices.

ConnorsRSI (CRSI) was created by my research firm about five years ago. As far as we know, it's the only quantified oscillator available for equity traders. You can find the formula for ConnorsRSI in the Appendix. For those of you on Bloomberg, it's programmed into the terminal in the Studies section.

ConnorsRSI applies Wilder's RSI and goes further. It's much more difficult for a security to reach extreme levels with ConnorsRSI than it is for Wilder's RSI and historically when they've reached these extreme CRSI levels, quantified edges have existed.

4. Math

There is no advanced math used in this book. It's not necessary.

5. Stops

No stops were used in the testing. This was intentional and is consistent with our long-term belief of stops. I've added to the Appendix our position on stops for you to read. I've

also made suggestions in the book on how to protect your positions in ways better than stops.

6. Buying Fear, in a Systematic Quantified Way, Is One of the Few Built-In Edges That Exist in the Marketplace

The greater the fear, the greater the mispricing of securities, which means the greater the edges are for you. In this book, we're going to identify when the fear is in place, why it's in place, how to systematically trade it when it's in place, and we're going to fully quantify this fear going back as far as 25 years of trading in some cases.

7. Fear Is Hardwired in Humans

This has been proven over and over by science. There are many dozens of scientific studies done with MRIs showing that the brain changes when fear is present.

"When people are frightened, intelligent parts of the brain cease to dominate," Dr. Bruce Perry explains, quoted in an article published on the *Time* magazine website. "When faced with a threat, the cortex responsible for risk assessment and actions cease to function. In other words, logical thinking is replaced by overwhelming emotions, thus favoring short-term solutions and sudden reactions."

Please remember this. It's an important part of the reason the trading edges and biases you're about to learn exist.

8. Not All Fear Is Equal

This means that the behavior associated with fear can be better pinpointed, potentially leading to greater returns based upon the overall market condition. This is important to remember, and the data proves this out.

a) Fear in bull markets in equities and equity ETFs tends to have a much shorter duration than fear in bear markets.

Healthy bull markets quickly shrug off fear and then push prices higher. Simply adding a 200-day simple moving average and trading in the direction of the moving average many times proves this out. As I mention a number of times in the book, it's why it's hedge fund legend Paul Tudor Jones' favorite indicator.

- b) The behavior of prices leading into the days ahead of the signal plays a role. Mounting fear is a whole lot better than one day of fear. In military terms it's why multiple days of shelling is far more effective than one day of shelling. Humans naturally get worn down from multiple days of pounding. In trading, when they get worn down from losing money over multiple days, they get mentally fatigued, they begin making irrational decisions, and the buyers on the other side often get advantageous pricing (they get added edges in their favor).
- c) Overnight fear leading to prices gapping lower is better than the times prices don't gap down. This is especially true after a security has sold off for several days. No one likes losing money multiple days in a row and then waking up to a stock or ETF gapping even lower, further increasing their losses.
- d) Intraday fear (and, even better, panic) is one of the best times to buy the fear. Investors and traders have less time to rationally react to sell-offs and they oftentimes panic, especially if that security has been under siege heading into the trading day.

9. Types of Fear

As you would expect, fear of loss in trading is the predominate driver. Fear of losing money — and in the case of the money managers, fear of losing their job — drives buying and selling decisions.

Fear of loss is in fact one of the main drivers in life — loss of money, loss of losing a loved one, loss of love, loss of acceptance, the list is long. Many professionals agree that the fear of loss is the single biggest factor as to why people don't follow through with their dreams.

There's also the Fear of Missing Out (FOMO). I've seen this personally. It drives people crazy when others are making money in the market and they are not. Studies done after 2010 found that people minded less about losing money in 2008 as long as what they lost was in line, or less than, what others lost.

They also found that people who had lost less than others in 2008 but made less in 2009 were angrier than those who lost more in 2008 and gained more in 2009. Even though the net effect was the same, they felt cheated that others had larger gains in 2009! The Fear of Missing Out plays a role in two of the strategies in this book. FOMO creates the edges and we'll be there to take advantage of them when buyers irrationally run up prices.

10. The Structure of the Instrument

The two volatility strategies in the book have performed so well primarily because of the way VXX was built. VXX is structurally inefficient. I'll walk you through this in greater detail when we get to that section. If you can find a structural inefficiency in the marketplace, you can then build strategies to exploit the inefficiency.

11. Risk Management

We'll also touch upon ways to protect positions, especially the trades where fear and greed overshoot. Fixed risk positioning allows you to predetermine and control your risk more efficiently than those traders and investors who take on open-ended risk.

What you will see is very high historical test results across multiple liquid securities. You'll see signals being correct as high as over 90% of the time since 1993, and you'll see a strategy that has averaged double-digit percentage gains as fund managers and unsophisticated retail investors buy historically overpriced insurance in order to protect their portfolios. With this in mind, we need to protect from the historically minority of times the strategies didn't win.

12. What About Fundamentals?

Fundamentals are very important for investing. They're much less important for short-term trading. Fundamentals usually don't change intraday or daily. Prices and sentiment do. This is why we primarily focus on those two, and you'll see throughout this book that it's quantitatively backed.

The bottom line: Fundamentals matter most in the long run. Price and sentiment matter most in the short run.

13. As Far as We Are Aware, This Is the First, and Only (as of 2018), "Short-Term Quantified Behavioral Finance Book" Ever Written

I'm sure others will follow and possibly go even further than we did. As of now, though, you'll gain access to quantified behavioral information that's never been published before in one book.

14. A Note About This Book and My Writing Style

My writing is brief and to the point. That's the way I was taught. I prefer to read something succinct than to read something that makes the same point in hundreds of pages.

I've written many books over the past 23 years, including one that was selected by *Stocks & Commodities* magazine as one of the "10 Trading Classics Written in the 20th Century" (*Street Smarts* with Linda Raschke). All of these books were relatively short in length — they more importantly focused on quality. I have multiple objectives in sharing the knowledge in this book with you. One is of course to show you quantitatively where behavioral short-term edges repeatedly exist. Another is to do so succinctly.

15. Overfitting Data

As quantitative analysis becomes more mainstream, I'm seeing a new generation of quantitative traders and researchers who have data science backgrounds. Those who are not classically trained in our industry are applying data techniques that are successful in other fields. Some of these techniques as they apply to financial engineering are very good. Others are bad.

One of the techniques I don't like that is currently being applied more and more is taking a large data set and trying every possible combination until good results come out.

Overfitting is exactly as it sounds: Take an abundance of data and mine through it to find variations that work. Never mind if the parameters don't make sense. The data knows better. "Trade the data!" is their mantra.

For example, let's create something that's not remotely logical. Let's assume that if the Red Sox win by three runs on a Thursday night, buy the S&P 500 on the open the next morning and sell on the close. Let's say hypothetically this actually has made money 80% of the time over the past 30 years, providing traders with large returns. We then test it changing one variable: the total number of runs they win by. When the Red Sox win by four runs, it doesn't work. It's 50% correct and loses money. And when they win by only two runs it's only 45% correct.

The data says to trade this. When the Red Sox win by exactly three runs on a Thursday night you buy the S&P on Friday morning and you sell it on the close. It's been 80% correct for 30 years!

There are now two large problems...

- 1. The rules are nonsense.** Why in the world would the S&P 500 rise after the Red Sox won on a Thursday by three runs? Boston fans are happy they won, so they're buying? Well, if that's logically true, then the Yankees fans are not happy and they are selling. But even if a win by three runs by the Red Sox led to higher prices the next day, why three runs? Why didn't it work for two runs, or four runs?
- 2. The data has been overfitted.** I've met data miners and optimizers with very few years of professional trading experience who will say, "This is what the data says and this is a valid trading strategy!" And the rest of us who live in the real world will say, "Uh-huh."

Dr. Marcos Lopez de Prado writes about this topic in his great book, *Advances in Financial Machine Learning*. Dr. de Prado is an expert in this field. He founded Guggenheim Partners' Quantitative Investment Strategies Business where they managed over \$13 billion. I recently attended a conference he spoke at and he discussed the many pitfalls of systems testing, including overfitting, as it applies to finance. I recommend you read his book. In the meantime, unless a strategy takes advantage of an arbitrage opportunity, a structural inefficiency opportunity, or a repeatable behavioral opportunity such as fear and greed, it potentially runs the risk of having been overfit.

In this book, we're quantifying repeatable human behavior. As we move ahead, you'll see over and over again that fear creates liquidity holes (lack of buyers), and sometimes when the fear turns into panic, not only do the buyers step aside at reasonable prices, the sellers don't care at what price they sell their securities for — they just want out.

16. Why Do So Many Panic Sell-Offs Eventually Reverse?

I first read the following in an interview with legendary market strategist and technician Tom DeMark. He said real market bottoms (both short-term and long-term) don't occur because buyers come piling back in at the same time. They occur because the selling has lessened or stopped.

It took me years to fully grasp the magnitude of this insight. The bottoms that occur when fear is high occur after all the scared money is out. Sometimes the buyers immediately step in. Other times (and this is more often), the buyers gradually return after they see conditions to be *safer than they were*. These buyers are buying after the fear has subsided. And they're often paying a higher price and are buying when there are no longer short-term edges in place. Those who stepped in near the peak of the fear are the ones who are most rewarded.

In this book, we'll see numerous statistically proven ways to systematically buy the fear, oftentimes when fear is at its highest. We're often buying as the final sellers are selling out their stock or ETF, usually because they can no longer take the pain.

Warren Buffett created his wealth by following his own advice to "be fearful when others are greedy and greedy when others are fearful."

Let's move ahead and find the times to be fearful when others are greedy and greedy when others are fearful. Let's now find the times to *Buy the Fear and Sell the Greed*.

CHAPTER 2

RSI PowerZones

I'm going to open this book with a trading strategy that has accurately predicted the direction of SPY (the S&P 500 ETF) over 91% of the time since 1993. The strategy on average holds SPY from 3-7 trading days and buys when fear (and oftentimes a great deal of fear) is present. This fear can range from fear of buying because of an upcoming economic report or news event all the way to a market panic. When this fear occurs there are large, consistent edges for traders. Buyers disappear, sellers panic, and prices in the S&P 500 reach levels that have consistently risen year after year for the past 25 years.

This strategy epitomizes the buy the fear, sell the greed philosophy.

What are RSI PowerZones? RSI PowerZones are high-probability buying levels for ETFs, especially US Equity ETFs, when they are in an upward trend. When ETFs reach these levels, they become high-probability areas for traders to take advantage of the fear in the marketplace, buy the fear, and then sell at higher prices when the fear subsides. For a quarter of a century, better than 9 out of every 10 times this has occurred in SPY, it's been profitable.

Let's jump right to the rules (and they are simple). We'll then look at two trade examples to get into the guts of the marketplace and to allow us to see just what this fear looks like, why it is occurring, who is doing the selling, and what happens when the selling is over.

Here are the rules for the RSI PowerZones:

1. SPY is trading above its 200-day simple moving average. This tells us we're in a longer-term uptrend.
2. The 4-period RSI of SPY closes under 30. I'll also share with you the test results of putting on the position when SPY is under 25 (this means the market is more oversold). Buy SPY on the close.

3. Buy a second unit if the 4-period RSI closes below 25 at any time while the position is open. This is essentially doubling the position, usually at lower prices.
4. Sell when the 4-period RSI closes above 55.

Here are the test results from 1993-2017 for SPY applying RSI PowerZones of 30 and a doubling of position under 25:

of Trades: **202**
Percent Correct: **90.59%**
Winning Trades: **183**
Losing Trades: **19**
Average Gain Per Trade: **1.73%**
Average Hold in Trading Days: **4.95 days**

Here are the test results from 1993-2017 for SPY applying RSI PowerZones of 25 and a doubling of position under 20:

of Trades: **147**
Percent Correct: **92.52%**
Winning Trades: **136**
Losing Trades: **11**
Average Gain Per Trade: **1.89%**
Average Hold in Trading Days: **4.84 days**

That's it. Four simple rules to trade SPY. And since 1993, 90.59% of the signals these four simple rules have produced have been profitable on average within five trading days.

How could four simple rules produce such incredible results? Let's go deeper:

1. Is this overfit based on today's lookback? No. In fact, thousands of traders first learned about this strategy when I originally published it in 2003.

I then published it again in my book *High Probability ETF Trading Strategies* in 2009. Tens of thousands of traders got access to the strategy from purchasing the book and many times it was shared by others.

2. Are 30/25 or 25/20 magical levels? They are not. Both are very good and robust, and there are additional variations for you to study further.

3. Were the results lopsided by the tremendous mean reversion inherent in stock prices in the 1990s? They were not. In fact, from 2009-2017, the results got better. Yes, the strategy was in the public domain for years; it was widely disseminated, the money management industry shifted from active investing to primarily passive investing, and I can give you a whole list of other ways markets changed, yet the results got better.

From 2009-2017 the RSI PowerZone Strategy saw 96.77% of its signals being profitable. 60 of the 62 signals were profitable!

Let's pause for a minute and look at this from the top down. Let's really begin to understand what's going on here.

1. This behavior is not only seen in SPY. We've looked at it multiple times over multiple years (first from 1993-2003, then from 1993-2009, and now from 1993-2017). We did this on 20 of the more liquid ETFs available for you to trade. They range from US Sector ETFs to Country Fund ETFs from Europe, Asia, and Latin America. The percentage of the time these markets rise when they reach their RSI PowerZones is on average above 81% since each has begun trading. This is a global phenomenon.
2. I used to believe that the magic was in the combination of the 200-day and the 4-period RSI. Even though they play a role, they're simply technical indicators. The real driver is the behavioral underpinnings that are occurring when equity ETFs reach the PowerZone levels. *There's fear in place*. And fear, as you will see over and over again throughout this book, is where edges come into play. The greater the fear, the greater the edges. We're going to go deeper into this as we move ahead together in this book.
3. The selling into strength, meaning when the 4-period RSI closing above 55 is also behavioral. Prices soon rise during these times because fear is subsiding. They were afraid to buy a few days earlier for dozens of possible reasons — some being as simple as an upcoming economic announcement to others being caused by the media creating a short-term hysteria (we'll look at one of these in the examples). The overwhelming period of the time the exits get triggered at higher prices is because investors now believe **"they are safe."** They may be safe, but they also are buying when the inefficiency in pricing caused from the fear has disappeared. Those who bought the fear have won the large majority of the time when they did so and then sold profitably when "the coast is clear."

Let's now look at two high-profile sell-offs that occurred, each triggering RSI PowerZone signals. You'll see in each that fear-induced selling led to lower prices in SPY. Buying the fear and then selling when the fear subsides repeats itself over and over again, especially in equity ETFs.

The longtime best-selling book *Influence: The Psychology of Persuasion*, written by Robert Cialdini, Ph.D., is one of the very few books consistently recommended by Warren Buffett. I attended a presentation years ago that Cialdini made to 200 CEOs of rapidly growing private companies. At that event Cialdini told the audience that, at that time, *Influence* was “the only book” Buffett had recommended many years in a row at his annual shareholders meeting in Omaha.

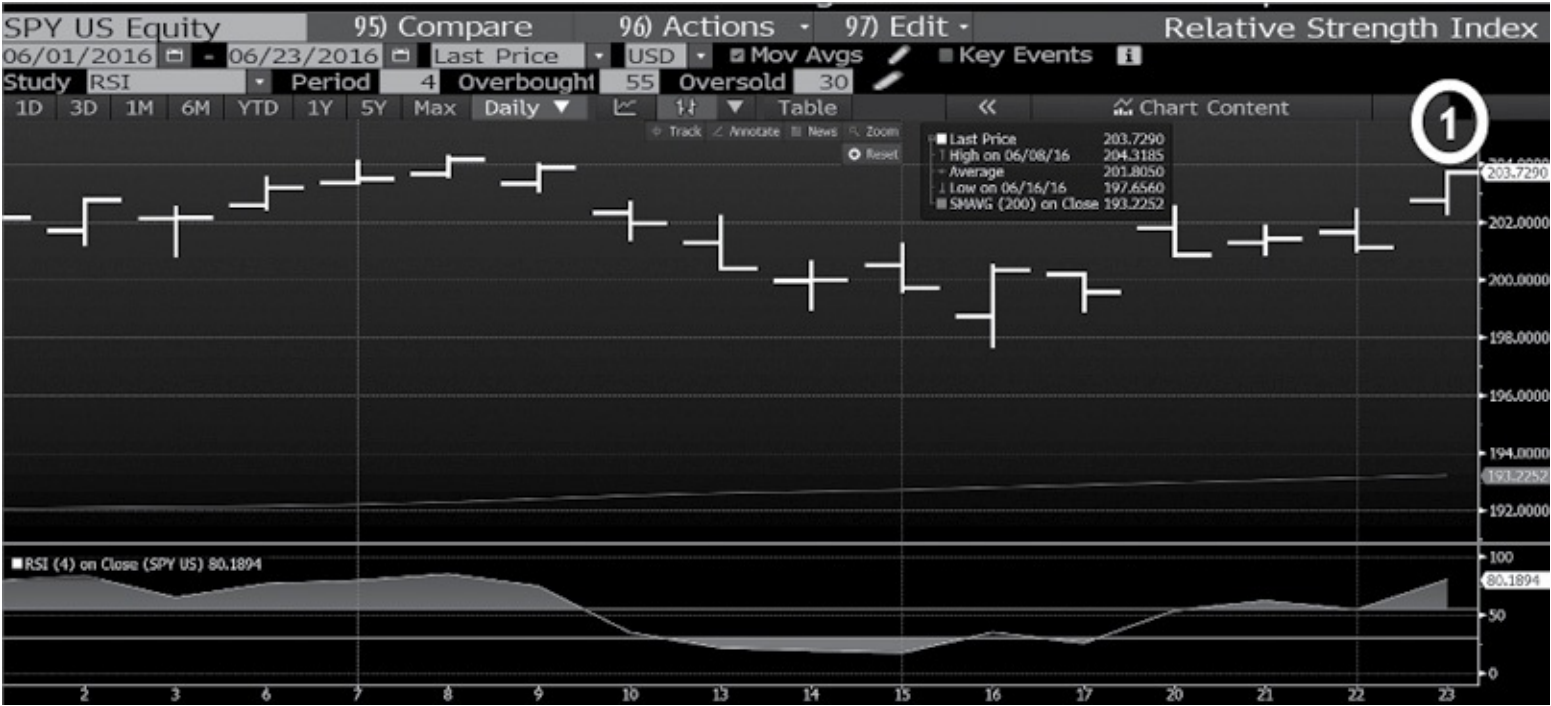
The book *Influence* has nothing to do with investing. It has everything to do with human behavior. And Buffett is a master of understanding human behavior. It’s a key part of how he created his wealth, and we’ll look at this further later in the book.

According to Cialdini, one of the six principles of *Influence* is “Authority.” When an expert speaks, people usually listen closely. The more someone is perceived as an authority, the more likely his or her advice will be followed.

On Wall Street, Global Head of Macro Quantitative & Derivatives Strategy and Senior Analyst Marko Kolanovic from J.P. Morgan has influence. He is considered one of the most influential authorities on quantitative analysis, especially as it pertains to how derivatives will impact the market. He deserves his elite status because he and his team have made some great short-term predictions over the years.

With influence comes power to move markets and Marko Kolanovic has the power to move markets. Here’s a good example of how events combined with a widely disseminated report from someone with tremendous influence induces fear and creates repeatable high-probability trading edges.

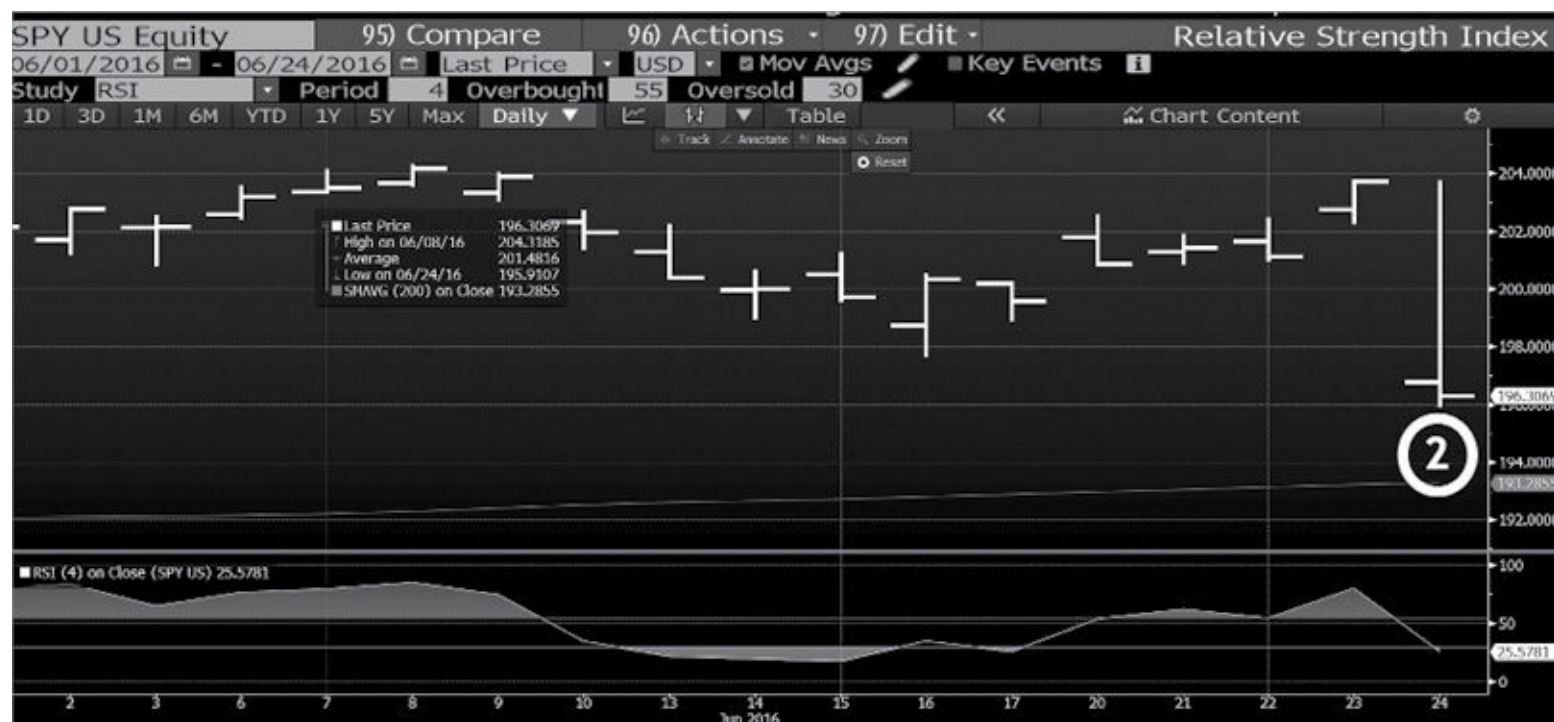
Example 2.1. SPDR S&P 500 ETF Trust (SPY)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

1. A nice quiet month in June 2016. SPY is well above its 200-day moving average and S&P 500 prices close near all-time highs. It's going to be an enjoyable summer!

Example 2.2. SPDR S&P 500 ETF Trust (SPY)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

2. Friday, June 24. Out of nowhere the citizens of the UK vote themselves out of the European Union (EU). Brexit strikes the global markets, including the United States. The market plummets with a massive one-day move to the downside. The 4-period RSI moves from 80 to 26 (a very large one-day move for the 4-period RSI).

At the same time, one of the most respected and highly followed strategists, Marko Kolanovic, sends out a report that gets widely disseminated throughout the world that \$100 billion of equity selling will likely follow over the next few days from systematic strategies that have high derivative exposure. He also sees a greater chance of a US recession because of Brexit.

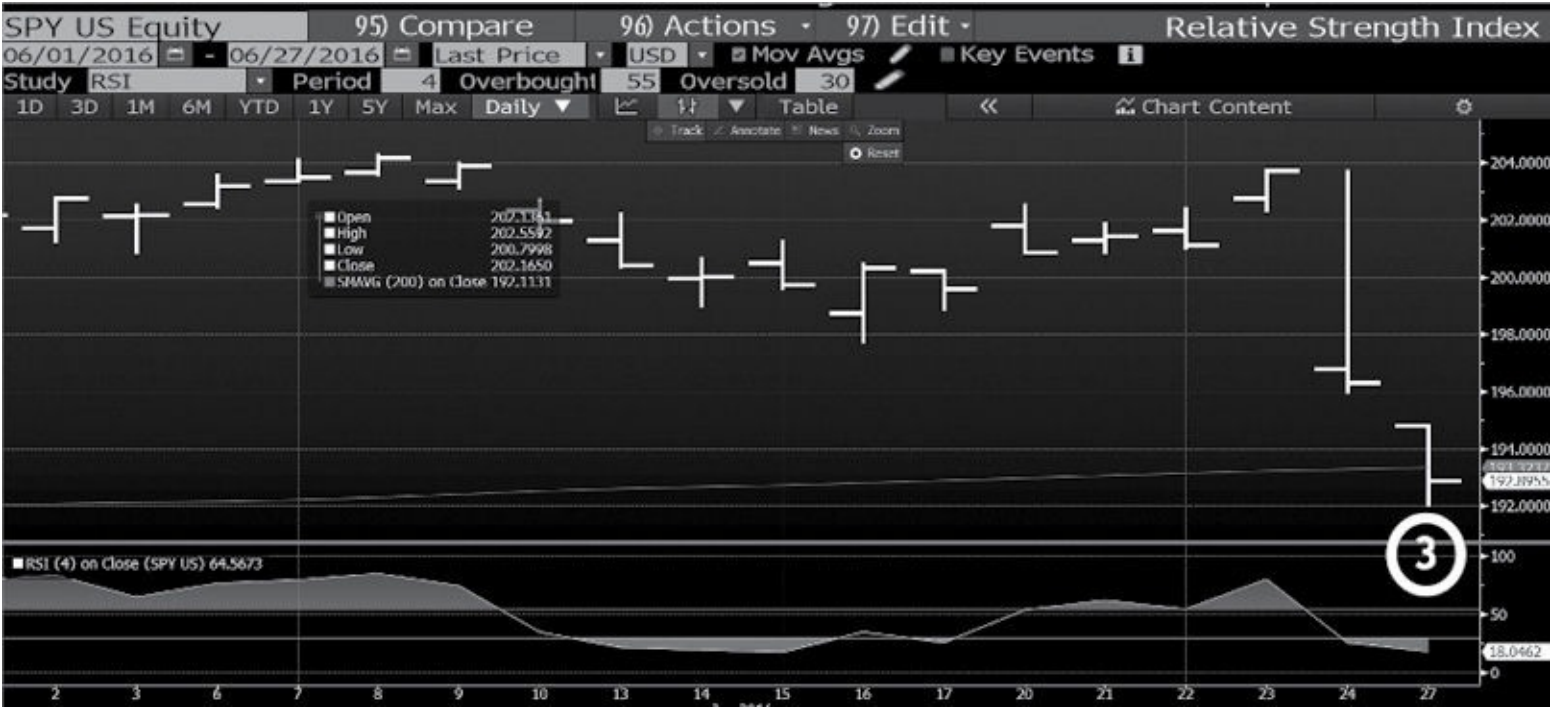
What a great way to start the summer of 2016. Tens of thousands of professionals are also dealing with their spouses who are asking them, “What the hell is wrong with you this weekend?”

If you’re disciplined and fully systematic trading the 30/25 version of the RSI PowerZone Strategy, you’re long at the close. If you’re using the 25/20 version, you probably feel like you just dodged a bullet.

In either scenario above, if Kolanovic is right, the market is going to sell off next week and you’re either doubling your position or are entering an opening position. No matter what, according to the rules, you’re taking the other side of one of the most influential professionals on Wall Street who has his Ph.D. from NYU in theoretical high energy

physics. And not only are you on the other side of his position, you're on the other side of some of the largest trading firms in the world that follow his advice!

Example 2.3. SPDR S&P 500 ETF Trust (SPY)

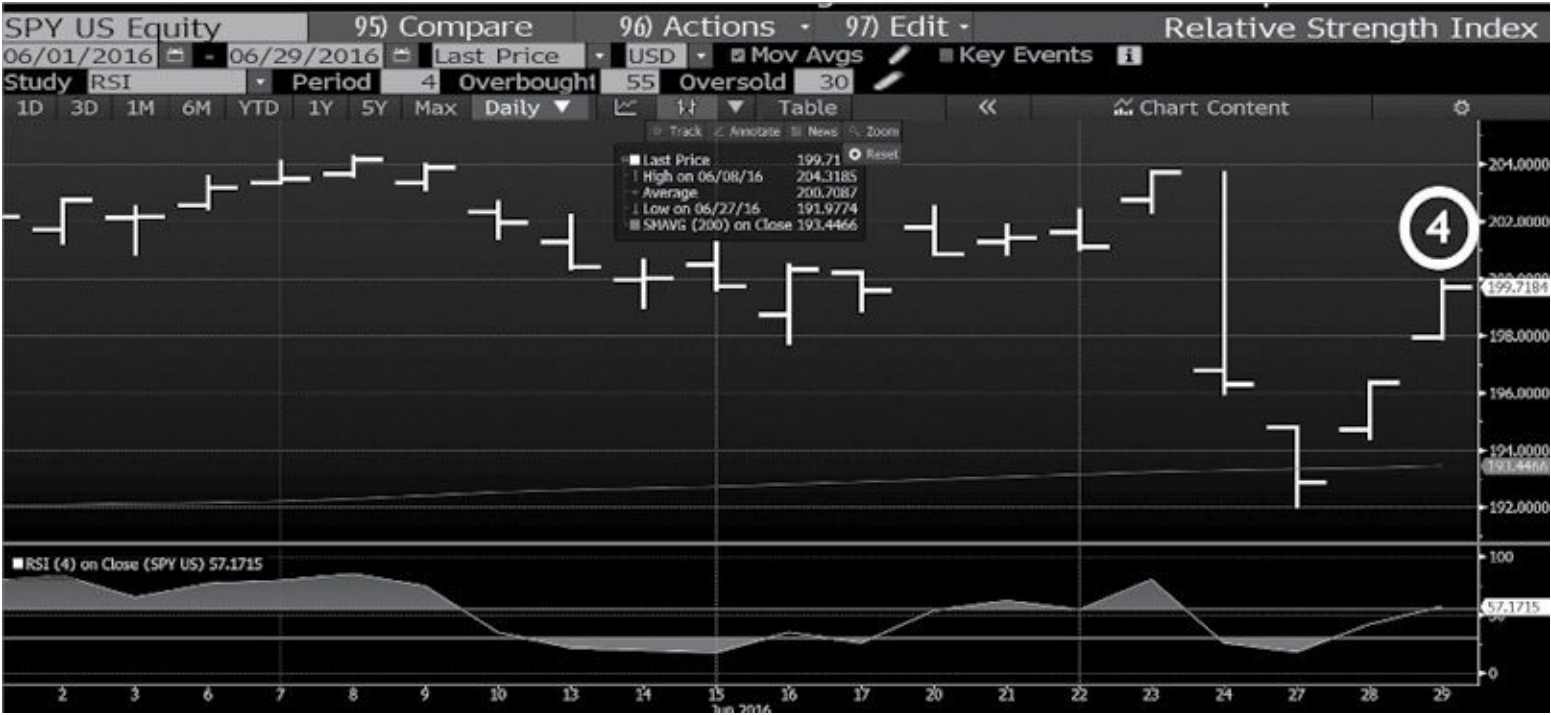


Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

3. Monday, June 27. Kolanovic’s scenario is playing out (and his report is viral along with nearly all financial media outlets discussing it). SPY sells off further, having lost over 5% off its value intraday versus where it closed on Thursday. There’s panic!

The 4-period RSI closes at 18 and you now have a decision to make: take the trade or listen to the majority including the professional majority and not take the trade. A systematic quantitative trader with no emotion takes the trade. You have a decision to make. Buy the fear? Or follow the herd and take a pass (or even short the market because of the increased risk of a recession). No one said this is going to be easy.

Example 2.4. SPDR S&P 500 ETF Trust (SPY)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

4. Wednesday, June 29. As has occurred over 90% of the time over the past quarter of a century, the correct move was to take the trade. The S&P 500 stages a broad rally and prices close significantly higher. The 4-period RSI closes above 55 and it's time to lock in your gains.

Emotionally this is a hard (and even brutal) trade to take. The logical move is to “step aside and wait this one out.” If you believe in buying the fear, though, you had the perfect scenario. A quiet market heading into the first summer weekend of the year that everyone is looking forward to is jolted by an unexpected sell-off on Friday (and likely a day that many professionals either took off or left early to get an early jump on the weekend). The market closed near the bottom of its daily range due to heavy late-day selling and one of the most respected and influential strategists puts out a report stating that there’s likely going to be another \$100 billion of selling coming in over the next few trading days. Plus, there’s now an increased likelihood of the US heading into a recession.

Thinking about this rationally leads one to the conclusion of either stepping aside or even selling short. It does not lead one to rationally conclude to buy.

Edges in large markets exist because of human emotions, especially fear. There was fear here and it was exacerbated by the media taking this report viral. It’s why prices dropped so low on Monday — everyone influenced by the report and the events got out. They all sold at the same time. And once they got out, it brought prices in the S&P to levels that have seen higher prices over 90% of the time within a few days.

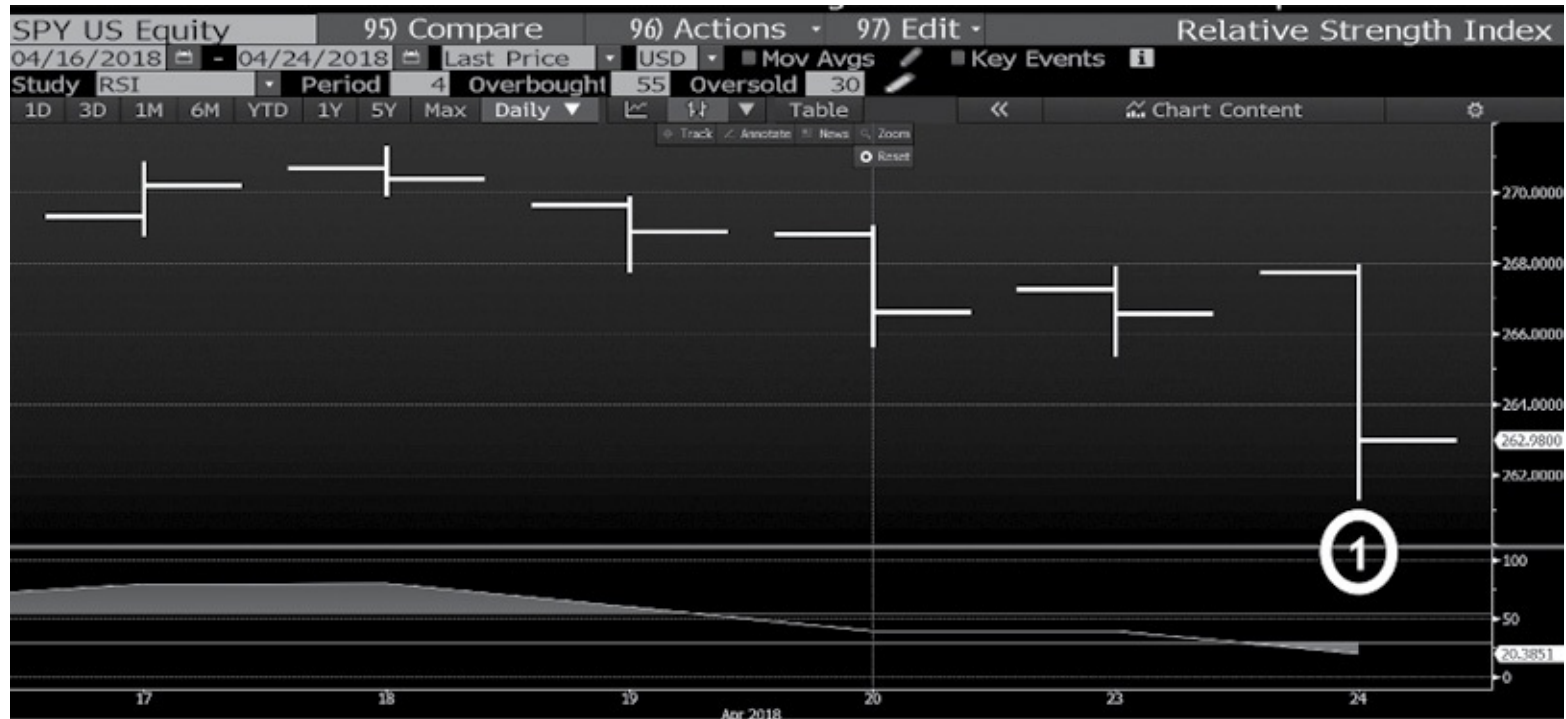
I have tremendous respect for Marko Kolanovic. He is brilliant, as is his team. In the long run, taking the other side of him and his team is likely a losing proposition. They’re smart and they often win. What does happen, though, is that they influence markets up and down and because of their influence, many professionals follow them, along with the media frequently hyping (and often overstating) their analysis. This cycle leads to exaggerated moves in both directions, and as you can see in this example, it led to a short-term panic, which led to a tremendous buying opportunity.

Quantitatively, based on 25 years of test results, these panics that take prices into the RSI PowerZones, primarily in equity ETFs, have been optimal times to buy. It’s not perfect, but prices often enter these zones when fear is high and selling is persistent. This fear leads prices to levels where one has been well compensated when buying the fear.

Let’s now look at another example.

April 24, 2017. For months market strategists have been warning about the 10 Year Treasury breaking above the 3% level. (One of the most influential — there's that word again — “bond kings” actually drew his line in the sand at 3%.) These concerns have rippled around the world. Should the 10-year rates get above 3%, it was predicted to have a calamitous effect on equity prices.

Example 2.5. SPDR S&P 500 ETF Trust (SPY)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

1. Yields moved above 3% and the well-predicted sell-off was triggered. Financial news analysis firm Benzinga (a site I respect) reported the following:

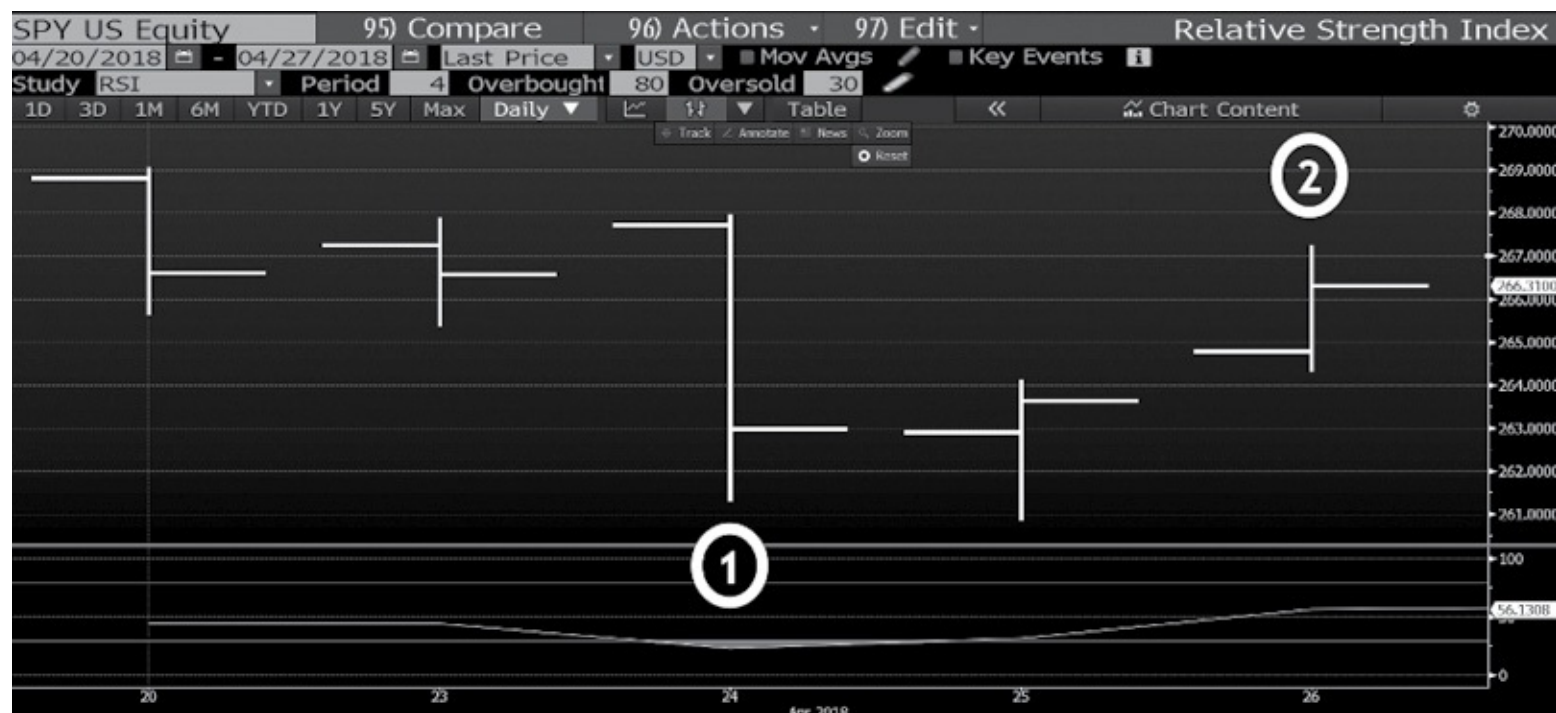
The 10-year Treasury yield reached the 3-percent psychological level Tuesday morning, triggering fears that inflation could be ramping up too quickly. Rapidly rising interest rates and inflation can eat into corporate profits. In addition, higher yields on fixed-income investments tend to sap money out of the stock market, particularly from dividend stocks.

To make matters worse, the yield curve — the difference between short-term rates and long-term rates — has been flattening. In the past, the yield curve has often inverted prior to U.S. recessions.

Let's pause for a second. The warnings about the 10-year breaking above 3% have been made for months by some of the most respected and widely followed firms and analysts on Wall Street. Inflation eating into corporate profits, higher rates sapping money out of the stock market, the yield curve becoming inverted leading to a recession. This is not the type of news that makes a rational investor say to themselves, **"This is great — I can't lose!!!"**

The sell-off took the 4-period RSI below 30. It also created a lot of additional fear, flushing out scared money and creating a liquidity trap as buyers stepped aside. You can see from the chart below the selling lasted one day. Not weeks, months, or years as was predicted — one day.

Example 2.6. SPDR S&P 500 ETF Trust (SPY)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

2. The fear created the opportunity. The RSI below 30, while the market was above its 200-day, acted as a great entry point as it has over 90% of the time since 1993, and those who bought the fear were rewarded. SPY rallied, the Dow rose nearly 300 points, and two days later exit signals were given to lock in the gains.

Since 1993, in SPY we have seen this type of behavior with the 4-period RSI closing under 30 with a 25 scale-in 202 times. 90.59% of those times prices closed higher exiting when the 4-RSI closed above 55. When applying PowerZone readings of 25/20, the times where fear is often even higher, this has occurred 147 times and has seen closing exit prices higher 92.52% of the time.

Lessons Learned

1. Nearly every sell-off has a reason. The larger the reason, the better. The more widespread the reason, the better. The more influence big-name firms and analysts have in creating the fear, the better. "FEAR IS GOOD!!!"
2. The RSI PowerZones with the 4-period RSI readings under 30 and a close above its 200-day simple moving average with a scale-in buy should the selling continue has done an impeccable job of quantifying where the fear has been and when the "stronger money" has stepped in and taken prices higher over the short term over 90% of the time in the S&P 500 going back a quarter of a century.
3. These consistent short-term edges are not being created by technical indicators. They're created by behavioral factors, especially when there is widespread fear. We then want (and need) to quantify the behavior using tools that have consistently pointed to when fear is high and selling has been substantial. Once this fear passes, the buying then resumes the large majority of the time, especially in long-term bull markets. Publishing an indicator level such as a 4-period RSI reading under 30 is going to have no impact on fear. We originally published these RSI levels 15 years ago and again 10 years ago. From 2009-2017 the historical test results got better, thanks to the long bull market and the way the media fuels the flames today. Fear is always going to be there. Buying this fear is where you find pricing inefficiencies, especially in equity index prices.
4. I'm going to remind you of this throughout the book: 91% correct on a lookback basis means that 9% was wrong. Yes, I know it's greater than 9 out of 10 times, but you have to respect the fact that no timing indicator is ever perfect. As we move ahead in the book, I'll discuss various ways to limit risk in positions.
5. Most importantly: Fear is inherent in humans. As we discussed in Chapter One, George Patton was brilliant enough to recognize that fact almost a century ago. Patton said that fear is inherent in all warriors and it has been throughout history. I feel it's safe to say if the bravest warriors have experienced fear throughout history, the same can certainly be said for money managers and traders.

CHAPTER 3

CRASH

“One minute you’re up a half a million dollars in soybeans, and the next, boom, your kids don’t go to college and they’ve repossessed your Bentley.”

Louis Winthorpe III
Trading Places

This is a story about greed. It’s a story that gets played many days on Wall Street, especially in bull markets. It’s a story about making a lot of money and needing to make more. And it’s a story of people watching other people making a lot of money, and they panic because of the behavioral phenomenon known as FOMO: the Fear of Missing Out.

Greed at extremes often creates absurd prices for securities, especially when it’s short-term greed when rational people lose their senses and mindlessly buy. As all of market history has shown, though, what goes up must come down. And a lot of money has been made from smart money taking the other side of greed. CRASH is our strategy to (legally) take money from greedy people.

CRASH is a strategy that “sells the greed.” It’s a short-selling strategy that in spite of a raging bull market over the past nine years, has consistently identified and pinpointed when greed has been extreme in individual stocks and in the majority of cases prices have reversed lower almost immediately.

The strategy is simple, elegant, and robust. It’s also psychologically likely going to be the hardest strategy for you to trade.

Taking a CRASH trade is going to be very difficult for most traders because the stocks it shorts are usually moving parabolically. And many of these are being driven higher by speculators, the news media, social media, and massive short-covering. These stocks are very often “story stocks on steroids.” And story stocks on steroids brings out the greediest worst in investors, traders, and speculators. All rationality gets thrown out the door.

Sometimes these buyers are right and the stocks continue to go even higher. But more often they're wrong — incredibly, over 70% of the time they're wrong when they're buying these stocks after they have run so far up.

Who wins when these irrational buyers are buying for no other reason than greed? The traders smart enough to know that the greed is extreme, and they'll gladly sell into these short-term, greed-induced bubbles.

In his book *Hedge Fund Market Wizards*, legendary author Jack Schwager interviewed a professional trader by the name of Jimmy Balodimas. Most of the traders Jack interviewed over the years for his books had compelling stories. Jimmy Balodimas' story, though, was one of the most compelling because he was the only trader who up to that point had made most of his money "exploiting traders' greed." No matter how strong a bull market was, Balodimas was consistently able to step in at the right times and sell into the greed. Bull markets steamroll most short sellers, but at the time of the interview, Balodimas had reportedly made many millions over the years selling to greedy irrational buyers at the right time. As Jack inferred, most professionals would be skeptical that someone could consistently do that. Not only did Balodimas do it, Schwager's son had witnessed it for years while working directly for Balodimas.

If you read the interview one thing stands out. Balodimas thrived in overbought, greed-induced situations. And he wasn't afraid of becoming even more aggressive even when prices moved against him and the greed became even greater. Selling greed was how Jimmy Balodimas made his living.

Most of us don't have the innate skill that he has and have the same conviction to sell short at extremes. We're going to need to go further and apply strict disciplined quantified rules that identify these times.

This book is a combination of behavioral finance being measured through quantitative test results and that's exactly what we're doing here with CRASH. The behavioral aspect is very important because it creates the edge. There's a "mob mentality" associated with most CRASH setups. This mob mentality creates the extreme overbought conditions. From there disciplined rules go into place combining behavioral finance with quantitative evidence.

Let's now look at the rules for CRASH. Then we'll look at the test results and then the trade setup examples. The examples will likely open your eyes. They'll also prepare you to better understand the strategy and, more importantly, what to expect when you're trading the strategy. Few if any books have been written that have you shorting these type of

stocks (it's usually the opposite). The quantitative results confirm, though, that over the past decade, these stocks, as a whole, have been solid short-selling candidates.

Here are the rules:

1. The stock's closing price must be greater than \$5/share.
2. The stock's average day volume over the past 21 trading days must be at least 1,000,000 shares a day. This assures there's liquidity in the stocks.
3. The stock's 100-day Historical Volatility must be at least 100%. This assures we're focusing only on the wildest stocks to be shorting.

We have also tested lower Historical Volatility levels (both 60% and 80%) and the test results were solid and are included for you to see. As we moved up to higher Historical Volatility levels, the test results improved both on a percentage correct basis and on an average gain per trade basis. This is likely the behavioral component becoming more apparent as stocks with extremely high volatility readings tend to further attract irrational behavior. I've provided you with additional test results in the Additional Knowledge section for you to see this further.

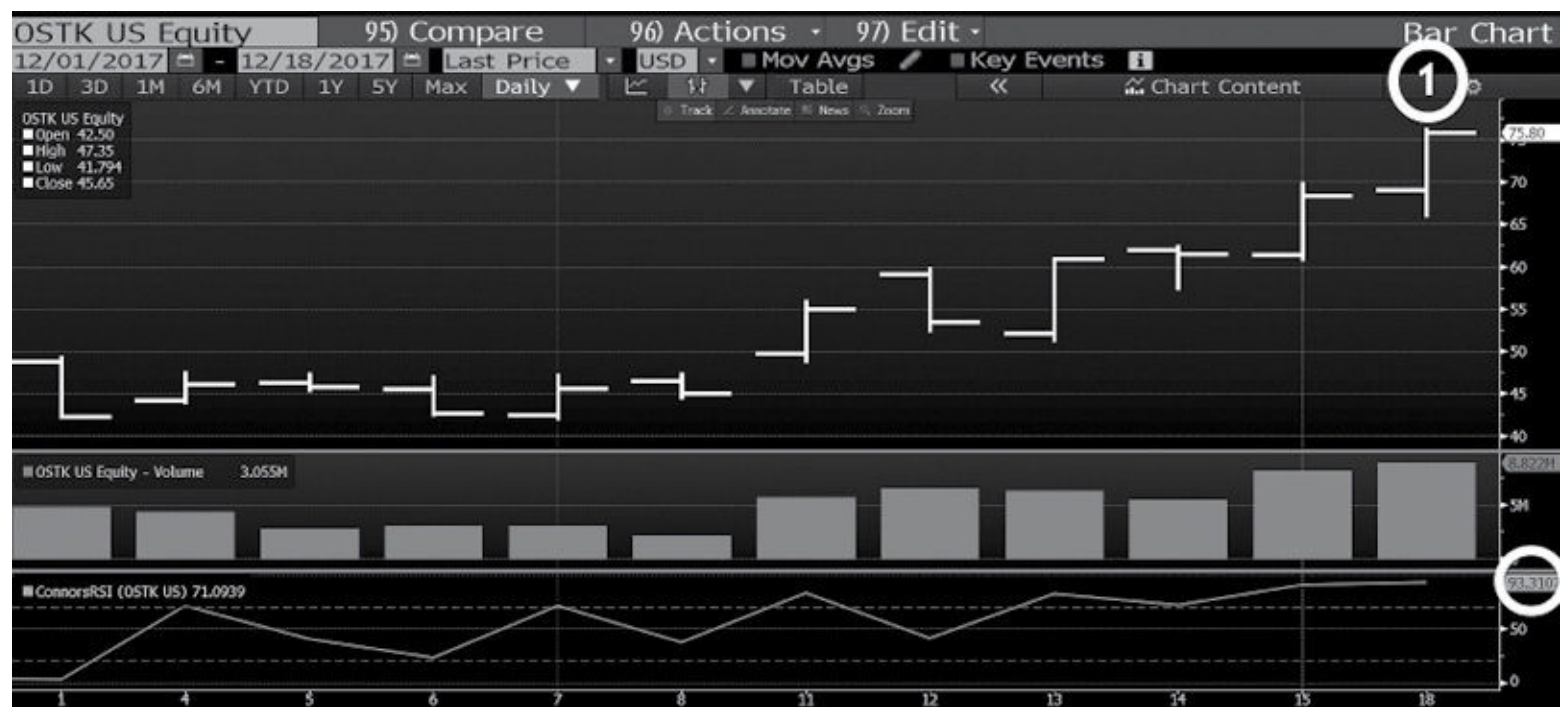
4. The stock has a closing ConnorsRSI (CRSI) reading of 90 or greater. Based on the way we wrote the CRSI algorithm years ago, we wanted to make it more difficult for a stock to reach the more extreme levels in order to make those readings higher-quality setups. When a stock or ETF gets above 90, it means it's extremely overbought. It doesn't necessarily mean a reversal is imminent. It simply tells us that the buying has been persistent and, as a whole, the security has a higher likelihood of reversing over the short term.
5. Short the stock the next day on a limit order higher. We tested 3% higher and 5% higher. The greedy have become greedier, FOMO becomes a further factor, and short sellers are getting squeezed. It's time to sell short.
6. The exit is when the stock closes with a CRSI reading closing under 30 (also included are test results with closing CRSI readings under 20). This allows the bubble to burst and the greed money turns into scared money as panic selling oftentimes ensues. **This is typical bubble-bursting behavior quantified.**

Here are the test results with a CRSI exit of 20 and 30 from 2007-2017:

Table 3.1. CRASH Strategy with CRSI Exit of 20 and 30 Results

% limit	ConnorsRSI Exit	# Trades	% of Winners	Avg % Profit/Loss	Avg Days Held	W. Avg % Profit	W. Avg. Days Held	L. Avg % Loss	L. Avg. Days Held
5%	30	527	71.73%	5.00%	4.15	11.68%	3.04	-11.97%	6.97
3%	30	719	71.49%	3.96%	4.32	10.53%	3.07	-12.53%	7.44
5%	20	510	73.33%	6.99%	11.85	16.68%	8.71	-19.65%	20.48
3%	20	685	70.22%	4.41%	12.41	15.35%	8.62	-21.40%	21.33

Example 3.1. Overstock.com, Inc. (OSTK)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

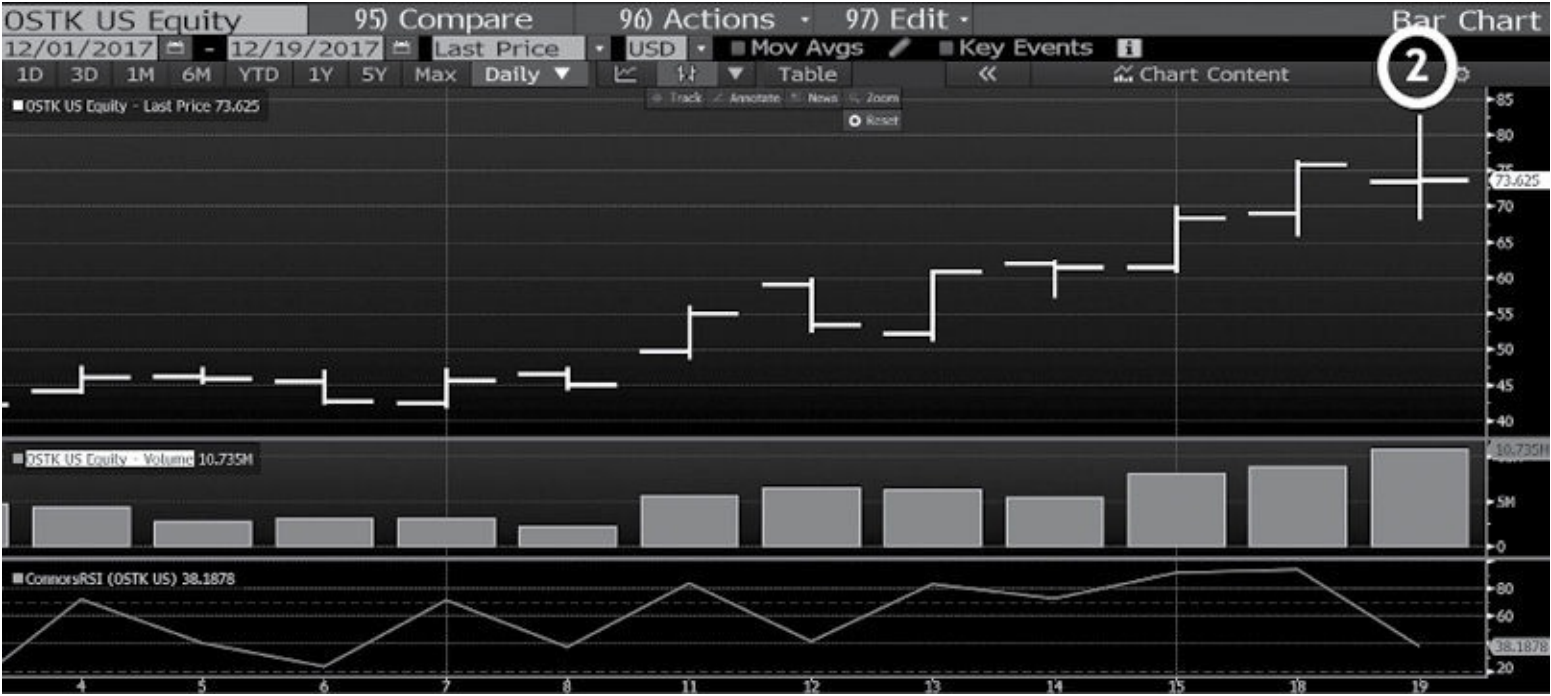
1. Cryptomania fever is in full force! Overstock, a company not known for doing business in conventional ways (that's an understatement), is back in the news again and its stock has risen over 70% in eight trading days. The company owns a Blockchain payments startup and all things crypto and Blockchain are hot! A day earlier, its Blockchain Unit Zero began a \$250 million token sale and traders and investors can't buy enough of OSTK.

On top of that, short sellers are panicking. A week earlier, the NASDAQ Stock Exchange reported the short interest in OSTK had risen from 32.4% to over 40% of the float. The stock's 100-day HV is a very high 102.5% (not shown), the ConnorsRSI reading is at 93, and you have the perfect storm for CRASH.

On an aside, I remember this day clearly. My then-19-year-old son was home from college break and his high school buddies came to our home that evening to visit. When I came out to greet them, one of them told me he was "invested" in Bitcoin (he works at Trader Joe's in his spare time). Before he finished his sentence at least three others jumped in and said they were in it too. (One said he was pretty sure he was in it — he had someone who "handled Bitcoin" for him.)

Right there neither you, me, nor anyone else in the industry needed CRASH indicators or any other indicators to know the short-term end was near for Bitcoin and anything related to Bitcoin. If you're looking for a good place to take the other side of someone's trade, four of my son's high school buddies is a good place to start.

Example 3.2. Overstock.com, Inc. (OSTK)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

2. The next day OSTK rises another 5% intraday, taking the stock up over 75% higher in under two weeks before reversing late in the day. A CRASH signal is triggered on the way up on a limit at 79.59.

Example 3.3. Overstock.com, Inc. (OSTK)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

3. All good things must come to an end and the bubble bursts in Bitcoin. It immediately plunges the next day and it takes down “all things crypto” along with it. OSTK closes at 69.50, giving CRASH double-digit percentage gains for a one-day trade. The “you can’t lose” owning Bitcoin crowd goes on to watch both Bitcoin and OSTK lose more than 50% of their value following the CRASH signal...

Example 3.4. Bitcoin USD (BTC)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

The top bar in this chart represents the day my son's high school buddies visited my home. If you have children with ambitious young friends who have uncovered the keys to sure-thing investment wealth, you may want to be on the other side of those trades.

Biotechnology stocks and buyout rumor stocks are always the psychologically hardest to trade on the short side because speculation is rampant. These are the drivers, though, that create irrational prices and historical short-term trading edges.

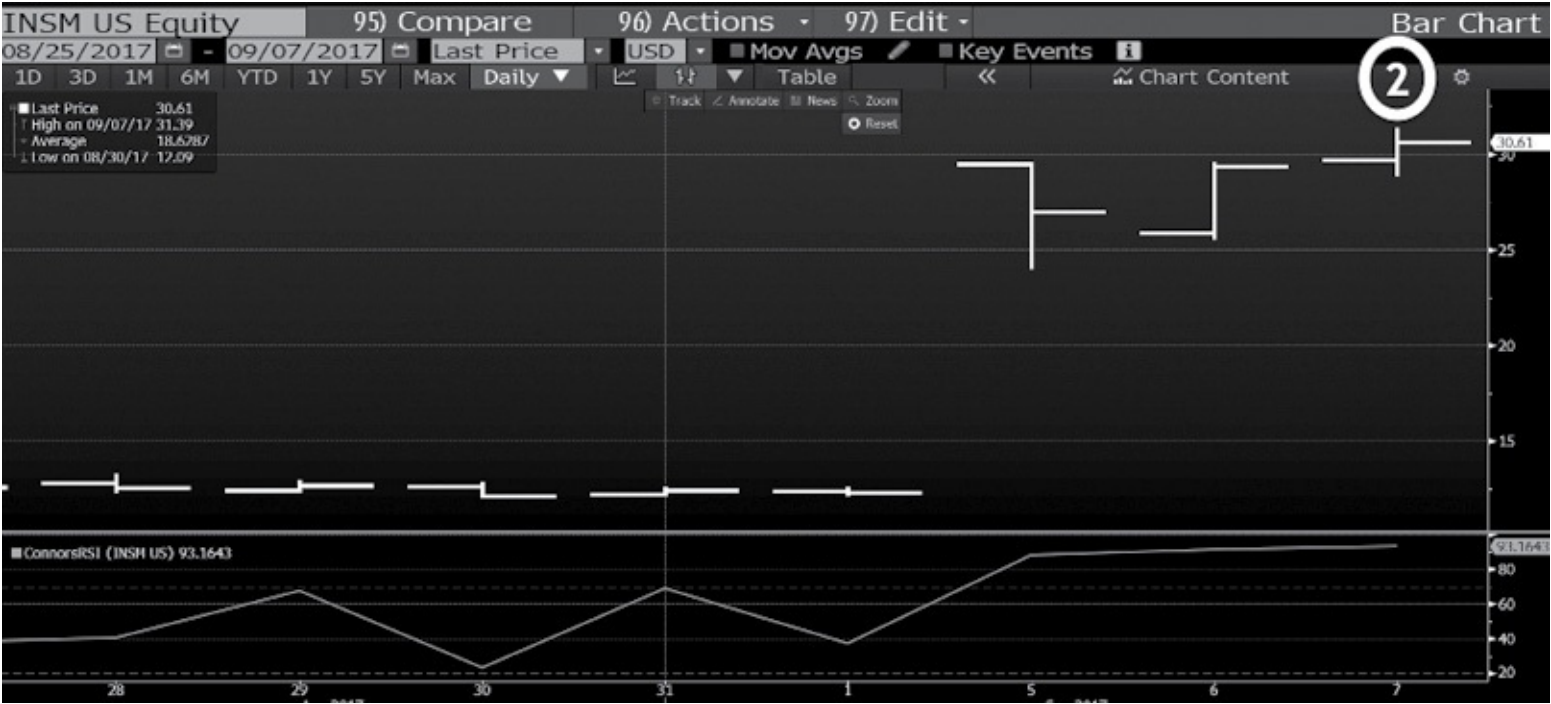
Example 3.5. Insmmed Incorporated (INSM)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

1. In this example, INSM got FDA approval to help clear lung infections. The stock soared on the news.

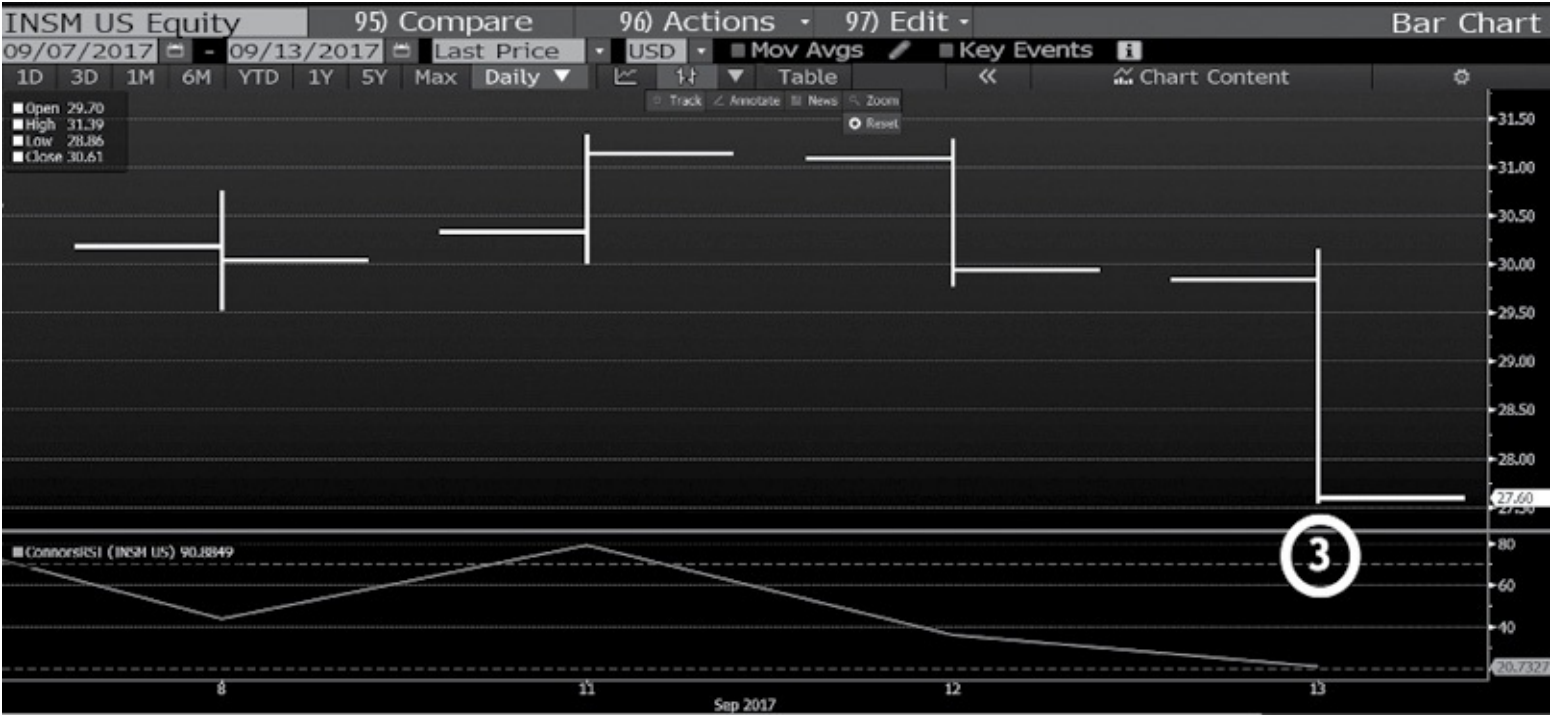
Example 3.6. Insmmed Incorporated (INSM)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

2. In this case, it’s simply a matter of the large upward move being too much. This type of excessive movement is often caused by future-looking speculation, along with plenty of buyout rumors. Sell short on the limit.

Example 3.7. Insmmed Incorporated (INSM)

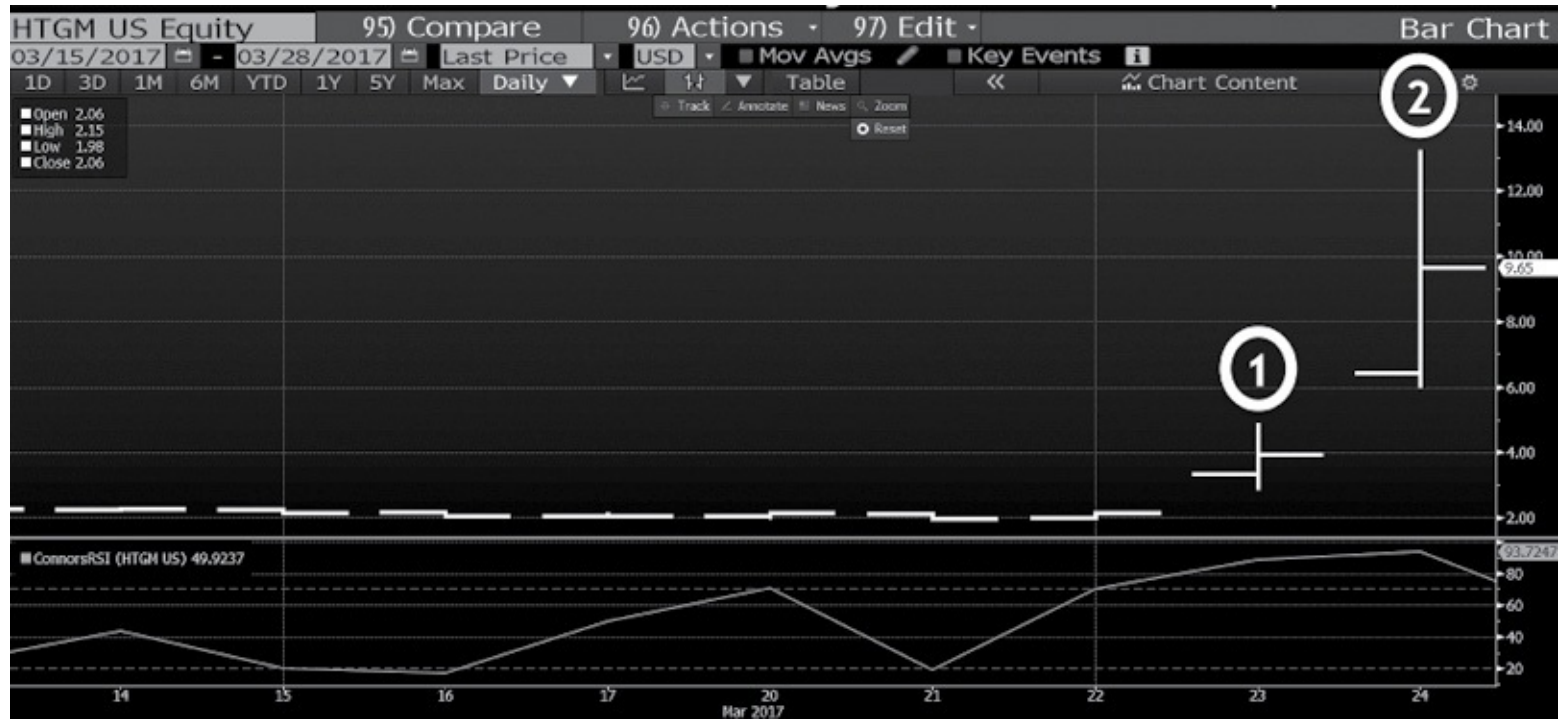


Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

3. Profit taking eventually comes in along with the late buyers who paid as high as \$30+/share now having losses, which furthered the selling. INSM loses over 10.5% of its value in a few days and profits are locked in.

HTG Molecular Diagnostics (HTGM) is an instruments and services company for molecular profiling applications, which has never had an operating profit.

Example 3.8. HTG Molecular Diagnostics, Inc. (HTGM)

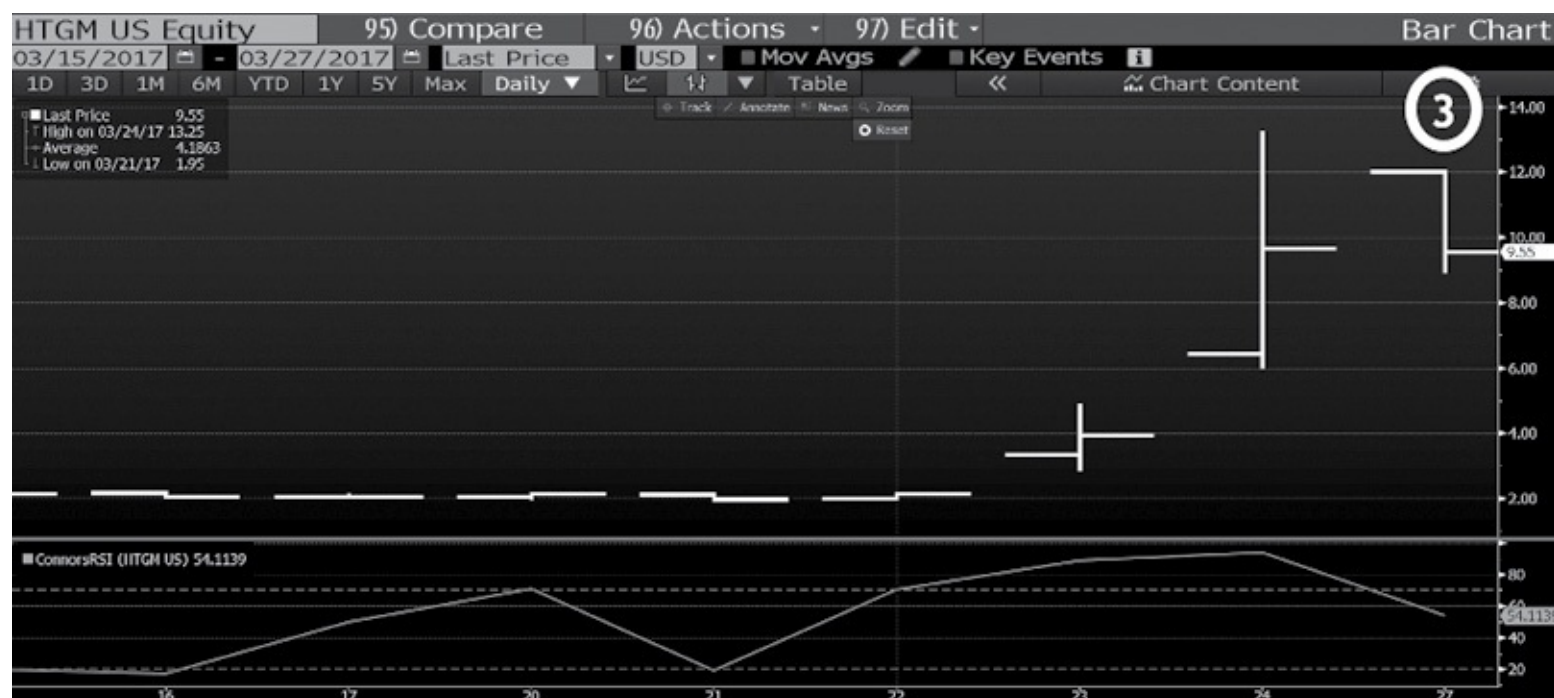


Source: Bloomberg Finance L.P.
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1. On March 23, 2017, the company announces that a new technology they owned was feasible with another firm's technology. Discussion boards go wild speculating with everything from sure profitability to them being bought out. CRSI closes at 88, so we don't have a setup yet.
2. The next day the stock surges 66% higher on buyout speculation after having been up over 200% intraday (crazy!). CRSI closes at 94 and a short signal will trigger on a limit the next trading day if the stock is up 5% intraday.

Speculators and shareholders now have the weekend to dream about how rich they're going to get. And the FOMO crowd has the weekend to realize they're missing out on a once-in-a-lifetime opportunity. They know they need in!

Example 3.9. HTG Molecular Diagnostics, Inc. (HTGM)



Source: Bloomberg Finance L.P.

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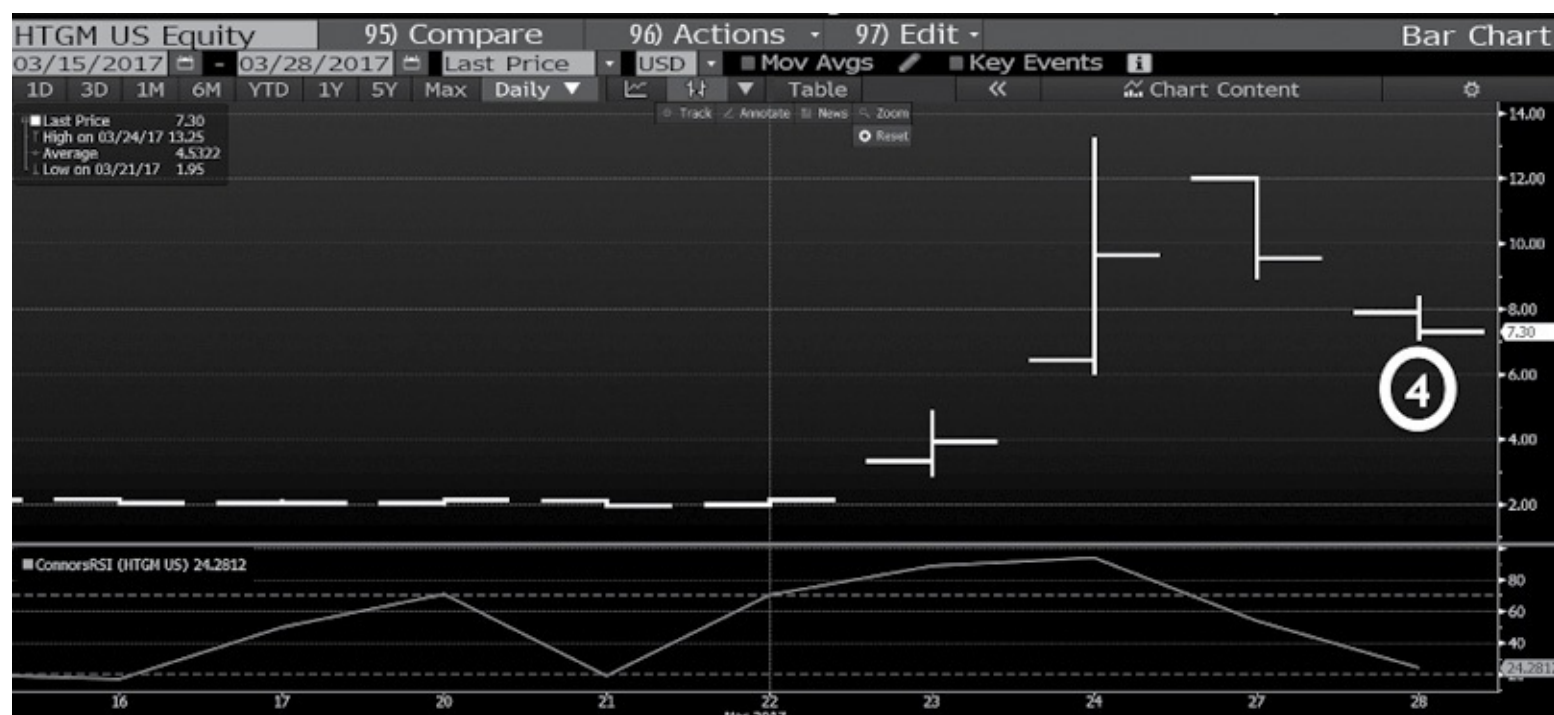
- Monday morning. The stock gaps up over 20% as speculation for the company is at fever pitch. The CRASH signal is triggered on the open and then the bubble bursts.

The behavior seen here over the weekend was originally discussed in the late 1980s by Larry Williams, one of the great pioneers in technical analysis. Larry created a very successful futures strategy based on this weekend gap effect he named "Oops."

The Oops strategy took the other side of Monday morning gaps in the futures market. The reasoning behind this reversal strategy was psychological. Buyers (or sellers) create pent-up emotion over the weekend, oftentimes due to some news event, and they would often overshoot prices on Monday morning. Fear (or greed) was at its most extreme on Monday morning and oftentimes those gaps reversed sharply.

Here with HTG Molecular, traders had the weekend to further work themselves up. The professionals on the other side of the trade are not dumb. They know this buying is coming and they step aside, letting prices open as high as possible. They then sell into this massive buying demand. This is exactly how it played out here in HTGM heading into Monday's opening.

Example 3.10. HTG Molecular Diagnostics, Inc. (HTGM)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

4. The next day the stock gaps lower as the selling continues and HTGM closes at 7.30, nearly 40% below yesterday's opening. Time to lock in the gains as the CRSI reading is under 30. As of the writing of this book in 2018 (more than a year later), HTG Molecular Diagnostics has yet to achieve a quarterly profit, it has not been bought out as was speculated, and it is trading more than 70% below its CRASH levels seen in March 2017.

An additional note for your knowledge: Even though we've just looked at two examples that saw significant longer-term losses in their stocks after CRASH triggered (OSTK and HTGM), CRASH is a quantified behavioral short-term trading strategy that sells greed. It is not a sell short and hold to zero strategy.

We're looking to enter when the greed is the highest (and you can see this especially in the Overstock and HTG Medical Diagnostics examples) and lock in the gains when the stocks reverse. Yes, some do go a lot lower, but many also move back higher. We're not looking to be in these positions for extended periods of time. The goal is to systematically wait for the buying to get to an extreme before the selling comes in. The CRASH strategy has successfully done this over 70% of the time since 2007.

Additional Knowledge

1. From 2007-2017, the S&P 500 Total Return Index (SPXTR) rose 10 out of the 11 years. Since 2009 the S&P 500 has been up nine consecutive years (with dividends included). Yet CRASH has seen over 70% of its signals being profitable with net positive average gains per trade during this time in the face of this strong upward bias.
2. As I mentioned earlier, we also tested lower Historical Volatility levels (both 60% and 80%) in order to see the robustness of the strategy. Due to the lower volatility of the individual stocks, the test results created more trades and slightly lower edges compared to the highest volatility stocks.

Table 3.2. CRASH Strategy with a Minimum HV of 60% and 80% and a CRSI Exit of 30 Results

HV(100) Entry	% limit	ConnorsRSI Exit	# Trades	% of Winners	Avg % Profit/Loss	Avg Days Held	W. Avg % Profit	W. Avg. Days Held	L. Avg % Loss	L. Avg. Days Held
60	3%	30	2420	68.02%	2.04%	4.9	7.79%	3.3	-10.18%	8.28
80	3%	30	1296	69.68%	2.96%	4.59	9.19%	3.23	-11.36%	7.74
60	5%	30	1500	69.87%	2.88%	4.75	8.80%	3.28	-10.84%	8.15
80	5%	30	892	70.96%	3.87%	4.39	10.14%	3.15	-11.47%	7.42

Table 3.3. CRASH Strategy with a Minimum HV of 60% and 80% and a CRSI Exit of 20 Results

HV(100) Entry	% limit	ConnorsRSI Exit	# Trades	% of Winners	Avg % Profit/Loss	Avg Days Held	W. Avg % Profit	W. Avg. Days Held	L. Avg % Loss	L. Avg. Days Held
60	3%	20	2333	68.07%	2.74%	12.57	11.21%	8.74%	-15.32%	20.72
80	3%	20	1240	69.11%	3.40%	12.49	13.18%	8.75%	-18.47%	20.84
60	5%	20	1467	69.87%	4.32%	12.16	12.80%	8.70%	-15.34%	20.21
80	5%	20	866	71.82%	5.44%	11.99	14.59%	8.79%	-17.90%	20.14

The higher HV filter obviously leads to less trade setups (it's a stricter requirement) and gets the highest returns. The lower HV requirements lead to more trades and slightly

less results. As a whole, the crazier the stock is, as measured by its 100-day Historical Volatility, the greater the imbedded emotion (this is intuitive). We ideally want this emotion to be as wild as possible because we see statistically this is when the largest reversals to the short side have occurred.

3. CRASH stocks are runaway stocks to the upside. There are usually stories (often crazy stories) as to why these stocks are never (ever) going lower again. Breakthrough technologies, must-have products, unlimited growth potential, buyout rumors (plenty of those!), and a lot of conjecture (including news and information manipulation being done online) drive these stocks to unsustainable levels. Eventually, though, as you see from the test results, the majority come back to earth (they CRASH) and the winners are those who shorted them at the correct time.
4. We know of no other short-selling strategy that has been 70% correct over the 11-year period from 2007-2017 with average gains per trade so high. Many of those years saw the overall market rise substantially, yet CRASH has correctly predicted lower prices over 70% of the time. Now imagine what may happen when a sustained bear market occurs.
5. The test results do not include the ability to borrow, nor the use of protection. In order to address both, this strategy may be the best strategy in the book to buy deep in-the-money puts (delta -80 or higher puts). The reasons why are fivefold:
 - a) You cannot predetermine your dollar risk on any short position no matter what the strategy is. You can predetermine your dollar risk with puts — it's the total cost of the puts. If liquid options are available, you may want to use them with this strategy. This also protects you from the open-ended risk when CRASH (or any short strategy) is wrong (and as you can see from the losing trades, the times CRASH is wrong, it's at times very wrong).
 - b) With puts you do not need to worry about the ability to borrow the stock.
 - c) Deep in-the-money puts will move fairly closely on a point-by-point basis replicating the movement of the stock, especially if the stock drops significantly. Delta -80 puts eventually move closer to delta -100 puts the deeper in the money they become and the nearer to expiration they get.
 - d) Because these stocks are so volatile, the spreads on the options tend to be wide. Orders may need to be worked. The good news is you're providing liquidity to the other side of the trade. This means that as the stock's price is running higher,

its put prices are often dropping. You are providing liquidity by being a buyer of the puts as they plunge lower in price.

- e) Look at the average gain per winning trade for the HV 100 stocks with a CRSI exit of 20. The average gain has been over 15% for these signals (that means on the trades that were profitable), and those stocks saw average gains of over 15% when the exit signal was triggered. These are obviously large gains for a few weeks of holding a position and clearly show just how irrational (and greedy!) the buyers are when they buy these stocks after they reach these extreme levels.

CRASH is the single best short-term short-selling strategy we've created. The behavioral approach of selling greed is quantified with this strategy, and unless human behavior changes, these opportunities will likely be here for years to come.

CHAPTER 4

Volatility Trading

“Victorious warriors win first and then go to war.”

Sun Tzu

A BRIEF HISTORY OF VXX AND THEN A QUIZ...

It's January 29, 2009. There's an announcement today that an exciting brand-new product with the symbol VXX will start trading. This exciting new instrument is one that is intended to “protect your portfolio” from the times when volatility rises.

Looking back at the wreckage of the previous year (2008), you sure wish you had owned something like VXX if you were long stocks. This product is a new form of insurance in the Exchange Traded Notes category — one that is always long VIX futures. With VIX futures, if you own them when implied volatility is rising, you make money. VXX in theory is supposed to do the exact same thing. Supposedly, if volatility rises as it had just done the previous year and you own this new product, it is believed you will have your portfolio hedged or, even better, you will make money — possibly a lot of money!

This is good news.

Now let's look ahead one year later. This brand-new product has gotten a lot of interest. But it's down over 65% since it was made public. OK, maybe it just had a bad 12 months. But the grumblings have begun.

Now let's look ahead almost three years later to the end of 2011. This exciting new product has now lost over 90% of its value since going public. And many articles have been written that say there's something wrong with this product. (Our research company was one of those firms that published a handful of these research articles.) But it's still

trading heavy volume. Every trade has to have a buyer and a seller. *So, who in their right mind is buying this???*

Now we'll jump ahead five years to the end of 2014. The product has incredibly lost over 99% of its value since inception. **It's been down every single year since 2009!**

And now let's look ahead to December 31, 2017 — over eight years after this product has been taken public. It trades an average of nearly 30 million shares a day. And it's now lost 99.99% of its value. Its price has been reset higher 5 times through 1 for 4 reverse stock splits. A reverse split is where a company increases the price by X times (VXX has always multiplied the price by 4) and decreases the amount of shares the holders have. For example, if the price of VXX is \$8 a share and you own 1,000 shares, on a 1 for 4 reverse split the price jumps up to \$32 overnight but you now own only 250 shares. Four times higher price and $\frac{1}{4}$ the number of shares.

You're probably asking why would a security like this keep reverse splitting? The main reason is to keep it trading. In this case if it didn't continuously reverse split, the incessant price erosion would eventually take it to under a penny. The main parties it helps include the issuers of VXX who are paid "management fees" and the brokerage firms who profit from commissions from the continuous daily activity.

Now that we know the price history of VXX, let's go back to 2011. You've seen what's happened with this product since 2009 and you've likely read some of the research out there stating that "VXX is built to go to zero!"

We'll now have a quiz:

Would you in 2011, after seeing what's happened to this trading product...

a) look for ways to go long this trading product?

or

b) look for ways to go short this product?

Which one would you like to be? The buyer of this product or the seller? Every trade requires a buyer and a seller. Which side would you prefer to be on?

I'd estimate about 99.9999% of us are voting for B.

Let's now jump ahead to 2014. We know the erosion in VXX has been relentless. It's down 99.9% since inception and it's lost money every year. You'd think no one in their right mind would buy this product. But that's not the case. Volume is rising in the product. It's actually becoming more popular! It's down over 99.9% and there are still buyers every day!!!

Part 2 of our quiz:

In 2014...

a) would you now likely be a buyer of VXX?

or

b) would you now likely be a seller of VXX?

Part 3 of the quiz:

And now let's look at this eight years later. It's 2017. VXX has reverse split five times since 2009, it's down 99.99% since it was brought public, and its average daily volume has reached all-time new highs during the year.

Instead of me asking the question, I'll assume you answered B. (If you answered A here, it's time to close the book.)

At this point in time, you obviously realize there's a major inherent directional bias to VXX and until proven otherwise, "THE PATH OF LEAST RESISTANCE IS DOWN!"

It's just amazing to me (and to just about every professional I know) that there are still buyers every day in this product. Not only has VXX not disappeared (as I calculated it would sometime around 2011), it's more popular than ever!

VXX was originally sold with good intentions as a hedge to protect your portfolio from rising volatility. It was a product that was brought to the market in response to the market sell-off of 2008. At its heart, though, it is a structurally flawed product that was in fact built (unintentionally) to go to zero. Yes, at times it has risen higher, especially in August 2011, the fourth quarter of 2015, and in February and March of 2018. But, as a whole, it's been a money pit for investors and investment advisors buying the product as protection and it's likely going to remain that way forever.

Before we look at two quantified ways to trade VXX (along with protecting ourselves from the few times it has short-term runs higher), let's first look at why VXX has lost its value. There are two reasons — one is discussed all the time, but the second reason is the key, in my opinion. This is the structural aspect of VXX — this allows you to understand why the erosion has occurred. Then once we look at the structural aspect, we'll look at the behavioral component — who is actually buying this product and why are they buying it? This will allow us to fully understand why it trades the way it does and then allow us to take two separate strategies: one a short-term panic strategy, and the second, a trend-following strategy, to climb aboard should the next nine years of VXX look like its first nine years.

UNDERSTANDING THE STRUCTURE OF VXX — WHY IT'S BUILT TO GO TO ZERO

Many people believe the main reason VXX loses its value is because of contango. I'll explain contango in a minute. The important thing to remember, in my opinion, is it's only part of the reason. The second reason is as important as the first and combined it gives us the formula as to why VXX erodes the way it does. Before we look at these two reasons, you should first understand how VXX is constructed:

VXX Construction

Reason #1: Contango

Wikipedia describes contango as follows: ***Contango** is a situation where the futures price (or forward price) of a commodity is higher than the anticipated spot price at maturity of the futures contract. In a contango situation, arbitrageurs/speculators (non-commercial investors) are “willing to pay more [now] for a commodity at some point in the future than the actual expected price of the commodity [at that future point].”*

Simply stated, if the VIX futures term structure is in contango, meaning the front month of the VIX futures are trading at a price less than the second month (and this has occurred the majority of the trading days for over two decades), then when VXX has to exit the front month and replace it with the next month, as it continuously has to do, it is selling low and buying high.

Here is a simplified example of this.

May VIX futures are at 14

June VIX futures are at 15

When VXX has to roll from May to June, they sell at 14 and buy at 15. Now imagine doing this trade the majority of the trading days since 2009 and you begin getting the idea that this is not going to be a winning situation over time.

As VXX has to continuously roll, it's oftentimes rolling for a loss. Yes, some days it rolls for gains because VIX futures do rise, especially ahead of important events or when there is fear in the marketplace. VIX is insurance for money managers and when they feel they need to insure their portfolio, VIX futures is one of the easiest ways for them to do so. But overall, this roll cost erodes the value of VXX and it's been estimated that the roll cost on average comes to about 4% a month. This means that on average one can expect VXX to lose approximately 4% a month from these costs.

Side Note - There's more to this because the formula behind it is mathematically complicated and this has led to various debates online about how "term structure" and "roll yield" exactly work. If you want to go deeper, start with this well written article by options strategist Larry McMillan titled *Misunderstandings about the VIX Futures Term Structure* (<http://www.optionstrategist.com/blog/2016/04/misunderstandings-about-vix-futures-term-structure>).

Let's now go further and look at this like a business. A business losing 4% a month is a business that's not going to stay in business too long. Only in this case, through the use of reverse splits and other people's money, VXX has stayed in business for nine years and running. And as long as it can find buyers for this product along with having the ability to reverse split it, it can stay in business for many more years.

Reason #2: The Volatility Premium

Reason 1 is the standard reason given why VXX erodes. Reason 2, though, is just as important (and discussed less often).

I'm going to walk you through this step by step so you have an understanding of the Volatility Premium:

1. VXX is made up of VIX futures.
2. VIX and VIX futures are made up of an estimate of how much volatility market participants believe will occur over the next 30 days. It's known as "implied volatility." If the VIX is at 16, it's implying volatility will move 16% (annualized) over the next month. When markets are quiet, the VIX will often move lower as the market is implying

volatility will likely be lower. When markets go wild, VIX spikes higher as market participants believe the daily movements of the S&P 500 will be larger.

In its most simple terms, VIX futures are considered insurance. If market participants believe there is a high potential for a market sell-off, they will pay more for VIX futures. This is no different than residents of South Florida seeing a massive hurricane bearing down on them in a few days. They're going to pay far more for insurance the days ahead of the storm, no different than portfolio managers will pay far more for insurance if they feel market volatility or a crash is in front of them.

The price of this S&P 500 market insurance changes constantly. It will often rise ahead of economic events, Federal Reserve meetings, global unrest, and any upcoming event that causes concern. Then oftentimes, after the event passes, especially if the worst fears are not played out (and this is often), VIX declines in value. Insurance demand lessens, as well as those who bought the insurance sell what they bought. This rhythm has been in place for decades and it's not likely ever going to change. Remember, **markets are made up of humans and humans have fear mechanisms hardwired in their brains.** Portfolio managers will buy insurance in the financial markets to protect their investors. They will also buy insurance to protect their jobs.

3. As we all know, most if not all insurance is overpriced. This is a generalized statement, but, as a whole, it's true. VIX futures is a form of insurance. And yes, as a whole it's historically been overpriced just like all other types of insurance.

One way to measure this is to take the difference between implied volatility vs. realized volatility. Implied volatility is the cost of insurance. Realized volatility is the actual volatility that has been produced by the actual daily movement of the S&P 500. Going back over 25 years, the majority of the time, realized volatility has come in less than what the market anticipated it would be. Stated further, implied volatility has been overpriced over 80% of the time over the past two and a half decades! Buying market insurance, the majority of the time, has not been a great investment.

So, why do professionals buy volatility knowing this? One reason, as I mentioned, is the insurance aspect of this — it buys protection, especially ahead of day-to-day events. The second reason is that it's often used as catastrophic insurance — if a 9/11 event occurs again (hopefully it never does), meaning an event that no one could have predicted, that portfolio insurance likely gets paid off. This is what happened in 2008. Buying insurance — buying 1-year VIX futures in late 2007 — looked really smart in late 2008.

What does this mean to VXX? It means that the underlying VIX futures it owns have been overpriced over 80% of the time. VXX is buying historically overpriced insurance.

And as time passes by and the realized volatility comes in less than the implied volatility, it loses money. VXX is holding onto an insurance policy that historically has been priced too high. And it has no choice. The products prospectus says it has to own these insurance policies no matter how overpriced they are.

PUTTING THE PIECES TOGETHER

So, let's now put the pieces together.

1. VXX owns near dated VIX futures. When it needs to roll out of the VIX futures, the majority of the time it pays more for the new futures than it sold. This is known as contango and on average it has cost VXX holders about 4% a month.
2. The VIX futures VXX owns have been inherently overpriced. As with most insurance, the price is too high. It has been overpaying for VIX futures the majority of the time. Continuously overpaying for anything erodes the bottom line.

There you have it. Monthly roll costs combined with trading in an overpriced product has led to losses of over 99.9% since the beginning of trading in 2009. Yes, the market may sell off and VXX will have its good times (as it did in August 2011 and some of the other periods I mentioned earlier). But as you can see, it's built to go to zero. And anything that's built to go to zero is something that one should consider finding ways to profit from on the short side.

THE BEHAVIORAL COMPONENT OF VXX

By this point you're probably saying to yourself, "Who in their right mind would buy this product?" Well, there are many buyers. I'm in no way inferring these buyers don't have their reasons. **But as a whole, these reasons are not data-driven reasons — they're behavior-driven reasons.** Let's look at them.

1. Advisors

I started my professional career shortly after college with Merrill Lynch in Boston. It was 1982, the Dow was in the 700s (on its way to the 600s) and I was the office's youngest hire to date. The saying goes that bull markets make geniuses of us all and in August 1982 Federal Reserve Chairman Paul Volcker made geniuses of us all by beginning the largest drop in interest rates in US history. Five years later, the Dow was over 400% higher and

yes, me at the ripe old age of 28 and everyone else in the industry during that period were geniuses.

During that time, and over the next seven years, I worked with literally hundreds of financial advisors (the title changed many times, but the job description remained the same). Many of the people I worked with did very well, many went on to have eight- and nine-figure net worths, and one in fact is a star on one of the major financial networks. As successful as many of these fellow co-workers became, there was one variable that was constant. *The knowledge gap between everyone was vast.* In most fields, there is usually a fairly tight correlation between knowledge and professional success. In the advisory industry (and I've lived this for nearly four decades), the knowledge gap is wide.

I worked with people who went on to build successful asset and advisory firms. These people worked hard, lived and breathed the markets, and created their success built around a philosophy of always learning.

On the other side of this, I worked with others who had the same type of professional success as the first group. These people were great at raising assets, a key component to building a successful advisory firm as they are paid on assets under management. But, when it came to investment knowledge — well, let's just say it was lacking. Lacking to the point where even subscribing to the *Wall Street Journal* was just something they had no interest in doing. Read a book on investing? Not for them. Discuss the market or upcoming macro events or study technical or quantitative strategies? The response was usually "Why?" I'm in no way saying this in a derogatory way. I'm saying this to get us to better understand who buys instruments like VXX and why do they do it. No judgment is being passed. I just want us to know who is on the other side of the trade.

I don't deal day to day with the broad array of advisors as I did many years ago. But based on my current conversations with professionals who do, it appears that it's "same day, different decade." A number of professionals who deal with advisors still tell me that many, many advisors care and they and their teams go the extra mile in making sure they know everything for the clients. This is the majority of advisors.

On the other side of the spectrum, there are still plenty of advisors who care — but their investment knowledge is secondary to caring about their ability to raise assets. The first group is likely never going to have their clients long VXX. **The second group is the group you want to be on the other side of their trades, especially at the right time.**

2. Uninformed Retail Traders and Investors

VXX is populated by retail traders and investors. Not all retail traders are dumb (in fact, academically many are smart). But when it comes to markets, they are not as smart (as a whole) as the professional volatility traders. If you spend time on certain sites that cater to retail traders that have message boards and you watch the discussions on VXX, you'll see exactly what I'm saying. Also, you can subscribe to Finviz.com, a really great data and information aggregation site. Put VXX in the symbol box and on the right-hand side of the quote page it aggregates live ongoing online conversations on every traded security including VXX. Watching the conversations and comments on VXX is eye-opening. I showed this to a professional trader friend of mine. His immediate response was, "How do I get on the other side of these people?"

Not all comments on these message boards are bad — in fact, some have great insight. But as a whole, these are not the professionals trading VXX — they're the retail clients and their level of sophistication is limited. And yes, you'll probably have the same reaction when you see these comments stream throughout the day. You'll be saying, "How can I get on the other side of these people?"

3. Fear-Driven Money Managers, Investors, and Traders

Now that VXX has become the de facto instrument to buy every time a market sells off or is anticipated to sell off, it's attracted the scared money who still believe it's going to protect their portfolio. In some days and even weeks it has, but it's down 99.9% in nine years, so unless we get a one-way massive bear market or if they change the structure of VXX, it's likely never going to do what the scared money expects it to do. They're fearful and they're buying an often-overpriced, structurally flawed product. At the right times, you want to be on the other side of their fear.

4. Short-Term Speculators Who Are Bearish

This is the same group as Number 3, but they understand that VXX has a long-term bias down. They are more hit-and-run traders looking to buy VXX and get a quick pop higher. They're not necessarily dumb money — they are buyers, though, and they play a role in the pricing and the liquidity.

5. Gamblers

I don't view quantitative trading and behavioral trading and investing as a form of gambling. We're not looking for action; we're looking for edges.

VXX is a volatility instrument and all securities with volatility attract gamblers who crave action. Gamblers are looking to score and the way to score with VXX is to be long because it goes up much faster than it goes down. Yes, most days it goes down, but on the few days it rises it sometimes does so significantly. Gamblers love this (it's like hitting the lottery or hitting a big slot machine). They're added to the list of buyers when asking the question, "Who in their right mind is buying this product?"

There you have the answer to "Who buys VXX?" I'm in no way passing any judgment on the character of these people. Most go in with good intentions, and like everyone else, all but the gamblers (who are primarily craving action) are looking to either make money or hedge portfolios.

Trading is a knowledge game. And it's also a "know your opponent" game. Knowing who is on the other side of your trade gives you a behavioral edge, especially if you understand why they are taking the trade.

Overview

We learned:

1. VXX is structurally broken. The majority of time its cost of carry of rolling the futures usually creates a 4% cost per month. It's built to go to zero.
2. VXX owns an underlying futures product — VIX futures — which has been inherently overpriced compared to its historical realized value.
3. We've identified the five main category of buyers of VXX. As a whole, this group is not the "smartest money in the world" except the few times VXX runs higher. Most of this money is uninformed, fear-driven money, or outright gambling money.

Let's now take our knowledge and put it to work in a VXX mean reversion trading strategy and a VXX trend-following strategy.

CHAPTER 5

Vol Panics

As we now understand from the inherent downward bias in VXX due to the way it's been constructed, it takes a good deal of fear to get VXX into an overbought condition. Essentially it's defying gravity because of the near constant contango combined with its underlying holding of VIX futures being perpetually overpriced. When it does get overbought, though, fear of an upcoming event or fear from an occurring event is high.

There are many ways to measure when fear is high. When applying a quantitative approach, you would like specific rules in place to measure the fear. We're not looking to "feel" the fear (though one can usually tell simply by watching or reading mainstream media outlets). We're looking to "precisely measure" the fear using structured rules that have historically told us that there is a great deal of fear in the marketplace, VXX is high, and historically shorting VXX at these measured levels has led to VXX prices declining over the short term the majority of the time.

With the VXX Strategies, I'll teach you two precise ways to do this. One holds a VXX short position for only a few days (it's very short term). The other holds it longer, allowing both for the fear to subside and for the structural aspects of VXX to play itself out over an average of a few weeks.

The test results begin at the inception of trading in VXX (January 2009) through the end of 2017. In the Vol Panics Strategy, we'll oftentimes see market panic in place. When they're scared, they're scared. They overprice VXX using it as protection (again, very poor protection). We're going to give them their protection, and as you will see, by applying the following rules, over 90% of the time you're profitably buying their fear.

Here are the rules for the Vol Panics Strategy:

1. VXX is trading above its 5-period moving average and its 4-period RSI reading is above 70. Sell short VXX.

2. Buy VXX on the close if it closes under its 5-period moving average.

If markets and VXX were efficient, this should be correct approximately 50% of the time. Some would say that because volatility reverts to the mean, the percentage should be a bit higher, likely 53-55% of the time.

In fact, this simple two-rule strategy has been correct 74.04% of the time occurring 104 times, or just under once a month on average over nine years. The average holding period has been just under four trading days.

Nearly 3 out of every 4 times that they've gotten concerned or afraid and elevated VXX higher as measured above, they've been wrong (and short sellers have been right) as prices quickly reversed down.

This is a good start. Can we do better? Let's see.

Let's take the above rules and become more aggressive as follows:

1. VXX is trading above its 5-period moving average and its 4-period RSI reading is above 70. Sell short VXX on the close.
2. Sell a second unit of VXX (doubling the position) if VXX closes higher than the entry day close.
3. Buy VXX on the close if it closes under its 5-period moving average.

Here are the test results from 2009-2017:

of Trades: **104 signals**

Percent Correct: **81.73% correct**

Average Gain Per Trade: **3.19%**

Average Hold: **Under 4 days**

Better than 4 out of 5 signals have been profitable. An additional day of fear as measured by higher VXX prices has led to an asymmetrical movement to the downside on the price of VXX.

Let's now look at a more conservative approach, one that scales into a full position. This means when VXX is oversold, instead of shorting a full unit you will dip your toes into the water and get short a piece of the unit. If VXX continues to rise, you will short more until you get up to a full unit. This can be done as follows using what is known as a 2-3-5 scale-

in approach and a 1-2-3-4 scale-in approach. These scale-ins will become apparent to you as you read the following rules:

1. VXX is trading above its 5-period moving average.
2. Its 4-period RSI reading is above 70.
3. Sell 20% of a full unit of VXX on the close. This means if a full unit size for you is hypothetically \$50,000 you will short 20% of \$50,000, which is \$10,000.
4. Sell another 30% of a full unit of VXX if at any time you're in the position and it closes higher than your entry price. This comes to \$50,000 times 30%, which equals \$15,000. Therefore, at this point you would be short \$25,000 of VXX or 1/2 unit.
5. Sell another 50% of a full unit of VXX if at any time you're in the position and it closes higher than your second entry price. This comes to \$50,000 times 50%, which equals \$25,000. Therefore, at this point you would be short a full position of VXX, which is \$50,000.
6. The exit is the same. We'll exit our position when VXX closes under its 5-period moving average.

Let's pause for a second and see what we've done. On day 1 VXX is overbought. We know historically prices have dropped nearly 3/4 of the time from here. But we're going to be more conservative with our position sizing and take a 20% position in order to allow us to be there if VXX continues to rise. If VXX continues to rise, it's becoming more overbought (fear is getting greater and attracting even more not-so-smart money). Prices started at an overbought condition with the RSI above 70 and now it's moved higher one, even two more days due to increased fear and, many times, outright panic.

Here are the test results:

of Trades: **104 signals**
Percent Correct: **90.38% correct**
Average Gain Per Trade: **4.29% on cash invested**
Average Hold: **3.32 days**

From the start of trading in VXX through the end of 2017, the price of VXX has declined over 90% of the time based on these fixed set of rules. These are large directional edges stemming from overblown fear combined with a security that is structurally inefficient.

Behavior plus understanding the structure, combined with structured rules has led to large directional biases like this for VXX traders for years.

Let's look at one more way to scale-in to VXX. This is done with a 1-2-3-4 scale-in.

Here are the rules:

1. VXX is trading above its 5-period moving average.
2. Its 4-period RSI reading is above 70.
3. Sell 10% of a full unit of VXX on the close. This means if a full unit size for you is hypothetically \$50,000 you will short 10% of \$50,000, which is \$5,000.
4. Sell another 20% of a full unit of VXX if at any time you're in the position and it closes higher than your entry price. This comes to \$50,000 times 20%, which equals \$10,000. Therefore, at this point you would be short \$15,000 of VXX.
5. Sell another 30% of a full unit of VXX if at any time you're in the position and it closes higher than your second entry price. This comes to \$50,000 times 30%, which equals \$15,000. Therefore, at this point you would be short 60% of a full unit of VXX, which is \$30,000.
6. Sell another 40% of a full unit of VXX if at any time you're in the position and it closes higher than your third entry price. This comes to \$50,000 times 40%, which equals \$20,000. Therefore, at this point you would be short a full position of VXX.

If your scale-in gets this far (and historically it has only occurred a small handful of times) it's usually perceived as the end of the world (or at least that's what the media is portraying it as). Fear is rampant and VXX will be extremely overstretched. The cost of insurance will often have skyrocketed at this point.

7. The exit is when VXX closes under its 5-period moving average.

Using the 1-2-3-4 scale-in exiting under its 5-period moving average has led to the following test results from 2009-2017:

of Trades: **104 signals**

Percent Correct: **97.12% correct (100 winning trades, 4 losing trades)**

Average Gain Per Trade: **4.86% on cash invested**

Average Hold: **3.32 days**

If you're looking to significantly extend the length of time you're in the trade to potentially capture even greater moves, you can change the moving average exit to a 4-period RSI exit under 20. This means instead of exiting when VXX crosses under its 5-day moving average, you exit when it crosses below a 4-period RSI reading of 20. By doing this the trades are held longer as panic subsides, and then complacency often kicks in for a while.

Here are the test results applying the same entry rules and exiting when VXX has a 4-period RSI closing under 20.

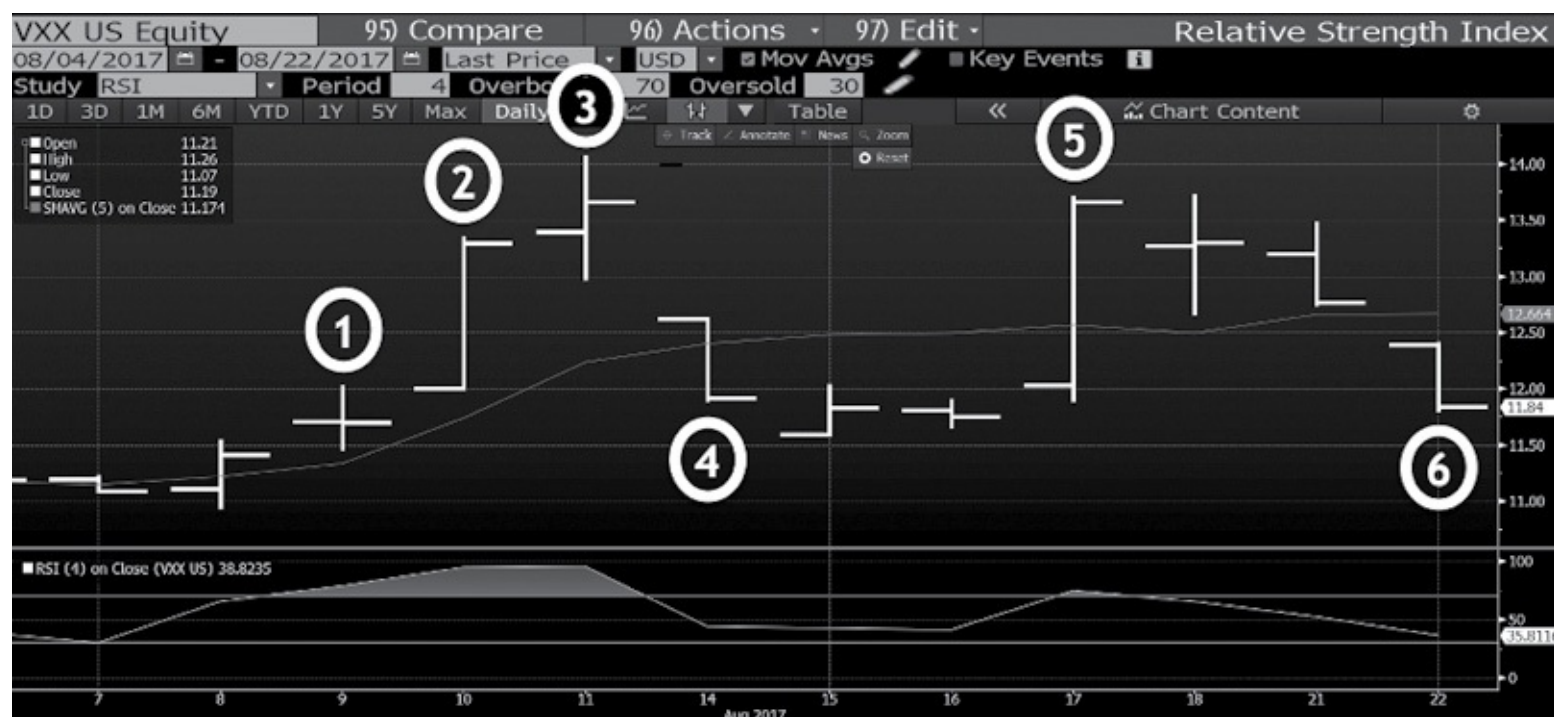
Table 5.1. Vol Panics VXX 4-Period RSI Closing Under 20 Test Results

Scaling	Exit	# Trades	% of Winners	Avg % Profit/Loss	Avg Days Held
None	RSI4<20	52	90.38%	7.41%	19.15
Double Up	RSI4<20	52	90.38%	13.72%	19.15
2/3/5	RSI4<20	52	92.31%	10.28%*	19.15
1/2/3/4	RSI4<20	52	94.23%	11.17%*	19.15

*Based on cash invested.

Let's look at a handful of Vol Panics examples.

Example 5.1. iPath S&P 500 VIX Short-Term Futures ETN (VXX)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

1. There is saber rattling going on as nuclear tensions between the US and North Korea increase. The cost of portfolio insurance begins rising and VXX has a 4-period RSI reading above 70. Sell short one unit (10%) of VXX at 11.70.
2. The next day President Donald Trump announces to the world that North Korea's nuclear threat to the United States would be met with **"fire and fury"** and the stampede to buy nuclear war insurance is on. The VIX spikes 47%, stocks have their largest one-day loss since May, and, as would be expected, few if any analysts or money managers are saying to their clients, "You know, now that Trump is threatening to blow North Korea off the face of the earth, I feel today would be a good day for you to go all-in!"

The 4-period RSI of VXX jumps up to 94 (a reading above 90 is extreme) and we'll short a second unit of VXX (20% at the closing price of 13.29).

3. President Trump tweets (welcome to the new world of news dissemination) **"military options are locked and loaded"** and there is now full-blown fear in the marketplace. Insurance costs continue to rise, as does VXX volume. Bloomberg reports that the past two days of volume in VXX is an all-time record (374 million shares). It's a very good assumption that the five types of VXX buyers we identified are active participants on the long side today.

With everyone panicking and the cost of insurance rising, guess what: You have a third scale-in signal — it's another day to "buy the fear."

4. There's a well-known adage that "risk happens fast." There's a lesser-known-yet-as-true adage that "risk comes off fast." And that's exactly what happens here. The rhetoric dies down and VIX loses over 25% of its value from the previous trading day's high, quickly taking VXX back down under its 5-period moving average and well under its closing prices of the two larger size scaling-in days. Lock in your gains in the short VXX position.
5. August 17. It's a well-known fact that "volatility has memory," and a few days later the cost of insurance via the VIX spikes over 20% as the saber rattling flares back up. VXX has a 4-period RSI above 70, which confirms this fear, and there's a short VXX signal triggered at the close at 13.66.
6. Three trading days later VXX again closes under its five-day moving average and gains of over 13% are locked in.

From 2009-2017 the 1-2-3-4 scale-ins like this have accurately predicted the price of VXX 97.12% of the time.

Example 5.2. iPath S&P 500 VIX Short-Term Futures ETN (VXX)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

1. The Dow plunges over 300 points as reports surface that former FBI Director James Comey has written a report alleging President Trump had asked him to drop an investigation of a former national security advisor.

At the same time steel stocks have continued their large sell-off as reports come in that the Trump administration will not be able to deliver on their promise on trade protection. Curt Woodworth, an analyst for Credit Suisse, tells Bloomberg News that “the Trump administration is somewhat in disarray.” The market has risen in anticipation of Trump protecting US companies via tariffs and should he not be able to deliver, stock prices will likely fall sharply.

Yesterday the Dow was trading at near record highs. Today, Trump has allegedly committed an impeachable crime and his administration is in “disarray.” The price of insurance as measured by the VIX jumps nearly 50% from 10.65 to 15.59 as a panic ensues.

If you get a chance, look at the last hour of trading of the day. I remember being in front of the screen that day while having CNBC and CNN on in the background. (CNN is the perfect source to give you the worst-case scenario regarding Trump and you get to see how much over-the-top fear they can induce.) The media was playing “can you top this” with their hyperbole and there was a selling frenzy induced by the hysteria. Bids kept dropping and any buying was immediately met with more heavy selling. This was the perfect Vol Panic.

2. The panic doesn't last long. Two days later order is restored. Trump isn't impeached, his economic agenda is not in as much disarray as reported, and VXX closes under its 5-period moving average, signaling it's time to lock in the gains. The Dow goes on to finish the month near all-time highs.

Additional Knowledge

1. A word of warning. These are incredibly high directional test results. With that said, THIS DOES NOT MEAN VXX HAS TO DROP. No security has to automatically go either higher or lower. What we are measuring, though, is fear and applying an instrument that feeds on fear. You always need to respect the market and apply risk management techniques to your positions.
2. You may be asking if using VIX as a signal trigger to trade VXX would be even better. We did test this. The test results were good but not as good as VXX to signal VXX. The percentage correct was slightly lower, as were the other metrics. I believe this is likely because of the way VXX is constructed along with who buys VXX vs. who trades VIX-related instruments. The VIX instrument buyers are often institutional buyers, whereas VXX buyers as a whole are less sophisticated traders and investors as we discussed in the introduction to VXX.
3. Protecting from VXX (or any short position) moving 100% or more against you — this ties into the inherent problem of shorting that everyone faces. Any short position can move higher by an unlimited amount. **Stops will not help a trader if the large move occurs from close to open.**

Therefore, one of the better ways to trade VXX is to do it on a fixed risk basis. This means it can be done with deep-in-the-money puts (those with deltas -70 or more) or with shorting VXX and buying OTM calls (for example, 20 or 30 delta calls). A long -80 delta put is equivalent to shorting the stock and buying the 20 delta call. The difference, though, is that the 20 delta call will often trade with a tighter spread than the -80 delta put, potentially saving you some money. This will be offset a bit by the few days of borrowing costs to borrow VXX and this is where doing your homework on the execution costs side of trading comes in. Work with your broker if you need assistance with this — this is their job and they should be able to help you. If you're not satisfied with the answer, ask to speak with a supervisor. The brokers want your business, especially if you're an active trader, and they are there to serve you.

In my opinion, unless one is trading VXX with only a very small portion of their overall portfolio, constructing positions with options is a safer way to trade VXX. This is your personal decision because you know what's best for you.

4. What about the times VXX is not available to borrow? VXX from time to time is not available to be shorted (shares can't be borrowed). If this occurs, there are solutions.

Options are one solution. Therefore, you should be able to get positioned for any signal that triggers.

5. What about UVXY? UVXY is the 1 1/2 times version of VXX. It used to be the 2x version until volatility spiked in February 2018 and they reduced the leverage.

UVXY is VXX on steroids. Because of the added leverage the erosion that occurs is amplified. It's an instrument that is used by more aggressive traders (and gamblers on the long side) and it also has options. These options are not as liquid as VXX options (as of the spring of 2018), but they do trade fairly well.

Here is a chart of UVXY since it was brought to the market:

Example 5.3. ProShares Ultra VIX Short-Term Futures ETF (UVXY)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

And here is its somewhat incredible reverse split history...

Example 5.4. ProShares Ultra VIX Short-Term Futures ETF (UVXY)

UVXY US Equity

Settings

Dividend/Split Summary

ProShares Ultra VIX Short-Term Futures ETF

Range

2012

-

2017

☒ Adjust for Capital Change

Currency

As Reported

12 Month Yield

N.A.

Indicated Yield

N.A.

No datapoints to graph.

Last Price

USD 10.334

Payment Frequency

None

Type

All

1) Color Legend

☐ Show Comparative BDVD Forecasts

	Declaration	Ex Date	Record	Payable	Curr	Amount	Type
11)	06/27/17	07/17/17				1 for 4	Stock Split
12)	12/22/16	01/12/17				1 for 5	Stock Split
13)	07/01/16	07/25/16				1 for 5	Stock Split
14)	05/01/15	05/20/15				1 for 5	Stock Split
15)	01/10/14	01/24/14				1 for 4	Stock Split
16)	06/04/13	06/10/13				1 for 10	Stock Split
17)	08/28/12	09/07/12				1 for 10	Stock Split
18)	02/27/12	03/08/12				1 for 6	Stock Split

Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

Summary

VXX may be the best instrument available to buy short-term fear and the Vol Panics Strategy with VXX has been a terrific short-term strategy that has correctly predicted the direction of VXX a very high percentage over time.

During the early years of VXX, I was concerned that the marketplace would become smarter and I do believe it has. But with VXX attracting the buyers that it does, along with the way it is structured, combined with the most important fact, as we've seen, that it is a trading instrument driven by fear, these large directional edges may remain in place for many years to come.

CHAPTER 6

VXX Trend Strategy

We're now going to expand our VXX trading strategy repertoire and take our knowledge and apply it to trading VXX on a trend-following basis. We know VXX continuously erodes due to its structural inefficiency and ideally we'd like to climb aboard as it trends lower in price.

Trend following is very different than short-term mean reversion trading. A good trend-following system will climb aboard longer-term trends and stay with that trend as long as possible. The type of mean reversion trading presented in this book holds positions on average from 3-7 trading days. Our VXX Trend-Following strategy holds positions on average much longer.

There are many good books on trend following, and if you would like to learn more I'd recommend you start with Michael Covel's books. Michael has studied trend following for over two decades and he's had access to many of the greatest trend followers in the history of trading.

Most good trend-following systems are correct between 30-35% of the time. The better ones may get to the upper 30s% to as high as 40%. Even though they lose more times than they win, their average gain is higher than their average loss (it has to be to make money). With most trend-following systems, one good trend can take care of long periods of choppy markets.

Now let's go back to VXX. It's obvious the longer-term trend has been in one direction. And we know that based on the way it was built, there's a high probability it's going to remain longer-term in one direction.

With VXX the goal is to get short at the proper time, climb aboard as it declines, and stay aboard as long as possible. We also want to exit early if there's a change in trend because if there is we can safely be in cash.

How do we do this? It’s fairly simple: with moving average crossovers. A moving average crossover simply takes two different period moving averages. When the shorter period crosses below the longer period moving average, you go short VXX. When it crosses above its longer-term moving average you exit and go into cash.

The rules are simple for the VXX Trend Strategy and the test results are solid, especially for a trend-following methodology.

- 1. When the 10-period simple moving average (SMA) (we also tested an exponential moving average, EMA) crosses under its 30-period SMA, short VXX. (If you use the exponential moving average, short VXX when its 10-period EMA crosses under its 30-period EMA.)
- 2. Stay short until the 10-period SMA (EMA) crosses above its 30-period SMA (EMA).

That’s it. Two rules. Trend following is often very simple and we are going to keep it that way with VXX.

From 2009-2017, this simple two-rule trend-following strategy has produced the following test results:

Table 6.1. VXX Trend Strategy Test Results

Moving Averages	# Trades	% of Winners	Avg % Profit/Loss	Avg Days Held	W. Avg % Profit	W. Avg. Days Held	L. Avg % Loss	L. Avg. Days Held
SMA(10)/SMA(30)	32	56.25%	13.02%	53.34	31.64%	81.00	-10.92%	17.79
EMA(10)/EMA(30)	30	50.00%	11.93%	59.97	35.06%	105.07	-11.20%	14.87

Here are a few examples of this strategy in action.

Example 6.1. CBOE Volatility Index (VIX)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

Example 6.2. iPath S&P 500 VIX Short-Term Futures ETN (VXX)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

The final four months of 2017 saw a significant drop in “realized” (actual) market volatility. “Implied volatility” as measured by VIX was continuously higher than realized volatility. What’s interesting is when you look at the top chart above of VIX prices, you’ll see a lot of sideways action along with days of implied volatility spikes. Yet, look at the next chart on VXX. The 10-period SMA crossed under its 30-period SMA in early September and despite VIX initially declining and then moving within a tight range (along

with a few sharp upward moves), VXX saw continuous erosion much like it has seen since 2009.

1. VXX was above 45 in early September when the short signal triggered.
2. The VXX Trend Strategy caught another move and VXX closed the year at 27.92, more than 25% lower.

This next example is shown to answer the question that is often asked: “Why don’t I just always stay short VXX?” It’s also why using puts instead of short VXX is suggested to predetermine your dollar risk.

Example 6.3. iPath S&P 500 VIX Short-Term Futures ETN (VXX)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

1. Earlier in the month a VXX Trend Strategy short signal was triggered.
2. It was exited when the 10-period SMA crossed above the 30-period SMA.
3. Over the next six weeks, volatility skyrocketed as the US Debt was downgraded by Moody's. VXX nearly doubled in price. By having a predetermined exit with the 10-day moving average moving above its 30-day MA, a small loss was incurred but a large loss was avoided.

Example 6.4. iPath S&P 500 VIX Short-Term Futures ETN (VXX)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

1 and 2. The above is another example of the VXX Trend Strategy capturing a large move to the downside as this structurally inefficient product erodes in value. As you have seen, this erosion has been in place since 2009.

Highlights

1. The percent of trades that were correct is the highest we have ever seen in a trend-following strategy. (There may be others out there that are higher, but if there are we have not seen them.) Remember, 30-35% correct is often the norm for successful trend-following methodologies. VXX Trend has run 50% correct for the EMA version and 56.25% correct for the SMA version.
2. Moving average crossovers have a mixed track record on most securities. We've done testing on them since the 1990s, and intensive moving average and moving average crossover research and testing was done as far back as the early 1960s, initially by Merrill Lynch followed by many other large firms. Up until 2009, though, there have been few if any liquid securities that have shown the type of built-in downward-trending characteristics of VXX. As the test results have shown, large moves to the upside have been contained using this approach (plus fixed risk put options further contain the risk), while at the same time the 10-day-30-day crossover has caught a number of substantial moves to the downside to generate large gains.
3. VXX trends for long periods of time. The average holding period is just under three months (53 trading days for the SMA and 60 trading days for the EMA). Contango, overpriced insurance, and the times that the five buyers of VXX step aside enhance the trend to the downside oftentimes for many months.
4. In my opinion, as with the Vol Panics Strategy, this strategy lends itself well for deep-in-the money puts. The deep-in-the-money puts allow you to participate if volatility gets crushed (this is known as a "vol crush") and VXX collapses in price. It also allows you to predetermine the dollar risk of the trade ahead of time. VXX options are liquid and you can often get filled in the middle of the bid-ask spread. Because the average holding period is so long, three-month expiration options should be considered. Also, for those of you who are more experienced with options or trade options professionally, you know the many ways you can potentially optimize the structure of your options position along with potentially bringing in income while in these longer-term trades.

Again, the key here is to be able to climb aboard, sleep at night by having your total dollar risk predetermined, and let VXX do its thing.

5. With this strategy you are "buying the fear." You are providing insurance to the many advisors and investors who still believe that VXX is protecting them. They are long VXX, believing that it's going to properly protect them. You, through the ownership of the

puts, are short VXX. They're afraid of something. You're giving them the protection for whatever they're afraid of.

6. In the Appendix there is an additional way to trade this strategy with options. The strategy should primarily be applied if you believe an imminent volatility crush is going to occur. This means you believe that for whatever reason, volatility is going to collapse (this often occurs after the VIX has been elevated for longer periods of time, and also when there are one-day spikes caused by incorrect news as occurred in May 2017). The option strategy is known as Risk Reversal Credit Spreads and Risk Reversal Debit Spreads. It's a fixed risk, potentially asymmetrical return option strategy that lends itself well to the times VXX goes through long-term trends on the short side.
7. Additional information on VXX and volatility trading can be found on many high-quality websites. The following is a reprint of one of the issues of the *Connors Research Traders Journal* (you can subscribe free to this on the TradingMarkets.com website) written in April 2018. After publishing this, a number of readers suggested I add VIX Central to the list and you can find that site at www.vixcentral.com.

I hope you enjoy these sites.

From the *Connors Research Traders Journal*, April 2018

1. *Six Figure Investing* (<https://sixfigureinvesting.com/>)

I like this site — a lot!

Vance Harwood has built an impressive information site that pretty much provides you with everything you need to know about VXX and the many other Volatility Exchange Traded products. It's broad, it's very deep, and it's a wealth of information. The navigation takes time to get used to but on the plus side of this is that you may end up on a page you didn't expect and on that page there will be information you never even knew about. One of my favorite sites on trading volatility.

2. *The CBOE Website*: <http://www.cboe.com/>

There's a great amount of information here. Start with subscribing to their daily blog newsletter. It's filled with information on options trading and volatility trading.

3. *Russell Rhoads on Twitter*: @RussellRhoads

Russell is one of my favorite trading authors. He's also one of the senior professionals at the CBOE. He's in the heart of volatility trading and he does a great job with his Twitter feed not only teaching you about options and volatility trading, but also keeping you updated on current market moves and what they may mean. Also, Russell is an accomplished author — if you haven't read any of his books, the two I recommend are *Trading Weekly Options* and *Trading VIX Derivatives*.

4. VIX and More: www.vixandmore.blogspot.com

I really enjoyed this site and unfortunately it doesn't update anymore. The good news though is that it has a number of back-tested strategies that are still applicable today. If you're looking for unique ways to trade volatility that includes test results up through the time they were published, this site has them.

5. SSRN: <https://papers.ssrn.com>

Living in Los Angeles in the pre-internet years, I would often spend my market-related research time in the library basements at UCLA, USC, and Pepperdine University reading academic studies in the back issues of the *Journal of Finance*. Some of the best ideas at the time came from these studies. Back then there were large competitive edges having access to this knowledge.

Today, you don't have to leave your computer or even your mobile phone to gain access to those studies. What is lost in scarcity is gained in sheer volume with SSRN. If you enjoy learning from academic studies, there's a good chance you're familiar with SSRN. There are over 750,000 academic papers across 30 disciplines on SSRN. Using their search function, you can focus in on any trading topic that has a study (or studies) on it. A search on "Volatility Trading" will lead you to numerous studies and you can then sort them a number of ways including by most recent. An additional bonus to using SSRN is that you can basically search for any trading subject — I'm often surprised at what it brings.

A few caveats about these studies (and you may already know this):

1. Publish or Perish is inherent in academia. So keep this in mind when reading these studies. Are there gems in the study or was the study published because it had to be published in order to meet other criteria?

2. Some finance-related journals are better than others. Spend some time on SSRN and the better journals will become apparent.

3. *Be careful of how the study was done. This is an endless topic debated on in hundreds of other fields. I try to look carefully at how the test was done and for those of you with STEM-related backgrounds, you're especially qualified to do this.*

4. *As with all research, past results are not indicative of future results. With that said, if you see a study that is confirming your own trading observations it may be a green light sign for you to run with.*

All in all, SSRN is a worthy site, especially if you enjoy looking for new trading research ideas.

That's my list. As a whole, these sites combined provide you with daily insights, professional knowledge, new strategies to trade, and ongoing new research.

Summary

We have now looked at two more systematic quantified strategies to “buy the fear.” With these two strategies you're providing protection oftentimes to less sophisticated traders and investors. You're shorting a structurally broken instrument when fear is at its highest with the Vol Panics strategy and when fear is continuous with the VXX Trend Strategy. If you decide to use options, you will be predetermining your dollar risk, which is a key part of the risk management component of any successful trading strategy.

In 2011, I expected VXX to disappear. In my mind there was no way anyone was going to buy an instrument heading to zero. It's now seven years later and VXX is alive and well. As long as they continue to reverse split it and continue with its current structure, it will hopefully continue to offer you opportunities for years to come.

CHAPTER 7

Trading New Highs

In 2004, *The Journal of Finance* published a seminal study written by Professors Thomas J. George and Chuan-Yang Hwang titled “The 52-Week High and Momentum Investing.” George and Hwang wrote in their abstract that *“the 52-week high price explains a large portion of the profits from momentum investing. Nearness to the 52-week high dominates and improves upon the forecasting power of past returns (both individual and industry returns) for future returns.”*

They found that the 52-week high acts as an “anchor point” for investors. (The anchor point is one of the behavioral components of this strategy.) This anchor point brought the stocks to investors’ attention and once hit, they were less likely to buy and more likely to sell over the short term.

According to the study, oftentimes stocks make new 52-week highs in anticipation of upcoming good news. Once the news is released investors become slow to react (they had already bought, which ultimately led to selling pressure coming into the stock). This is the proverbial “buy on the rumor, sell on the news” philosophy that has been seen for decades by professionals.

After a pause or pullback, prices then ultimately resume their longer-term uptrend as the good news is further assimilated.

This behavior is behind professional trading desks knowing that it’s better to “work an order” versus chasing new 52-week high breakouts. The conclusion, backed by this study and backed by the data, is: Don’t chase the breakout; there’s time to buy. Together we’re going to go further with this.

We’ve been able to quantify this anchoring behavior found at 52-week highs the professors found in their study. We then placed exact rules around it, added a “fear element” to it, which further improved their findings and created what we call the “Trading New Highs” (TNH) strategy.

Let's go directly to the rules.

SETUP

1. The stock should close above \$5.
2. The 21-day volume moving average should be above 1 million shares a day.
3. The stock makes a 52-week high within the past 20 trading days. This is the high price of the day not the closing price.
4. The ConnorsRSI reading is less than 15. This is the added fear component to the strategy. The "buy the new 52-week high breakout money" now often have losses due to this pullback and these "breakout buyers" begin to go into "I'm getting very uncomfortable" mode.

ENTRY

5. If all setup conditions are met, buy on a limit the next day at X% below today's close. We tested X being 7% and we also tested 10%. Yesterday's "uncomfortable mode" turns into today's panic mode because the stock is plunging.

EXIT

6. Exit the position on the close if the stock's ConnorsRSI reading closes above 70.

Let's Look at This Together

What do we have here? We have a stock making a new 52-week high, oftentimes on good news (or anticipation of good news), and everyone feels good. This 52-week high is an anchor point according to Professors George and Hwang, a price level that is widely disseminated by the media and data companies, and it's known to many traders and investors.

The 52-week high often attracts "breakout buyers" who believe there is clear sailing ahead because of the breakout.

At the same time, professional money (knowledgeable money) begins selling into the buying, having been smart enough to buy ahead of the good news and they now begin locking in their gains.

In our rule-based scenario, we're looking for the selling to then become extreme, eventually creating a form of panic selling. Not only is the professional money selling, the

speculative breakout money who bought the 52-week high (**which is now a failed breakout**) together push prices significantly lower.

At this point the stock is extremely oversold. The company is usually the exact same company as it was when it hit its 52-week high within the previous 20 days. It's simply cheaper — much cheaper — and the smart money steps in again. Professors George and Hwang pointed out that 52-week highs are not often major long-term reversal levels. They're correct, because as we see here from our test results, these short-term deep pullbacks often lead to prices moving higher again, especially the week following the panic selling.

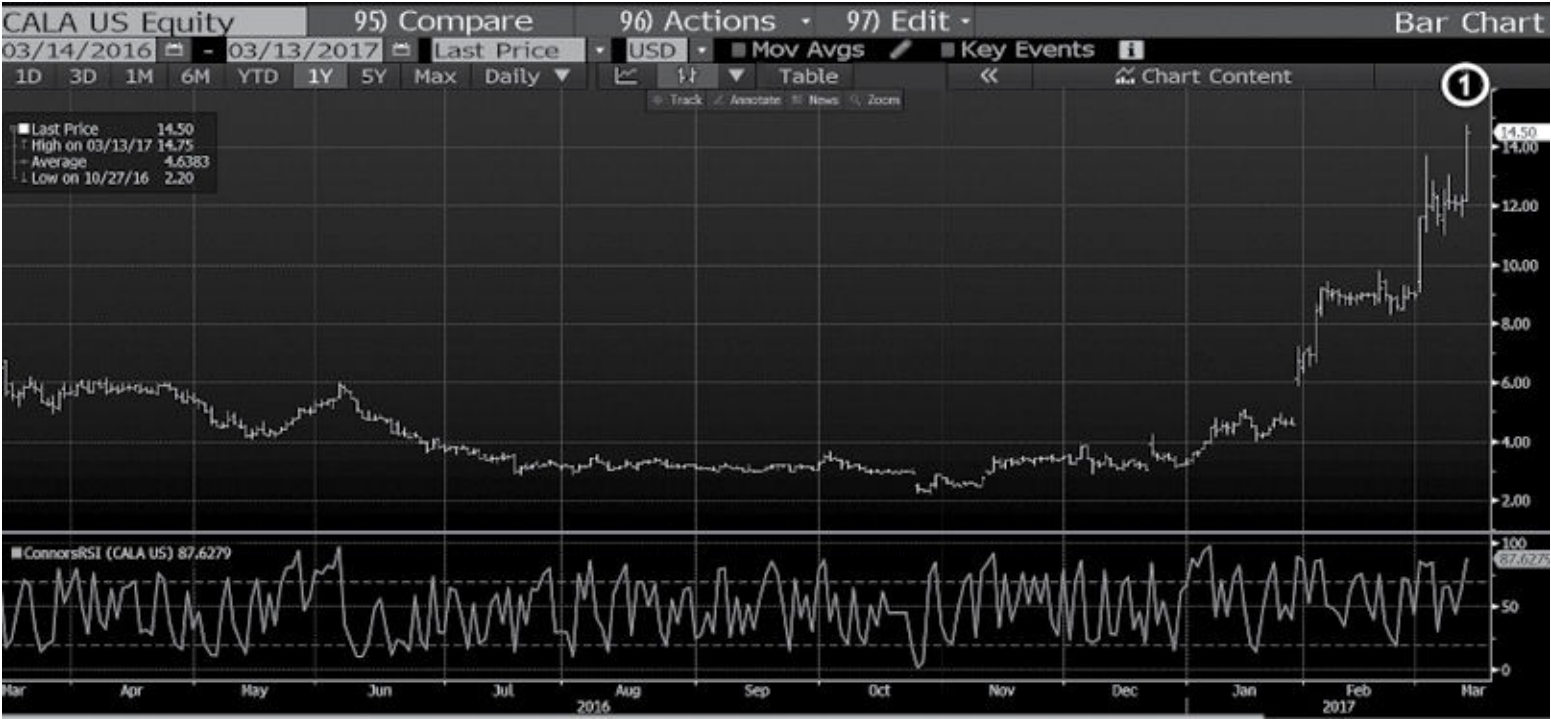
Let's look at the Trading New Highs test results from January 1, 2001 to December 31, 2017:

Table 7.1. Trading New Highs Test Results

Limit %	# Trades	% of Winners	Avg % Profit/Loss	Avg Days Held	W. Avg % Profit	W. Avg. Days Held	L. Avg % Loss	L. Avg. Days Held
10%	533	79.74%	6.47%	3.87	10.13%	2.96	-7.92%	7.44
7%	1367	77.03%	4.10%	4.03	7.57%	2.95	-7.53%	7.66

Let’s now look at two trade examples:

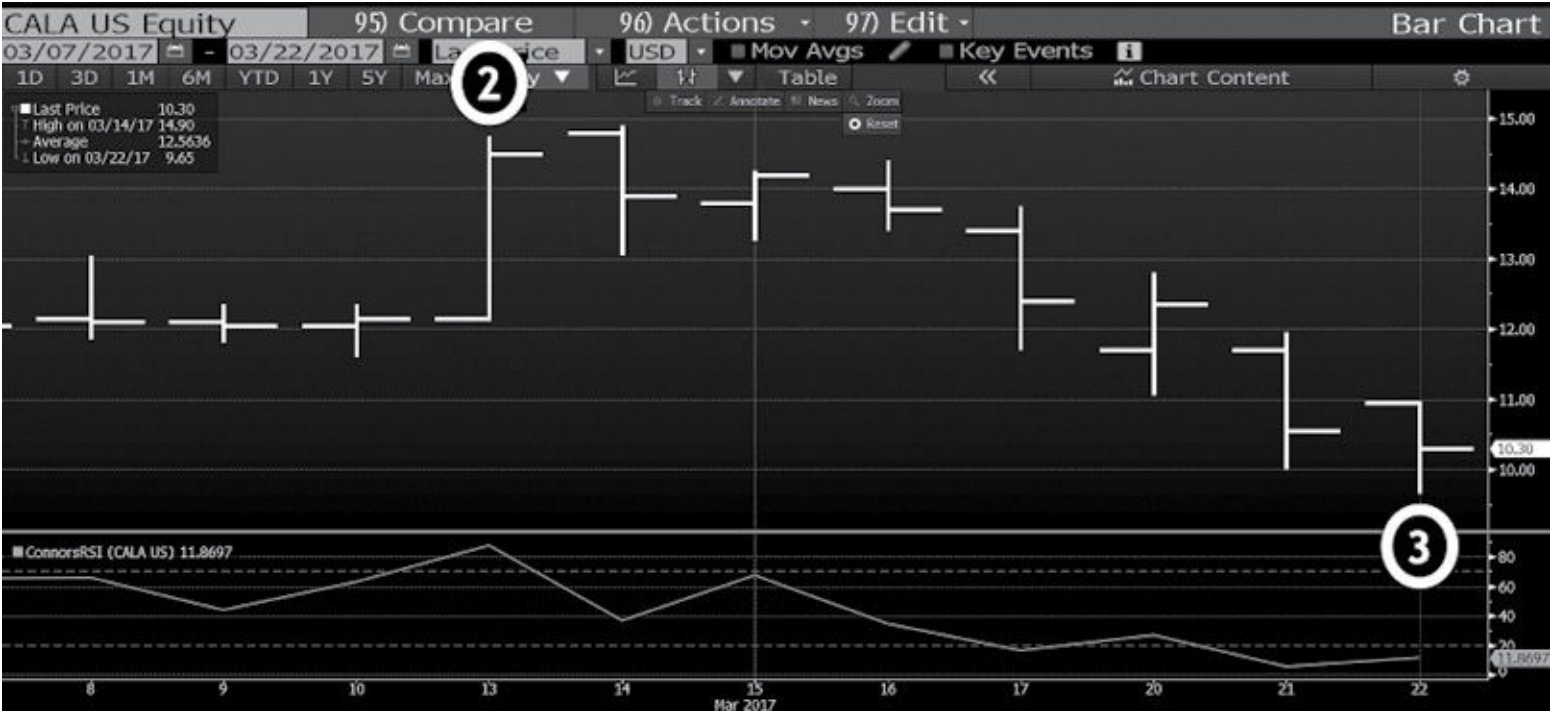
Example 7.1. Calithera Biosciences, Inc. (CALA)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

1. In March 2017, CALA, a biotechnology company, saw its stock make 52-week new highs as investors ran up prices ahead of the company’s annual earnings and, more importantly, an eagerly awaited update of its promising new drug pipeline.

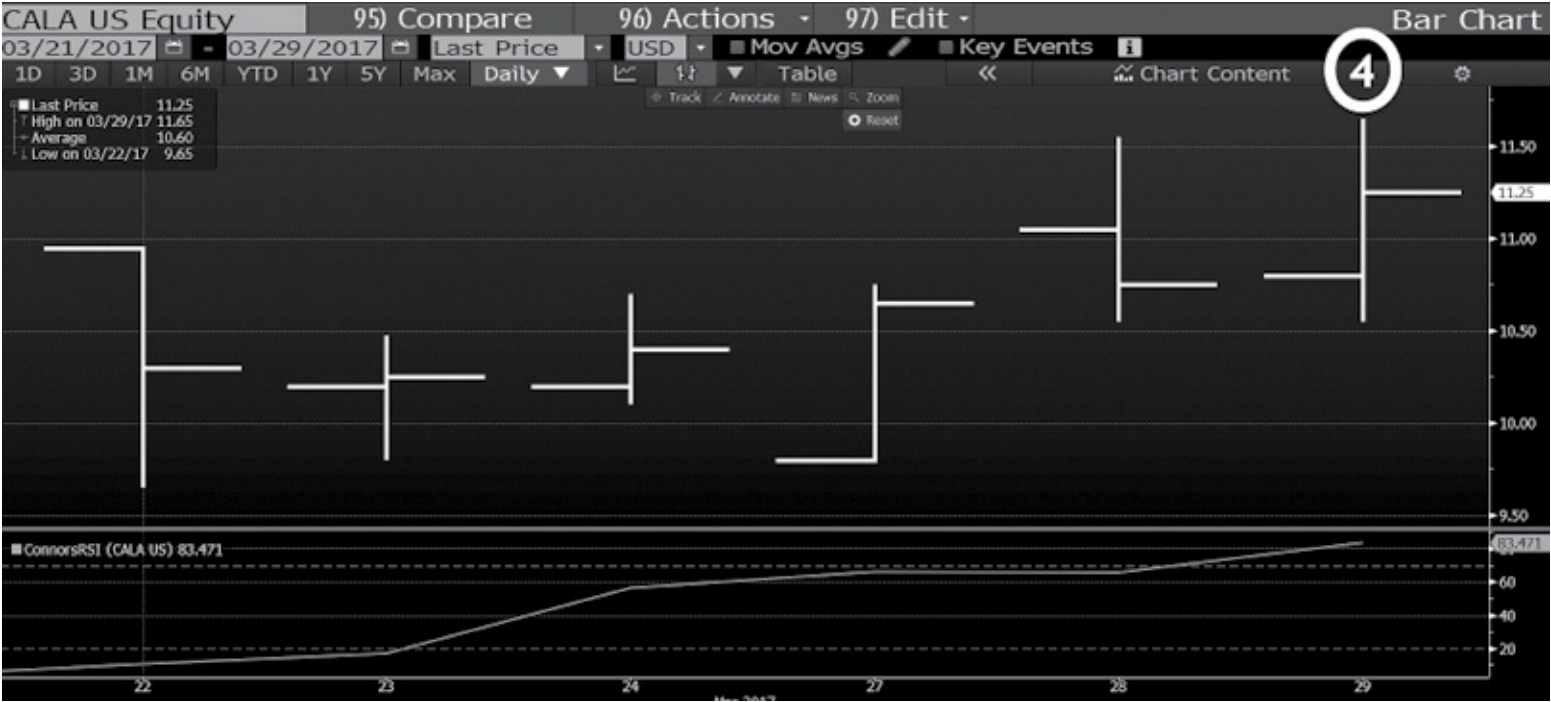
Example 7.2. Calithera Biosciences, Inc. (CALA)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

- 2. The company announces earnings and gives an optimistic outlook for its pipeline. The proverbial buy on the rumor and sell on the news will become clear-cut here.
- 3. The pullback begins and the behavior is in line with the findings Professors George and Hwang documented more than a decade earlier as prices pull back over the next seven trading days.

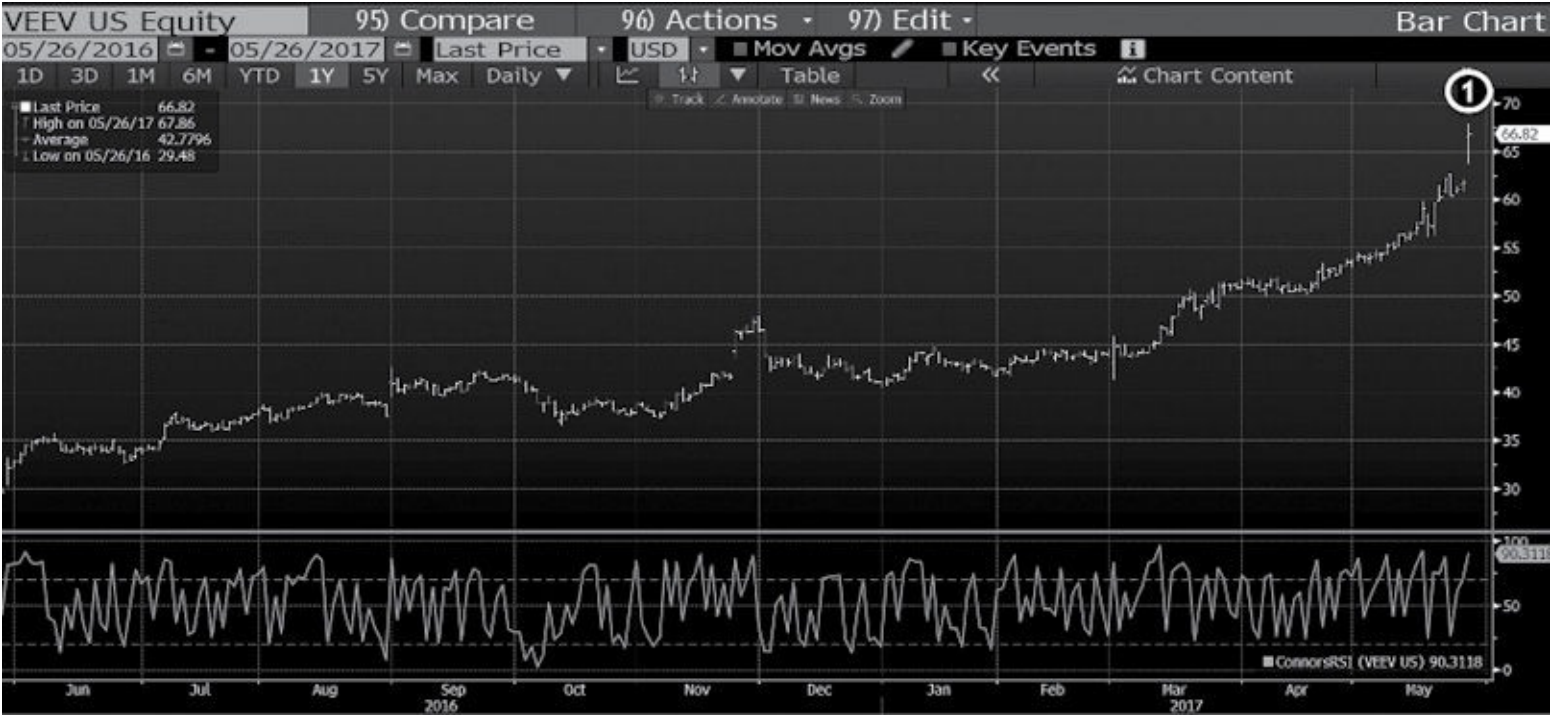
Example 7.3. Calithera Biosciences, Inc. (CALA)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

4. Good news starts coming back in again as the company begins achieving milestone goals and receives cash payments from a partner in reaching these goals. Buyers reappear, as has occurred over 75% of the time when this setup has triggered. CRSI closes above 70 with a gain of 14.5%. Profits are locked in.

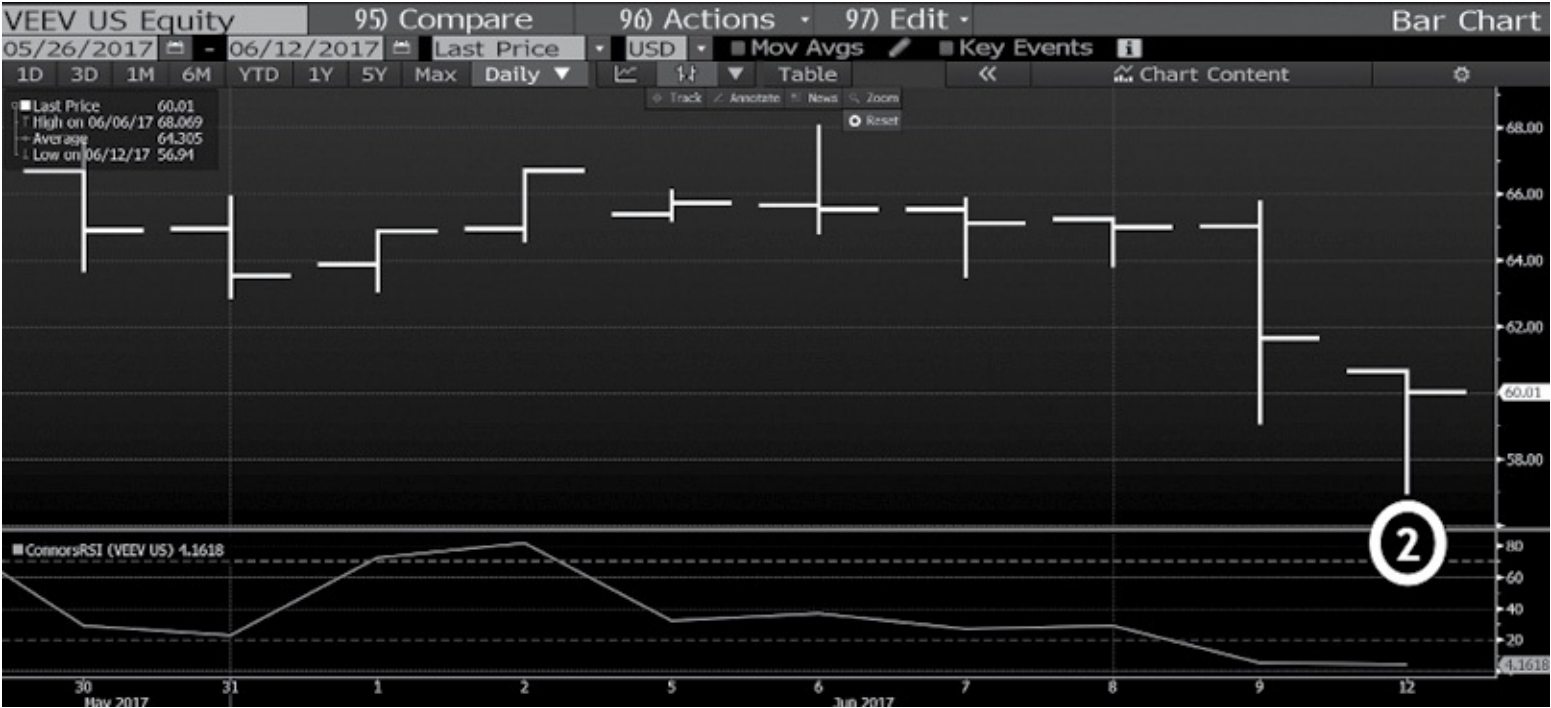
Example 7.4. Veeva Systems Inc. (VEEV)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

- 1. Heading into the May 26 earnings, Veeva Systems (VEEV) makes new 52-week highs. Earnings and the outlook are released and investors are pleased enough to drive prices to further new 52-week highs.

Example 7.5. Veeva Systems Inc. (VEEV)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

2. The days following the earnings release sees selling come into the buying and then the stock eventually capitulates. Buyers who bought on the news at the 52-week highs cannot be happy because they now have double-digit percentage losses and these losses become even larger intraday on June 12. The selling triggers a Trading New Highs buy signal at 57.34 near the day's low.

Example 7.6. Veeva Systems Inc. (VEEV)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

- Exactly one week later the buyers move back into VEEV and their buying triggers a signal to lock in the gains.

As we see with this example, after the news triggered the new highs, many long stock owners locked in gains. Within a few days, with profits shrinking for the remaining long holders, along with losses adding up for the new high buyers, the selling took prices to levels that have historically risen 77% of the time. Oftentimes, as was this case, speculative (and usually weak) money piled into the stock on the news and as sellers overwhelm the buyers, the sell-offs get overdone. The intraday sell-off after the pullback is often a sign of capitulation (panic) and a good opportunity to step in.

I can show you dozens of these examples. There are twists and turns to each but from a tops-down level they're basically the same. The 52-week high acts as the anchor point, news is oftentimes met with sellers who ultimately get panicked when the stock sells off. These sell-offs as a whole are not longer-term reversal points. They're actually great opportunities to buy these companies ahead of the institutional money who steps back in.

Additional Knowledge

1. The edges in this strategy are substantial. Average gains of over 4% in four days for 7% intraday pullbacks with 77% of the signals being profitable, and average gains of over 6.4% in four days with the 10% intraday pullbacks with 79.7% of the pullbacks being profitable is not something you see every day.

The professors' data-supported observation that 52-week highs have predictive powers, along with us adding a behavioral price component to it, meaning the CRSI reading under 15 pullback, combined with the intraday panic pullback, is a powerful combination to accurately predict short-term price movement.

2. We also looked at 39-week highs, 26-week highs, and 13-week highs to make sure that the price behavior we were seeing was not being created only by the short-term pullbacks. In each case of the shorter time period new highs, we saw far smaller edges compared to the 52-week highs. Professors George and Hwang's observation that the 52-week high is viewed by traders and investors as an anchor point remains intact.
3. The professors state in their study that "these studies [their study] present a serious challenge to the view that markets are semi-strong-view efficient." Most published academic studies that publish price anomalies make the same claim. In this case, though, their assertion appears to be well founded. The *Journal of Finance* is highly respected and this study has been widely disseminated over the past 14 years since it was published. Our test results start in 2001 and beginning in 2004 they walk forward after their study was published. Yet, we see that even after this information was published by them, adding a behavioral component (the extreme pullback) to their findings has produced significant high-probability short-term edges for many years.
4. There are many ways to apply this strategy and as with all the strategies, it's your personal choice on how to best trade this. The larger picture here is that the behavioral aspects of 52-week high anchoring combined with the addition of a fear component has led to substantial short-term edges in stock prices from 2001-2017.
5. Going forward, be careful blindly buying new 52-week highs. Yes, some do continue to rise. **But few if any studies have quantified positive short-term edges buying these highs.** The edges are created by patiently waiting for the pullback. In the Trading New Highs strategy this patience has been well rewarded for many years.

CHAPTER 8

TPS; *Fear and Greed Rising*

The more things change, the more things stay the same.

We're all wired to be excited by the new, new thing. New technology, new movies, new books, and especially new trading strategies. New is exciting. But often, some of the best things are right in front of us. They're readily available and are often better than something new and unproven. New doesn't always mean better.

There's a classic book written in 1890 titled *Acres of Diamonds* by Russell Conwell. The book remains on many people's all-time favorite inspirational book list. The message of the book is that one doesn't need to look elsewhere in the world to find riches. It's often right there in front of them.

Conwell's message has many meanings to many people. One of those meanings is that you don't always need the new, new thing. What's already available may provide you with everything you need.

Over a decade ago we introduced a strategy we titled "TPS." Since that strategy has been published I have seen many traders take it, improve it, build money management businesses around it, and ultimately allow it to become the core of the way they approach markets.

TPS epitomizes the fear and greed aspect especially inherent in equity indexes. It scales into fear as fear is rising and it scales out as fear is dissipating. It scales into greed as greed is rising and it scales out as greed turns back into fear.

At its best it buys the fear and sells the greed, especially with equity indexes around the world and especially with the main US equity ETF — SPY.

As with *Acres of Diamonds*, the TPS philosophy has stood the test of time after being published, in this case a decade that has seen great changes including the final stage of

the 2008-early 2009 market crash, a market melt-up, an unprecedented global credit expansion, a flash crash in May 2010, and in the United States a political regime change moving from Obama to Trump that is as extreme as any regime change our country has seen in its 242-year history.

As you will see with TPS, even though so many historical events have occurred and so many things have changed substantially since this fear and greed strategy was originally published, the behavior of TPS and its effectiveness has remained the same. Year after year, TPS has done its job of identifying when fear enters the equity index markets and then subsides, and when greed enters the equity index markets and then subsides.

The TPS strategy rules are simple. Like many of the strategies in this book, the edges have existed because of its fear/greed behavioral aspect. With this strategy you're buying the largest piece of the position when fear is at its greatest (the noise is loud!) and you're shorting its largest piece when greed is at its highest.

Let's now look at exactly what TPS is.

What does TPS stand for?

T-Time

P-Price

S-Scale-In

TPS identifies when an ETF is overbought or oversold and it then averages into the position as it becomes more overbought and more oversold. It then combines Time, Price, and Scale-In to enter a position as fear is rising, and on the short side, when greed is rising.

Before we get to the rules, let me first give you some background.

Beginning in 2006, we oversaw a private research group called Chairman's Club, which was made up of high-end professional traders from around the world. The group included people who ran hedge funds, traded for hedge funds, owned or traded for major proprietary trading firms, and also successful individuals who managed their own money. These people paid a substantial membership fee (five figures) to join and renew annually. They were among the better and smarter traders in the world and it was an honor for us to have the opportunity to work with them each year for many years.

TPS was first taught to the Chairman's Club members in 2008 and has since been expanded. As part of this expansion, literally tens of thousands of variations of TPS have

been published and presented to the Chairman's Club members. To put this in perspective, it would take thousands of pages (meaning an encyclopedia) for us to publish the complete TPS findings and the many ways to trade it. On top of that are the many additional ways traders on their own have expanded TPS. Some have privately shared those ways with me and obviously I can't, nor ever will, disclose those ways.

Therefore, it's impossible for TPS to be published in its entirety in one chapter. I felt, though, that it is important for you to see the concepts behind it and then you can decide whether or not you want to go further in trading, researching, and/or learning more about TPS. This is my way of saying that if you are fully embracing the concept of buying fear and selling greed, TPS is a strategy you can run with.

-

Here are the basic rules behind TPS. Please remember again that there are literally thousands of variations that can be traded with TPS.

Here are the TPS rules on the long side:

1. The ETF is above its 200-day simple moving average.
2. The 2-period RSI is below 25 for two days in a row. Buy 10% of your position on the close.
3. If prices are lower on the close than your previous entry price, any day you're in the position, buy 20% more of your position (you're averaging in).*
4. If prices are lower on the close than your previous entry price, any day you're in the position, buy 30% more of your position.*
5. If prices are lower on the close than your previous entry price, any day you're in the position, buy 40% more of your position.*

Using this 10%, 20%, 30%, 40% scaling-in approach (also known as 1-2-3-4), you now have a full position in a very oversold ETF.

*Do not take a new position anytime the ETF closes under the 200-day MA.

6. Exit on the close when the 2-period RSI closes above 70. Before we show you the test results, let's look at what we've done on the long side (on the short side we do the exact opposite).

Let's get into the heart of the rules in order to fully understand the behavior behind the price movement.

1. We wait for an up-trending ETF (above its 200-day) to become oversold by waiting for the 2-period RSI to close under 25 for two consecutive days. We then dip our toes into the water by only buying 10%. Oftentimes when an ETF gets to this level, selling has occurred and buying has begun drying up. It's usually due to concerns about an upcoming political or economic event, or there is an actual event occurring that is causing market concerns.
2. We then wait for the ETF to close lower and become more oversold another day and then commit only 20% more. More selling equals more angst.
3. We then wait for the ETF to become even more oversold as prices drop further and then we commit 30% more. At this point, angst is turning into pain. This now makes a minimum of four days of selling (in some cases the selling has been heavy). In this environment negative news is at the forefront.
4. We then wait for the ETF to become even more oversold another day and commit 40% more. This gets us to a full position. This does not happen often. There's been at least a full week of selling and often this day is a panic day. These trades will be the hardest to take because **it's going to feel like it's you on one side and the rest of the world, further induced by a hysterical media, on the other.** (Everyone I know who trades this strategy has said the same thing.) You are buying real fear here.
5. We exit when our position (no matter if we have 10% or up to 100%) moves higher, getting the 2-period RSI to close above 70. This occurs when the fear, panic, or hysteria has passed and as prices move higher, the fear subsides, making it "safer" for buyers to re-enter the markets.

Time, Price, and Scale-In combined get us in and out of the position.

The test results are from 2006 through 2017 for all non-leveraged, non-inverse ETFs that have had a minimum daily trading volume of 250,000 shares over the previous 20 trading days. **As you'll see, with the 2/3/5 version of TPS, it's been accurate 85.1% of the time. And for the 1/2/3/4 version, it's been accurate just over 88% of the time. For SPY, the dates of the test are from 1993 through 2017. For the 2/3/5 version, it's been accurate 91.47% of the time. And for the 1/2/3/4 version, it's been accurate 94.79% of the time. Rising fear very often leads to high probability trading opportunities.**

Let's look at the test results for all liquid ETFs from the rules above. Please note the Average Percentage P&L is based upon cash-actually-invested per trade.

Table 8.1. TPS Long Test Results - All Liquid Non-Leveraged, Non-Inverse ETFs

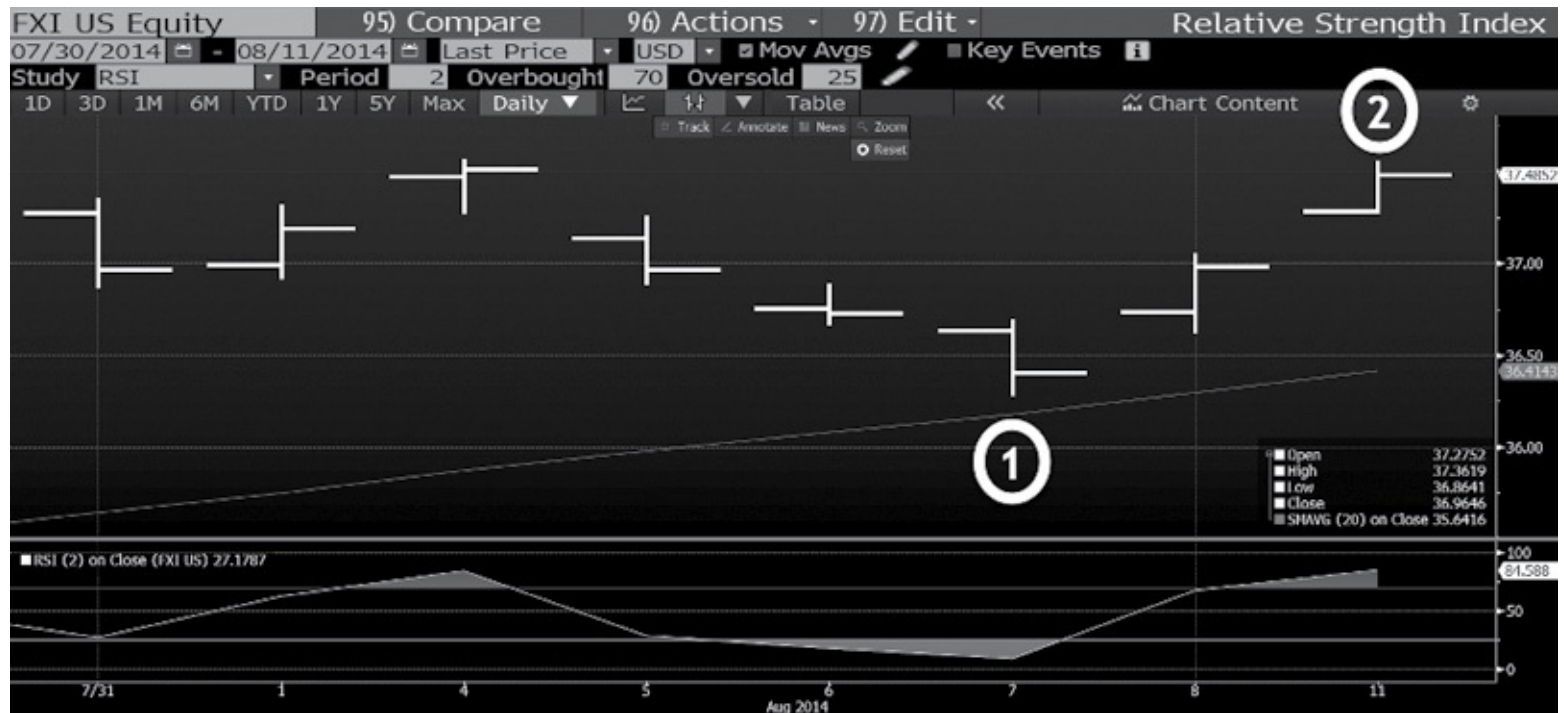
Trade Direction	Scaling	# Trades	% of Winners	Avg % Profit/Loss	Avg Days Held	W. Avg % Profit	W. Avg. Days Held	L. Avg % Loss	L. Avg. Days Held
Long	2/3/5	15,661	85.10%	1.06%	4.7	1.70%	3.72	-2.63%	10.27
Long	1/2/3/4	15,661	88.04%	1.18%	4.7	1.73%	3.87	-2.82%	10.81

Table 8.2. TPS Long Test Results - SPY

Trade Direction	Scaling	# Trades	% of Winners	Avg % Profit/Loss	Avg Days Held	W. Avg % Profit	W. Avg. Days Held	L. Avg % Loss	L. Avg. Days Held
Long	2/3/5	211	91.47%	1.14%	4	1.39%	3.42	-1.46%	10.22
Long	1/2/3/4	211	94.79%	1.21%	4	1.38%	3.59	-1.97%	11.55

This is an example of the TPS behavior seen in China — the main ETF for China is FXI.

Example 8.1. iShares China Large-Cap ETF (FXI)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

1. FXI is above its 200-day MA signifying a longer-term bull market. FXI pulls back and has two consecutive days of 2-period RSI readings under 25. Buy one unit.

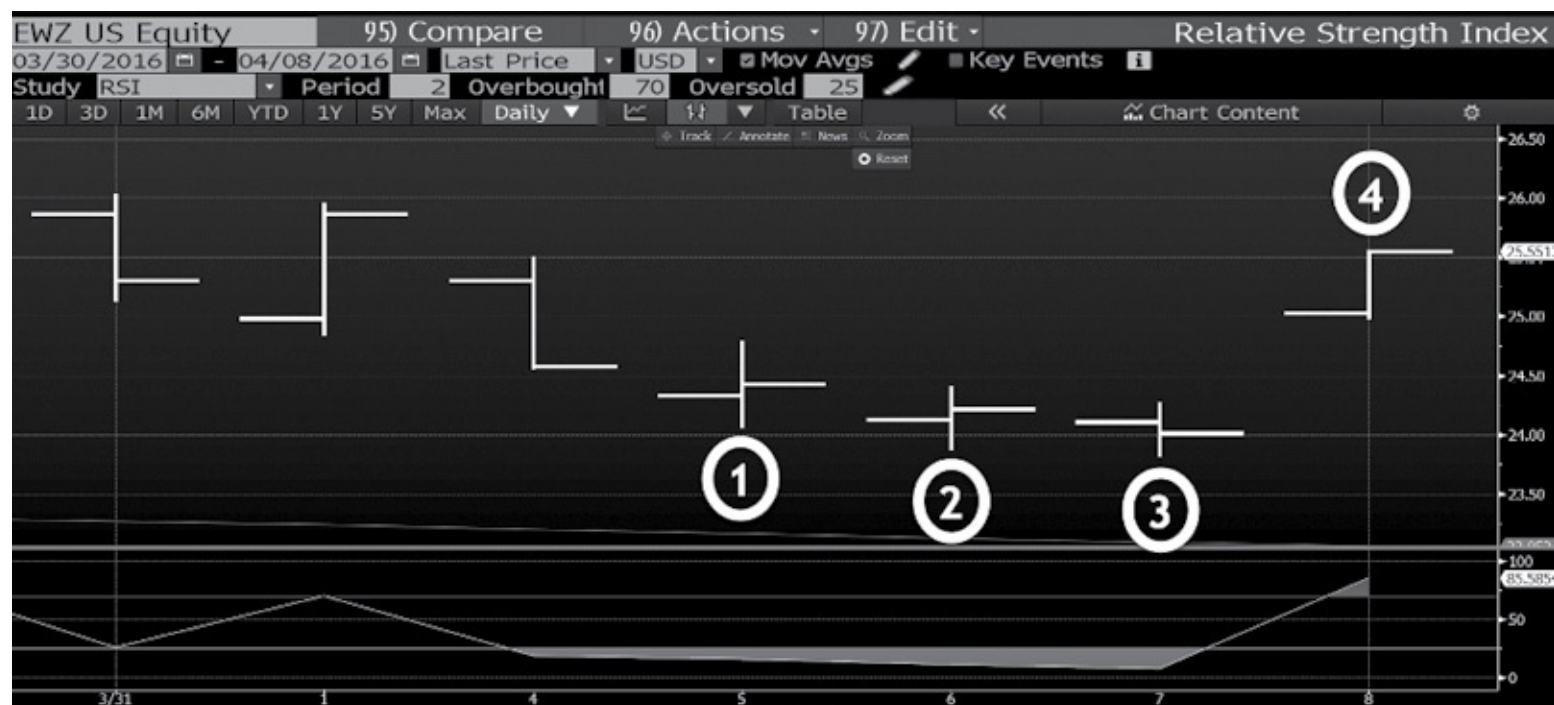
Let's pause here for a second and discuss this pullback. There's a well-known truism that "fear is a contagion." This means fear spreads. The military has known this for centuries and it's a widely accepted principle in many walks of life.

This is an example of fear being a contagion. Much of the world was selling off over the previous days, led by Europe. The Vanguard FTSE All-World ex-US ETF had dropped multiple days coming into this day. There were some concerns in China, beyond the omnipresent belief that any slowdown around the globe would impact China. As fear spread, it did so globally, giving traders an opportunity to go long China at an extreme pullback as the 2-period RSI was at single-digit levels.

2. As has occurred well over 80% of the time in major Country Fund ETFs, the fear subsides and buying returns over the next few days. The 2-period RSI closes above 70, telling us that the healthy edges of a few days ago from the global fear-induced

pullback have subsided and prices rise as this fear subsides. Time to lock in the gains and sell the position to those who feel safer buying higher.

Example 8.2. iShares MSCI Brazil ETF (EWZ)

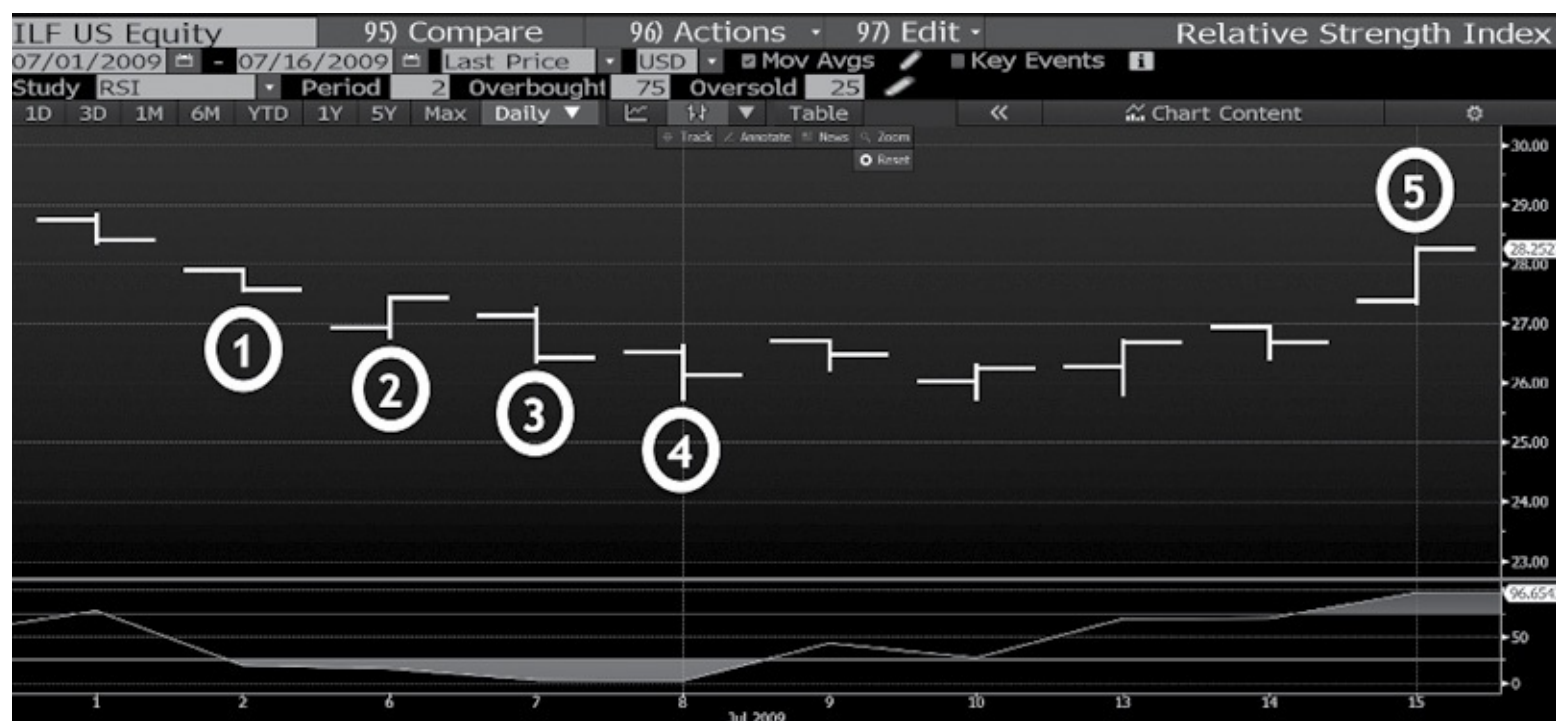


Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

1. EWZ is trading above its 200-day MA and begins selling off due to major political scandal concerns. The 2-period RSI closes under 25 for two consecutive days. Brazil's Congress is considering impeaching the country's president and fear of political upheaval leads to lower prices and greater fear. Price + political upheaval = buy the first unit.
2. The sell-off continues as buyers step aside due to the continued political uncertainty while sellers unload positions for fear of loss. Buy the second unit of the scale-in.
3. March 7, 2016. Fear rises even further as EWZ now sees its fourth consecutive day of losses and has lost 7% of its value for the week. At this point the "sensible" thing to do is either step aside or sell because the sell-off is turning into a rout. TPS signals its next entry, buying an additional 30% of the position at the close.
4. The next day EWZ rallies significantly as the panic was overdone. Buyers pay 4% more on the open versus the previous day's close and continue buying through the day as "things appear safer." Lock in your gains as the 2-period RSI triggers an exit signal.

Example 3: The Perfect Storm

Example 8.3. iShares S&P Latin America 40 ETF (ILF)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

The world markets were decimated in 2008 through the first three months of 2009. Fear was rampant and even as the markets began to bottom, volatility was extremely high as investors around the world were in no mood to lose any more money.

Let's learn something additional with this example on top of learning the TPS strategy. Markets coming out of long-term bear markets are some of the greatest markets to trade, especially on the long side. There's usually an abundance of cash on the sidelines due to the fear of loss being so high, the recency effect (people irrationally give more weight to recent events compared to less recent events), along with the potential for massive short covering from the many bears who are out there, all provide the rocket fuel to significantly move prices higher.

If you have the opportunity to study charts, look at price movements beginning in 2003 following the internet bubble crash and in 2009 following the credit crisis crash. The moves, especially in the higher volatility stocks, are eye-opening as a) fear subsides and investors begin piling back into stocks, b) the Fear of Missing Out (FOMO) kicks in, and c) the fear from the short sellers who are quickly seeing their gains turn into losses oftentimes have substantial losses, as their short positions are beginning to move against them.

These three fears combined at the same time are perfect storm times. When a bear market occurs in any equity market around the globe and then transitions into a bull market, you

can profit substantially from it knowing that these three fears combined have historically led to substantially higher prices. This is not only US centric. It's global because human emotions, especially their reaction to fear, are global.

We can see this occur here with iShares Latin America 40 ETF (ILF).

1. The market is above its 200-day MA, the first day of the 2-period RSI under 25.
2. The second day is under RSI 25. There's a buy of the first unit.
3. The second unit is purchased on the close as prices are lower than the previous entry.
4. The third unit is purchased on the close as prices are lower again than the previous day's close.
5. A few days of sideways movement and then "liftoff." Lock in your gains.

Buying fear, especially fear that builds up over time when prices are above their 200-day moving average, has led to high-probability trades globally for decades.

TPS Short

Incredibly, when you reverse the rules for TPS and trade the same universe of ETFs on the short side during their bear markets, you see historical test results almost as strong as test results on the long side. This gradual buildup on optimism and eventually greed during bear markets often precedes reality kicking in and seeing prices reverse lower again.

Here are the rules for TPS Short:

1. The ETF is below its 200-day simple moving average.
2. The 2-period RSI is above 75 for two days in a row. Sell short 10% of your position on the close.
3. If prices are higher on the close than your previous entry price, any day you're in the position, sell short 20% more of your position (you're averaging in).*
4. If prices are higher on the close than your previous entry price, any day you're in the position, sell short 30% more of your position.*
5. If prices are higher on the close than your previous entry price, any day you're in the position, sell short 40% more of your position.*

*Do not take a new position anytime the ETF closes above the 200-day MA.

6. Exit on the close when the 2-period RSI closes below 30.

Here are the test results.

Table 8.3. TPS Short Test Results - All Liquid Non-Leveraged, Non-Inverse ETFs

Trade Direction	Scaling	# Trades	% of Winners	Avg % Profit/Loss	Avg Days Held	W. Avg % Profit	W. Avg. Days Held	L. Avg % Loss	L. Avg. Days Held
Short	2/3/5	9,146	77.19%	1.52%	5.67	3.01%	3.87	-3.52%	11.77
Short	1/2/3/4	9,146	80.38%	1.79%	5.67	3.05%	4.04	-3.39%	12.34

Table 8.4. TPS Short Test Results - SPY

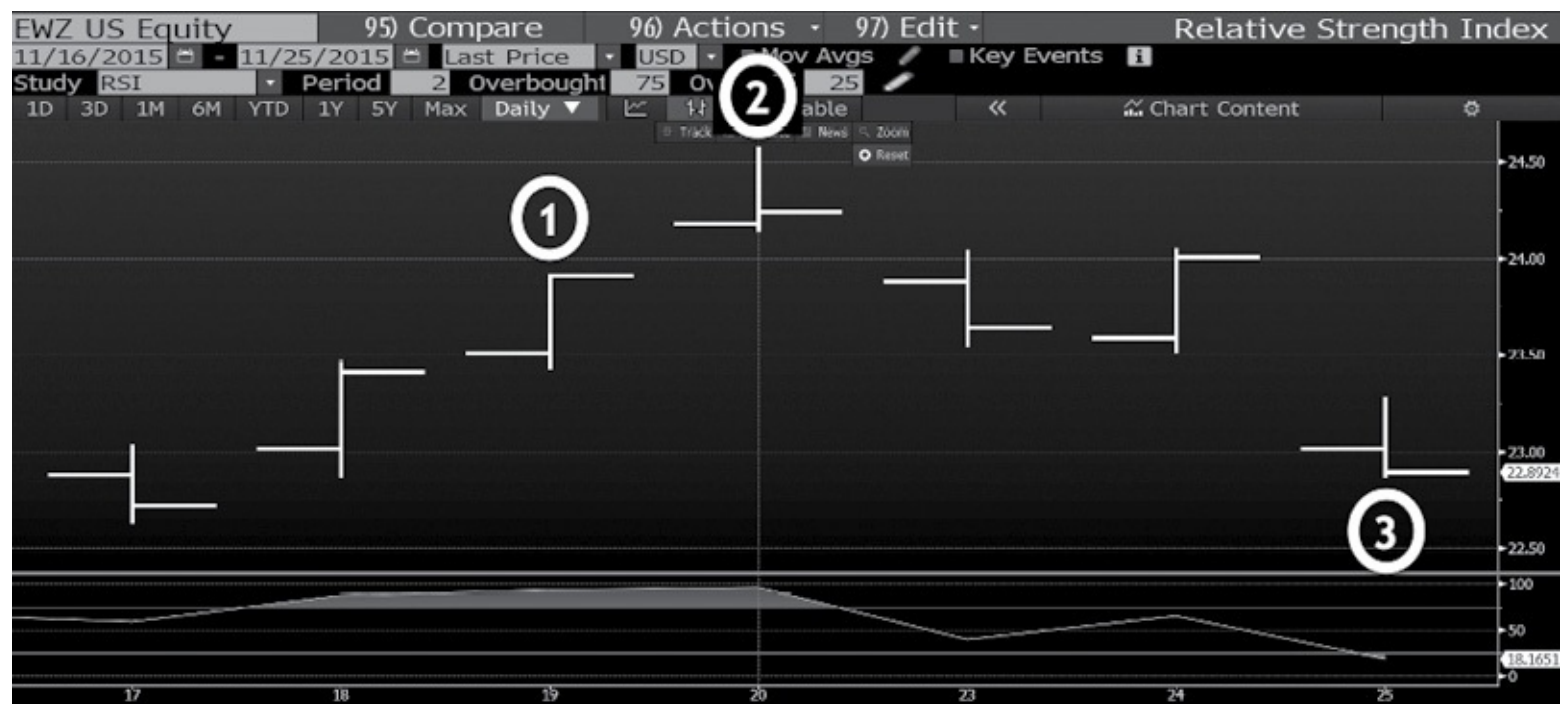
Trade Direction	Scaling	# Trades	% of Winners	Avg % Profit/Loss	Avg Days Held	W. Avg % Profit	W. Avg. Days Held	L. Avg % Loss	L. Avg. Days Held
Short	2/3/5	78	79.49%	1.29%	5.26	2.04%	4.13	-1.62%	9.63
Short	1/2/3/4	78	82.05%	1.46%	5.26	2.09%	4.23	-1.38%	9.93

From 2006-2017, most markets around the globe saw their overall prices significantly higher in 2017 than they were in 2006. In fact, most markets and ETFs saw the majority of their years closing higher than when the year started. Yet, using the 200-day moving average as a trend filter, we're able to identify when any of these markets are trending lower (and most investors are losing money) and then the fear and greed cycle plays out.

Rising prices under the 200-day moving average often lead to a) fear subsiding, b) Fear of Missing Out, and c) short sellers panicking. These three fears send prices higher and have provided excellent entry points for those looking to be on the other side of these fear-driven (and oftentimes greedy) investors. On the whole, using the 2-3-5 scale-in saw prices lower on average within approximately a week over 77% of the time, and using the 1-2-3-4 scale-in saw prices lower over 81% of the time.

Here is an example of what the setup looks like: EWZ.

Example 8.4. iShares MSCI Brazil ETF (EWZ)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

Brazil's markets have been ravaged due to political scandals. EWZ is well under its 200-day moving average (not shown) and a full-blown bear market is in force.

1. EWZ rallies and the 2-period RSI closes above 75 for two consecutive days. Sell short one unit.
2. The rally continues the next day, allowing us to add to the short position at higher prices.
3. All good things often (not always — often) come to an end when prices are under their 200-day moving average and this holds true here as the rally reverses and the buyers from the previous days see their gains disappear. Everyone who went long EWZ over the past five days has lost money again by today's close. The 2-period RSI closes at 18 and gains are locked in.

I intentionally showed you examples outside the US to reiterate that fear and greed in markets are inherent in all traders throughout the world.

TPS allows for fear to build up over time. The longer fear is in place, combined with you increasing your position size as the fear increases, has successfully led to prices moving in your favor the majority of the time. On the long side, applying a broad universe of ETFs, prices have moved higher almost 90% of the time. On the short side, prices have moved lower up to 81% of the time since 2006. **Fear building up over time is a powerful force, especially when it pertains to financial markets.**

Additional Knowledge

1. The concept of scaling into a position has been around for a long time. For our purposes on the long side we're scaling into positions as fear gets higher and exiting as fear subsides.

Interestingly, some long-term growth investors often use the same scaling-in approach as they build their positions. They'll buy a position in a stock that they believe is fundamentally sound and whose growth outlook is very bright. If the stock price goes lower and the fundamentals haven't changed, they'll buy more.

At Merrill Lynch I worked with one of the most successful people at the firm who handled the money for a number of the most prominent families and individuals in Los Angeles. From time to time I'd hear him tell these people, "We're going to buy this stock today and I want you to go home tonight hoping this stock goes down because we're going to buy more if it does." He had incredible conviction in his growth stock-picking abilities. Many of his clients were high profile — they included people who ran publicly traded companies, well-known individuals in the entertainment industry, and at least two individuals who went on to high-ranking positions inside the White House. I personally witnessed the wealth he created for them through his stock-picking ability combined with his ability to accurately buy more when prices went lower. He wasn't always correct, but over the years, because he was able to accumulate the majority of the company's shares at lower prices, his clients' profits were tremendous.

Investors use TPS to accumulate what they believe to be undervalued growth companies at even cheaper prices. They can be undervalued for many reasons — the key is that the investors are buying more shares at even lower levels. We as traders are applying the exact same approach over the short term. We're accumulating shares in ETFs, which are often mispriced due to fear, and buying even more shares when the fear gets greater. **As**

we have seen over and over again in this book, the greater the fear, the greater the likelihood the security is mispriced.

2. As you can see from the test results, the percentage correct numbers are very high. These are very short-term trades, and the average gains per trade are not as high as other strategies in this book because most positions don't get to full scale-in. The fear oftentimes subsides too quickly. What does stand out is the strategy's consistency and further reinforces that in bull markets fear has a rhythm to it in equity indexes both in the United States and around the world.
3. Because TPS has open-ended risk like all direct security ownership positions do, options should be considered. The options can be used for protection. In fact, one of the professional traders we worked with for many years before he retired used to buy "full position deep out of the money puts" at the same time he went long the S&P futures with this strategy. (This is not a futures trading book, so this reference is being mentioned for knowledge purposes only.)

As an example of how this is done with any S&P 500 instrument, let's assume you get a TPS signal to buy 20% of the position for a 2-3-5 scale-in. Let's assume it's 200 shares (or two contracts) to keep thing simple. It's likely early in a pullback and the fear isn't very high. Therefore, one could purchase 10 out of the money puts at the same time. You may ask why would he buy 10 instead of 2? Because the implied volatility in the puts is likely cheap. (Remember, implied volatility often rises as fear rises — in this case, that fear is much lower than when the next round of scaling-in occurs. Stated another way, the puts, as measured by implied volatility, are likely cheaper now than they'll be in the future should prices continue to sell off.) He now had his entire potential position covered.

Here are the highlights:

1. He slept well at night. He predetermined his dollar risk. He was trading a healthy seven-figure account and he went all-in with his buying power, which means that without the puts he was potentially taking on tremendous risk, especially if a black swan event occurred. Risk can be perceived many ways and he knew he needed his risk contained to the exact dollar. This is disciplined risk management and thinking.
2. Because he could predetermine his risk, he was able to take on a larger position size. He put to full use his buying power because he knew what his maximum dollar risk was. When he was right (and with TPS that was often), he made very good money.
3. The main drawback to this approach is that when there is only one scale-in (for example, you own either 10% or 20% of your position) and it immediately moves higher,

the position with the options may at best break even, but more realistically it loses some money. It's not fun being right and not making money (or losing money), but that was, in his mind, the cost to be able to get on the full protection in order to predetermine his risk and to put on sufficient size in order to achieve healthy dollar gains throughout the year when the trades were correct. The final scale-in are psychologically the hardest (remember, at that point much of it's you against the rest of the world). He did not want to ever have to be in the position of second-guessing himself and listening to the experts and the media telling why prices could only go lower. He already has his protection in place, he had predetermined his dollar risk, and all he needed to do was "see the trade, take the trade." All that crazy fear and noise from the outside were easily ignored.

4. This approach will work differently at different times. OTM put options are often bought as protection and at various times the cost will change based on a number of supply/demand factors, including who is in the White House. If you're proficient with options, you already know this. If you're new to options, I've listed a handful of options authors I recommend in order for you to expand your options trading knowledge.
5. TPS can be traded a multitude of ways including different starting RSI levels, the number of days under that RSI level, the scale-in variation, and whether or not to do it with options (plus the incredible flexibility options offer you).

Summary

TPS has so many applications and I've witnessed its effectiveness (meaning the scaling-in approach) so many different ways over many decades. The concept has been widely used and yes, some people have strong opinions why they don't like to scale into (or average into) positions (everyone has opinions on everything, good or bad). The main thing for you is that if you feel it's right for you, as so many people do, you have a statistically backed systematic methodology to trade fear and greed.

TPS is a "trading strategy," it's not a "portfolio strategy." This means it's not meant to replace or compete with buy and hold going long-only an equity ETF and mindlessly forgetting about it, "hoping" prices go higher over the years. It's there to be applied to identify when fear is rising. You are taking a piece of the position as the fear is rising and then knowing ahead of time that if the fear goes longer and becomes greater, you're well stocked with cash in order to buy a larger position in the ETF at lower prices in order to take advantage of the historical short-term high-probability edges. These edges have been created over and over again for decades due to the very predictable behavior of investors and traders during market stress. **Take advantage of these times because they're one of**

the few true quantified edges that have consistently been in place in equity indexes since the early 1990s.

CHAPTER 9

Terror Gaps

“In the midst of chaos, there is also opportunity.”

Sun-Tzu

I’m going to give you a three-step formula to find high-probability setups intraday when fear is high and oftentimes selling is rampant. These three steps were originally taught to me by a professional trader who created a good amount of wealth for himself and his family by trading with the philosophy of “buying fear and selling greed.” Even though he was not data-driven, he did have the instincts of knowing when there was fear in the marketplace and he took advantage of this year after year. He told me he made the greatest amount of his money when money managers, investors, and traders were “squirming and then panicking.” He said the ideal times (like TPS) were when these market participants were getting nervous over a period of days, then an intraday panic (meaning an intraday sell-off) would occur where they would lose their senses. When he saw this, he stepped in and bought in size.

These were the three market conditions he traded by in order to take the other side of fear. What’s so interesting about these conditions is that when we put precise rules around this and quantified it, the data showed he was absolutely correct.

Just as importantly, the rules behaviorally make a lot of sense. When these setups occur, it’s not a fun time for those long the markets, especially ETFs. The market participants who are long these ETFs oftentimes can’t take the pain while the ETFs sell off and they lose money. Then when this pain gets to be too much, they often panic, especially intraday. **This is the ideal time to step in.** By relieving them of their pain by taking the ETF off their hands, they’re now out of the position (and out of pain). In exchange, you bought at prices where historically they’ve risen the large majority of times over the next few days.

Let's look at how we took his concepts and placed them into three rules. We'll discuss them on a behavioral basis, and then we'll quantify them.

Here's the way he explained this to me:

Rule 1. You want the security (and in this case we're going to say ETF) to have sold off over the past few days. A few days of a security selling off makes people uncomfortable.

Rule 2. You want the ETF to gap lower. Not only have the owners of the ETFs lost money over the past few days, they wake up and immediately get hit with a further loss. It's not the best way to start the day.

Rule 3. You then want the ETF to sell off even further intraday. It's bad enough they've lost money over the past couple of days, and they've gotten hit even harder on the open. Now the ETF is plunging further. Chaos is oftentimes running through their mind and they go into protection and survival mode (otherwise known as terror mode). At this point they're most fearful and vulnerable and they are often simply selling irrationally. **You now have the ability to buy an ETF from scared money.**

When he explained this to me it made a lot of sense. ETFs are baskets of stocks and when the above occurs it's usually for one of two reasons. The sector is getting clobbered oftentimes accompanied by analysts and the media further invoking the fear, and/or the overall market is selling off.

The multiday pullback creates angst, the gap in the morning creates significantly more angst and fear, and the intraday sell-off creates a short-term panic. Behaviorally it's completely logical. This behavior now has to be placed into precise rules and then be quantified.

This is how we placed the rules around each one of his points:

Rule 1. The ETF closes with a ConnorsRSI reading of less than 5. This does not occur often, but when it does it often means the selling has been persistent (there's angst).

Rule 2. The ETF must gap lower. We know that not all gaps are equal. The larger the gap the better, but for this strategy any gap will do. Long traders hate when a position gaps against them. They hate the gap even more when it occurs after the ETF has been beaten down over the previous days. They're likely already panicked about the position coming into the day because it's been so beaten down (a CRSI reading of under 5 certainly reflects this). Now it gaps even lower on them. Not the best way to start the day, and this is usually when they're ready to kill their pain and begin thinking about exiting the position.

Rule 3. We're now looking for a further sell-off. We want fear to potentially turn into panic selling after the downward gap opening. This intraday sell-off oftentimes leads to terror. No one likes to lose money over multiple days then see their investment gap lower in price, and then watch that investment melt down additional percentage points intraday.

We tested multiple level intraday sell-offs. As you will see from the test results, the larger the sell-off, the greater the gains.

Rule 4. Exit when ConnorsRSI closes greater than 70.

As you'll see this setup has on average exited only three days later. The panic selling washes out most of the weak hands and then the stronger money steps in. As prices move higher some of the scared money that sold lower starts buying again (this is historically known as "sell low and buy it back higher when the coast is clear").

Let's look at the test results from 2006-2017 for the ETF Terror Gap Strategy tested on all ETFs, including leveraged ETFs, with an average daily volume of 250,000 shares over the previous twenty trading days.

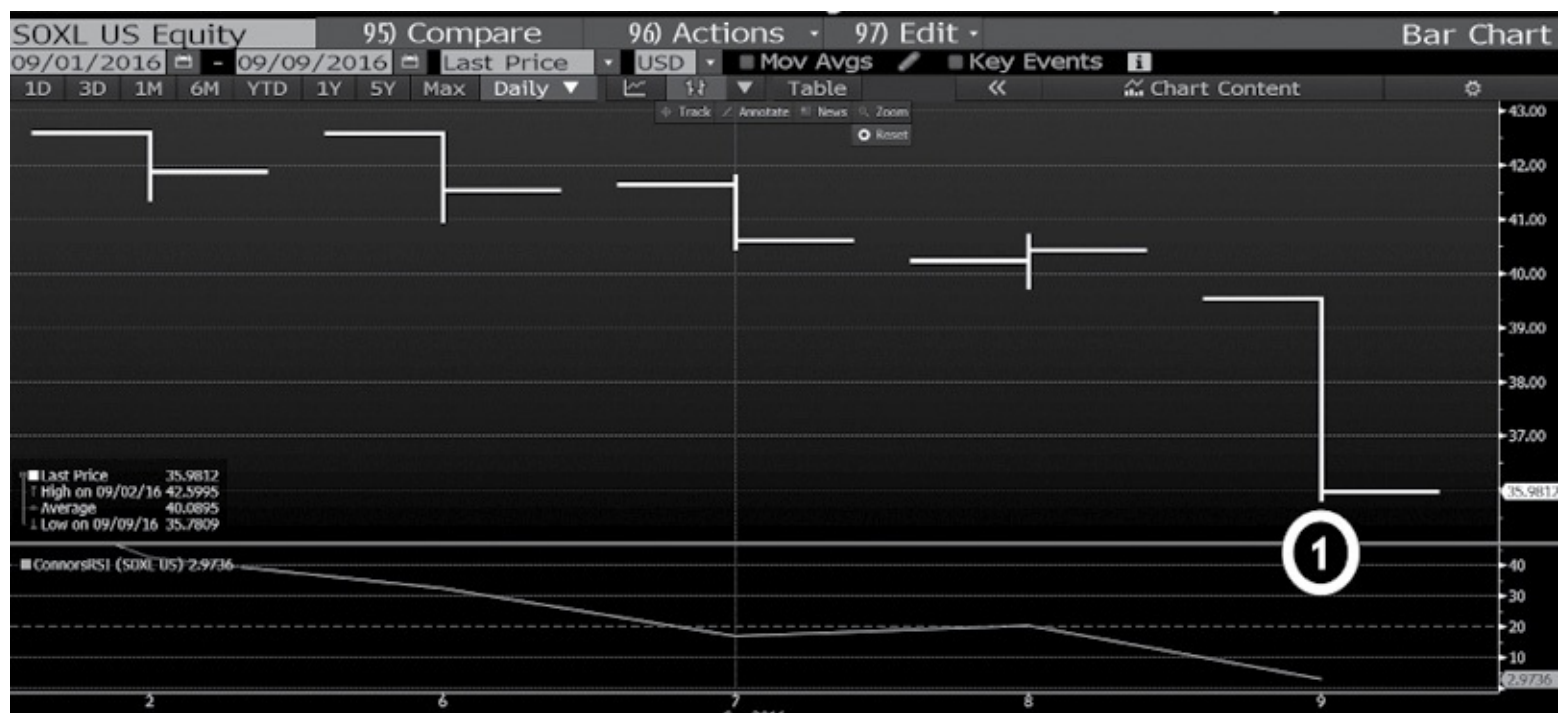
Table 9.1. Terror Gaps Test Results

Limit %	# Trades	% of Winners	Avg % Profit/Loss	Avg Days Held	W. Avg % Profit	W. Avg. Days Held	L. Avg % Loss	L. Avg. Days Held
2.50%	102	81.37	5.25	2.9	7.87	2.19	-6.21	5.89
2.00%	130	74.62	4.1	2.9	7.11	2.21	-4.73	4.85
1.50%	190	74.74	3.33	3	5.99	2.44	-4.53	4.52
1.00%	275	77.82	3.02	2.9	5.16	2.3	-4.47	4.84

1. Over 77% of the Terror Gaps were profitable over a 12-year period of time when applying a 1% pullback. Over 81% were successful when allowing for the sell-off to be at least 2.5% below the lower gap.
2. The average gain per trade is exceptionally high for ETFs (over 1% a day held). As we discussed earlier, the selling is very often terror-driven selling, and once this selling is exhausted the ETFs reverse sharply higher a large majority of the time.
3. The average holding period for Terror Gaps is on average less than three days. Panic selling turns into panic buying in a very short period of time. Once the CRSI reading closes above 70, you can feel free to sell the ETF back to the buyers who are no longer afraid. Sometimes prices will move higher, but as a whole we've tested multiple ways to hold positions longer and on a risk/reward basis we have found nothing better for this strategy than a CRSI reading of 70. This is very likely due to the behavioral component — you bought the fear where there were large edges and once the ETF moves higher the edges disappear. You were well paid for stepping in.

Let's look at two Terror Gaps examples (the second one is in the Additional Knowledge section).

Example 9.1. Direxion Daily Semiconductor Bull 3X Shares ETF (SOXL)

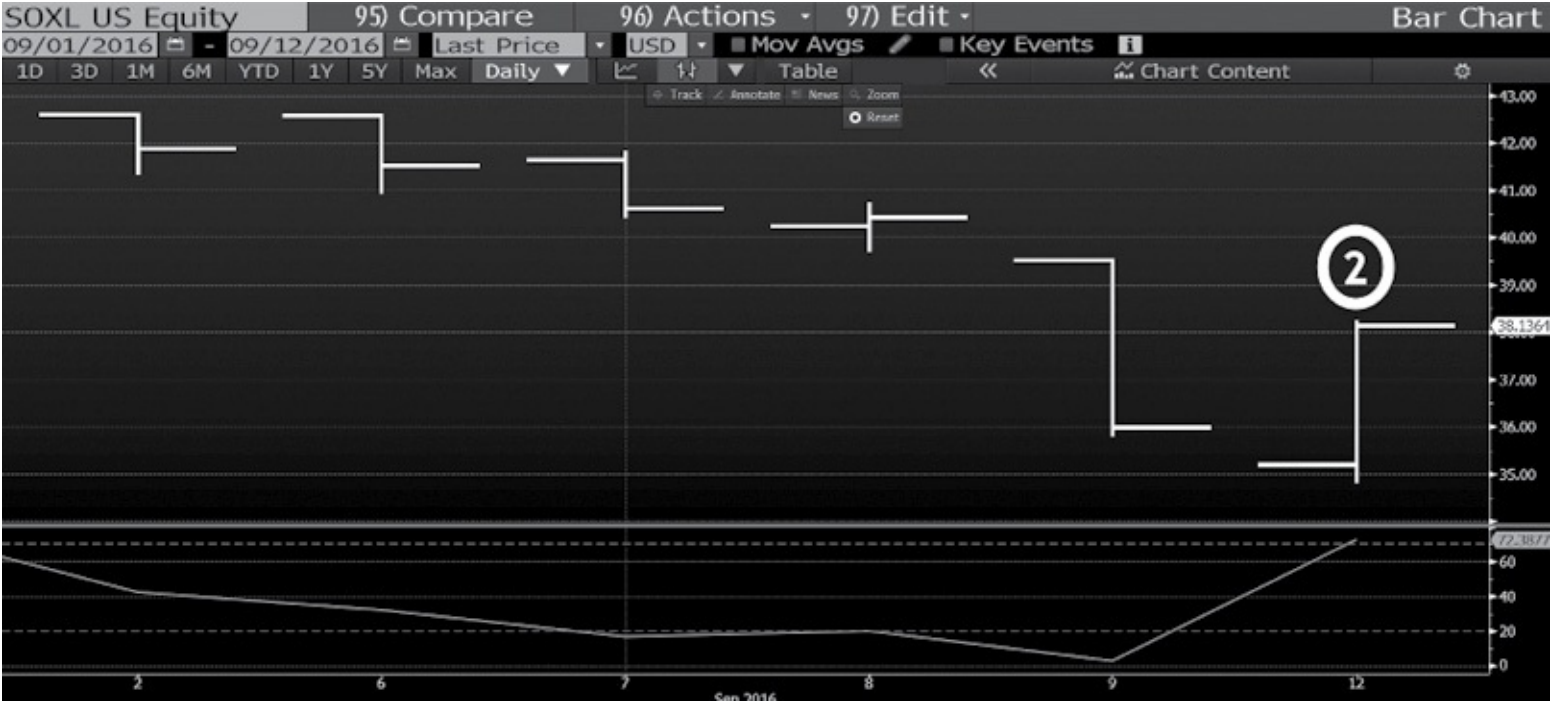


Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

1. On September 9, 2016 Federal Reserve Bank of Boston President Eric Rosengren creates a broad-based stock market sell-off as he warns that waiting too long to raise interest rates risks creating financial instability. The Dow plunges nearly 400 points as investors panic that rising interest rates will end years of monetary easing and soon lead to a recession.

SOXL, the Direxion Semiconductors 3X Bull Shares, gets crushed by over 10% because the market knows a recession always hits semiconductors hard. ConnorsRSI closes under 5 and we'll have a buy signal tomorrow if the semiconductors gap and continue to sell off after they gap.

Example 9.2. Direxion Daily Semiconductor Bull 3X Shares ETF (SOXL)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

2. Yesterday’s panic carries into the next day and SOXL gaps lower and sells off further. A Terror Gap buy is triggered at 34.84 and within minutes SOXL reverses higher and closes at 38.13 for a nearly 9.5% gain for the day.

Additional Knowledge

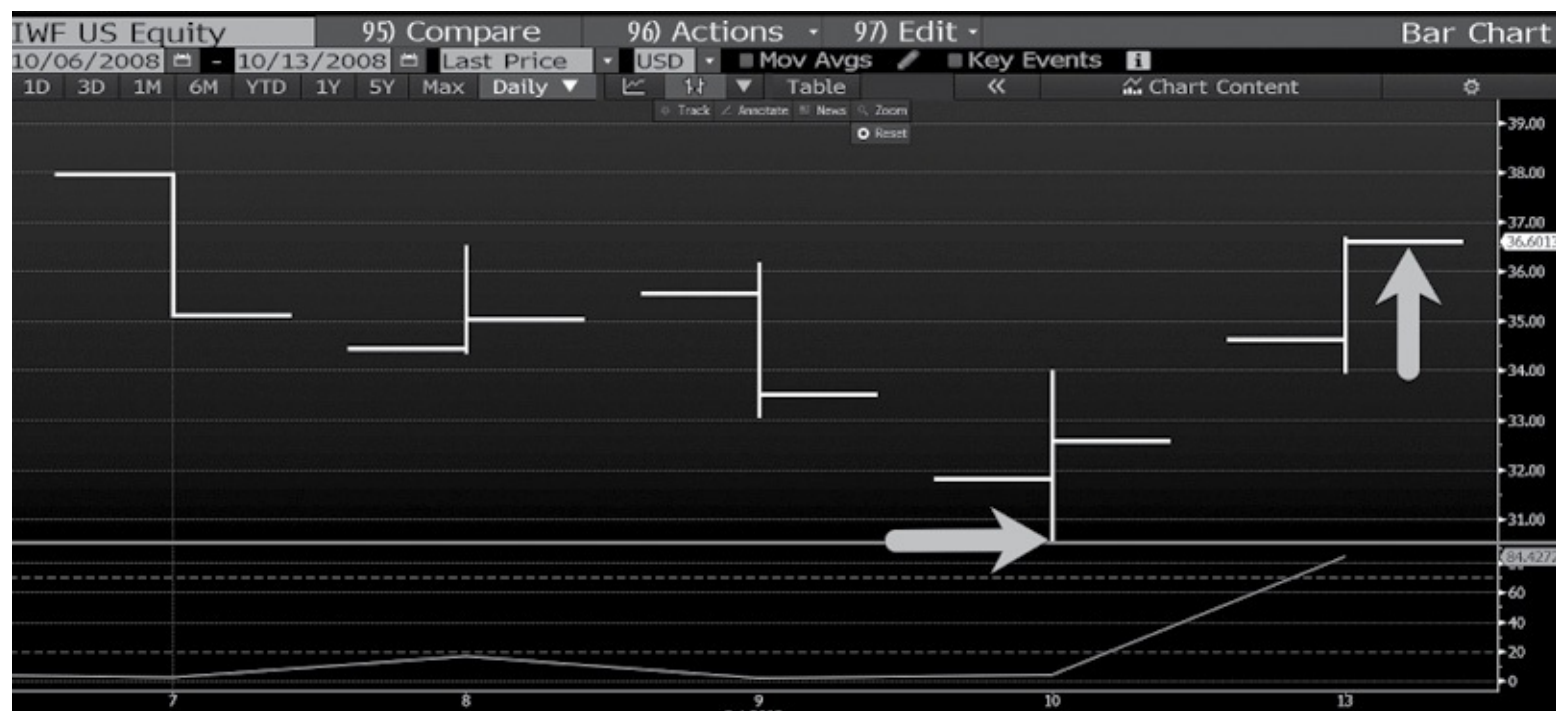
1. Please note: The overall test results from the dates tested were significantly higher than what you see above. The reason the test results above are lower is because we intentionally omitted from the test results October 10, 2008 and August 24, 2015 (the day the ETF markets went haywire, plunged, and then reversed). Here's why.

On October 10, 2008, many dozens of ETFs gapped significantly lower and sold off further. General Electric (then a market leader) had missed badly on earnings, Moody's stated it was looking into cutting Morgan Stanley's credit rating again (this was just after the Lehman bankruptcy had occurred), and there was massive selling caused by forced liquidations due to margin calls combined with massive mutual fund redemptions. By midday the S&P 500 had experienced its worst seven-day decline since 1996.

-
The same type of massive sell-off behavior was again seen after oversold conditions occurred on August 24, 2015. Dozens of ETFs had tremendous Terror Gaps before reversing higher.

When we looked at the historical test results, we saw that over 30% of the overall Terror Gaps signals occurred on those two days and the majority gained double digits after they rallied over the next two days. We knew we had a decision to make because **these two days potentially overstated the overall test results. It would have been easy for us to leave these two days in, but we felt that even though they had such a positive impact, a more realistic view (a more conservative view) was to not include those two days in the test results.**

Example 9.3. iShares Russell 1000 Growth ETF (IWF)



Source: Bloomberg Finance L.P.
Used with permission of Bloomberg Finance L.P.

The iShares Russell 1000 Growth Index (IWF) gained over 20% in two trading days. There were dozens of these Terror Gaps signaling that day, which saw double-digit percentage gains.

These two days were true terror days. Will events like the events of these two days occur again? Potentially, yes. We decided, though, to publish the test results without those two days in order to better reflect the remaining days. Possibly we are being too conservative doing this. Should those days repeat themselves, you can view those days as gifts.

2. From a larger picture, the behavior creating these edges is consistent with what we've seen throughout the book. Investors don't like pain and they often kill their pain by selling out of positions oftentimes near bottoms.
3. This strategy identifies the following sequence in the fear process:
 - a) A large buildup of fear over a number of days;
 - b) an overnight jolt with the ETF gapping lower; and
 - c) an immediate continuation of selling, creating chaos and even a panic, taking out weak long holders and moving the ETF into the hands of the stronger buyers.
4. The three behavioral pieces combined have led to a pattern that has repeated itself over and over for years and this is backed statistically. Fear builds, losses mount, losses

increase overnight, and then losses become even greater intraday, inducing panic selling. Once the selling is done, prices move higher, fear dissipates, FOMO kicks in along with shorts panicking, and prices move higher again. It's inherent behavior you've seen over and over again, quantitatively supported throughout this book. In this case, fear turns to panic and eventually to terror. Nothing increases short-term edges faster and greater than terror in the marketplace.

CHAPTER 10

Buying Fear, Selling Greed

Now that we've learned and studied each of the strategies, let's first recap what we've learned and then go even further.

1. Fear creates trading edges. Period.
2. The greater the fear, the greater the edges.
3. Greed also creates edges. It's not as powerful an emotion as fear so the edges are less prevalent, but they exist, especially when FOMO is present and also in bear markets.
4. People of authority influence larger numbers of traders and investors and often create the greatest fear. The media (both traditional and social) widely disseminate these individuals' opinions and then often overstate them, sometimes to the point of creating short-term panics.
5. Fear in bull markets more often leads to more consistent reversals in prices. Fear in bear markets is oftentimes justified and therefore is more volatile to trade and usually not as profitable as fear in bull markets.
6. Panic bottoms, both long-term and short-term, are not made from buyers all piling in at the same time. They're made when the last seller sells. These people, both professionals and individuals, can no longer take the pain of losing (in trading desk terminology "they puke it up"). The buyers who are there at that time, when correct, gain advantageous pricing. The security moves from the weak to the strong. It's a Darwinian process created by built-in human emotion.
7. Fear of loss is one of the strongest emotions hardwired in humans. It's what creates edges on the long side for traders.

8. Fear is a global phenomenon in markets as we saw in the RSI PowerZones. In the global markets it has accurately predicted the short-term prices of ETFs over 80% of the time going back many years. In the United States it has accurately predicted the short-term prices of the S&P 500 ETF (SPY) over 91% of the time. This is not by chance. It's because sellers come in, buyers step aside, and once the event passes or the fear dissipates, the buying reappears. It has been a repeatable process for a quarter of a century — a time when there have been many political, economic, and technological changes. What hasn't changed, nor will ever change, is human emotion.
9. FOMO as we saw with CRASH and with TPS short is real. With CRASH, most of the times these stocks lose all sense of reality. The Fear of Missing Out is so powerful (it's pure greed) it oftentimes leads to irrational short-term mispricing of securities.
10. VXX was built to go to zero. It's structurally inefficient. Nine years of consecutive losses tells you there's something wrong with it. In this book you now understand why it was built, how it was built, the inefficiencies in its construction, who trades it and why they trade it. You have two strategies to take advantage of this fear-driven structurally inefficient product. Add in trading it with options to predetermine your max risk along with putting yourself in the potential position for asymmetrical returns, and you have the makings of a winning formula.
11. New 52-week highs, followed by strong pullbacks, create a boom-to-bust cycle. Owners heading into those 52-week highs are making money. Oftentimes positive news comes out reassuring them, while at the same time attracting new buyers who now "logically see" the good news must mean prices can only go higher. When those stocks don't go higher and then they sell off, they create fear. At times that fear turns to panic and irrational selling occurs where prices capitulate intraday. Being there as a buyer at those times has seen prices trade higher on average within five trading days. By us applying a seminal academic study with fear-induced selling creates a systematic quantified way to trade new 52-week highs.
12. Scaling in and out of positions that are oversold due to fear while they're in longer-term uptrends has been a consistently positive-performing strategy in dozens of equity ETFs, especially the US ETFs, for years. The same scaling-in approach has been used by other successful groups in accumulating stock at advantageous prices. The TPS strategy scales in as fear increases and scales out as fear dissipates and things become safer for the buyers who often are paying higher prices. The same is true in bear markets. Greed created from the fear of missing out along with the fear of loss created by higher prices for short sellers drives prices up and then ultimately reality sets in, usually economic reality that a bear market is in place. This leads to lower prices and has allowed for

profitable short covering over 77% of the time over the past two decades, in spite of the overall global markets being significantly higher during that period.

13. Overnight fear leading to down gaps after a security has sold off for a number of days makes long holders extremely uncomfortable and fearful. Then add in a further sell-off after the down gap and most often they just want out. They can no longer take the pain. You now have the opportunity to take the pain off their hands at historically advantageous pricing.
14. Throughout this book we have seen through extensive testing along with live examples that buying fear and selling greed is one of the single best quantified ways to trade. In my opinion it's one of the few areas where true edges still exist for traders and I believe these edges have the potential to exist for years to come.

Markets change — they always do. Human emotion does not change. Identify when the emotions are at an extreme, especially fear and greed, systematize the process, quantify the process, and execute this process over and over again.

Additional Knowledge for Trading Execution

Trading execution — meaning how to best construct individual trades — the constructions of one's portfolio, risk management, market correlations, sector correlations, and many additional important elements that go into successful trading are all stand-alone book topics. People spend lifetimes studying these topics.

I'm going to briefly cover a few here because they're important, and I encourage you to go further on these topics including learning from the authors I've listed in the Appendix for you. I've also added additional trading information into the Appendix for you to learn from.

Risk Management

Learn how to trade options. If they are new to you, start at the beginning. If you understand them, get better. If you're a professional, go even further. Options offer you the opportunity to do so many things including shifting your risk from open-ended risk to fixed risk. They allow you the opportunity to achieve asymmetrical returns at the same time.

As short-term traders you now have the opportunity to trade hundreds of different options with weekly expirations. For S&P traders, you have expiration three times a week (Monday, Wednesday, and Friday). When I first started trading they only had quarterly expiration options. Think about this. The trading opportunities you learned in this book would have had to have gone out as long as three months to get the position on — and that's even if the security had options on them. You now have literally millions of possible combinations a year available to you. There are thousands of securities with options, with many strike prices, with multiple expirations.

This is the golden age of options trading and it's only going to get better. Why in the world would one take open risk in a trading position when you now have the ability to structure trades with fixed, predetermined dollar risk along with the opportunity to achieve asymmetrical returns? There are many good books that teach you options and that's a start. Then working one-on-one with the right options professional will take you further. Having an options coach who can teach you how to best construct your trades when there are large edges in place is a great investment that can be scaled-up for years.

Trade Execution

Learn ways to get better fills on your execution. The technology today on most brokerage trading platforms is the best it's ever been and it's only going to continue to improve as brokers continuously compete for traders' business. Attend your broker's workshops on execution. It's not boring — it's opportunity!

Quantify

In this book, I've mentioned examples of traders I have known along the way who have been able to instinctively buy fear and sell greed. Many of them acquired this skill by being floor traders or market specialists. For everyone else we want to quantify the behavior and the historical edges that have existed. There's no guarantee these edges will exist exactly the same in the future — what we do know is that edges over many decades get created from fear and greed. Quantifying this allows you to put disciplined structure in place.

Know Your Market and, as Importantly, Know Who Is on the Other Side of Your Trade

1. Know your market. Bull or bear — the 200-day is a good place to start and you have the support of one of the hedge fund legends who confirms this. Also, know where you are in days. This means asking, "What's been the behavior of the security coming into today"? RSI certainly guides you.

Even though we didn't isolate the types of news events that occur, global events that touch the world tend to create global fear and potentially greater fear than an upcoming Fed report. This too can be measured by RSI and it's also fertile research ground for you if you enjoy doing research in order to focus on the news events that matter the most. Also, the longer the news persists, the more uncomfortable people become. No one likes losing money day after day and reading about all the negative possibilities that may occur in upcoming days. They're often going to sell out to avoid the potential for more losses. When they're done, the buyers who are purchasing their stocks oftentimes do so at advantageous prices.

2. Who is on the other side of your trade? I discussed this in the first chapter, and now that you've seen it throughout the book it's worth discussing again because it's so important.

Fear, and to a lesser extent greed, creates quantified edges. When you're buying, you want the other side to be fearful. Fear of loss is the most powerful type of fear. Ask yourself, is their fear of loss in this position so high at this point that the other side can

no longer take the pain? Am I buying this position from someone who is scared? It can be a retail trader who fears losing more money because he bought higher, all the way to an institutional money manager who fears losing his job because the position is crushing his portfolio. In either case, they're selling out of fear. If the security is extremely oversold it's likely due to their fear. Buy their fear and when things look safer and prices are higher, sell it back to them and others at higher prices. This is a repeatable process. With VXX we went deep with who is on the other side of the trade. The good news is you can do the same with many other securities.

None of These Strategies Will Be Easy to Trade

That's why the edges exist. And as hard as each will be to trade, a few will be even harder. Vol Panics and CRASH will likely be the hardest. Who wants to short volatility when all hell is breaking loose? And who wants to short stocks that are running away to the upside while at the same time greed is spreading like wildfire all over the internet?

Going further, who can consistently buy in the face of bad news? Or wants to step in ahead of a major news event? It's easier to take a pass, as most others do, and wait for the event to pass. This waiting, though, creates the liquidity hole when buyers disappear. This causes prices to drop not only because of selling. It's also because the "rational money managers" are stepping aside. And because they're supposedly acting in a "rational manner," prices often pull back even further, creating historically high-probability edges and buying opportunities. This cycle repeats itself over and over.

News Nearly Always Precedes These Extreme Short-Term Conditions and Opportunities

In fact, having witnessed this for years, I'm the most concerned when these signals trigger without any news. I've never been able to quantify this, but I know that when prices move to extremes without news, it's often not a good sign. There's potentially something bigger out there. Ideally you want news stories to accompany the price. The bigger the story, the greater the fear and greed, and the greater the edges become.

Buffett's Philosophy of "Be Fearful When Others Are Greedy and Be Greedy When Others Are Fearful"

After completing this book, I began to reflect on just how similar the behavioral aspects of these strategies are to the way Warren Buffett invests and the way he runs Berkshire

Hathaway.

Buffett is an investment genius. You don't need me to tell you that.

Buffett is also a financial engineering genius. No one doubts this, and I've read academic studies that have discussed how he, with the proper use of leverage and cash flow, has maximized his returns over decades.

In my opinion, though, what Buffett is most brilliant at is understanding human behavior. **He fully understands that people become irrational in times of stress.**

Buffett lives in a much longer time frame than we do. He sometimes waits years before he pounces. When he pounces is a function of price — how cheaply can he buy a company. And like a number of the strategies in this book do, he especially pounces in times of great stress.

He knows that he's getting advantageous prices because of the fear this stress induces in the sellers. His investment in Goldman Sachs near the bottom of the market crash of 2008-2009 is an example of this brilliance (and I can name many more examples like this).

He has an uncanny way of buying when there's blood in the streets and he has the advantage of structuring his investments in such a way that he minimizes the risk on his side by feeling the fear (and pain) on the other side. The more pain they're in, the more he squeezes. I know this is not the way he portrays himself (nor likely wants to be portrayed). But when you look at his investments, his timing, and his structuring of his investments, no one epitomizes the "buy the fear, sell the greed" better than he does.

The daily operations of many of his businesses are run the same way. A number of years ago he spent a good deal of time discussing Berkshire's reinsurance business. Two years earlier severe weather storms hurt this business line. Of course, when bad weather occurs, the recency effect kicks in and the demand for future insurance skyrockets. With greater demand also comes the ability to charge higher premiums. The following year the weather "magically" returned to normal. Profits for his reinsurance line soared. And how did he describe it? He said, "We got lucky with the weather."

You and I know they didn't get lucky with the weather. They were playing with long-term probabilities in their favor. Plus, they were getting paid a premium price to insure something that was modeled out looking back over decades of data. They were efficient (one could say ruthlessly efficient) in taking advantage of the fear people had about the potential for upcoming storms. His business correctly bought the fear and then profited when, in his words, they "got lucky."

No one can do what Buffett has done. But we can learn from him, especially when it comes to his understanding of human nature. Buffett has done this on a broad scale over many decades. He's combined a behavioral formula with a financial formula that's led to tremendous success.

In this book, the same formula has been applied. I'm in no way inferring you or anyone will achieve Buffett-like success following this formula. But, human behavior doesn't change. The same fear that exists not only in markets but also in all walks of life, has existed for centuries, and it exists today and will exist tomorrow. These fears are repeated over and over again. Your job as a trader, should you wish to take this path, is to execute the times when fear and greed are at their greatest.

As you see from the strategies in this book, these fears get played out in many different ways over many different trading days, weeks, months, and years in the markets. It's a repeatable behavioral process combined with quantitative analysis.

Finishing Where We Began

At the beginning of this book I made you an offer. Now that you know the offer is real, we can discuss its details.

Here was the offer...

The Offer

Now let's assume I make the following offer to you: I offer you the chance to trade directly against traders and investors who are having the exact symptoms listed above, including the feelings of going crazy and impending doom. And as a bonus...

1. You Can Trade One-on-One Against Them...

With VXX, for example, there's a high likelihood that one of the five types of people who buy VXX will be on the other side of your trade. It's you against them. You are shorting a security, either outright or with fixed risk options, that has been built to go to zero. You know who they are — it's you against them. In the larger ETFs, it's oftentimes you against a group of money managers who are either dealing with redemptions and getting forced out of the positions or with discretionary "active" managers who know if they underperform the averages they'll potentially get replaced by a passive index fund. In each of these cases above, they're dealing with fears, and according to the data-driven historical test results, their fear provides you with the edge should you decide to buy the times they fear. The advantage goes to you.

2. You Get to Determine the Times You'll Trade Against Them...

Many traders feel they need to trade daily or even intraday. They'll often fly by the seat of their pants trading three bar breakouts, Fibonacci levels, news stories, story stocks some stranger is recommending on one of the financial networks — the list is long. This behavior is common and is part of the reason why so many traders fail to make money.

You, on the other hand, get to determine the times you're going to trade, and when you find the times that are most advantageous to you, you'll step in and often be taking the other side of these impulsive, discretionary traders. You are in full control here.

3. You Get to Construct the Trade However You Feel Is Best for You...

You can take the other side of the trade in any manner you like. For example, if there is a long signal in the S&P 500 you can buy SPY, buy the S&P 500 futures, buy SSO which is the 2x version of SPY, you can buy SPY calls, you can buy SPY debit spreads, you can sell SPY put spreads, you can buy SPY and protect it with an OTM put, you can place on a risk reversal credit spread, you can place on a risk reversal debit spread, you can put on a ratio spread, and with all the options, you can decide the strike expiration dates and the strike prices that best put you in the position of maximizing your gains while minimizing the risks. On top of this, there is a long list of additional ways to take this trade. While they're panicking because their long position is plunging against them, you're in full control dictating the terms on how you want to best take advantage of the trade.

4. And You Get to Do This Multiple Times Every Month for the Rest of Your Life. On Top of This, You Know That Had You Done This Over the Past Quarter Century, You Would Have Won Anywhere From Over 70% of the Time to Up to 97% of the Times You Did This.

Depending on which strategies you decide to trade and how you construct your trading plan, there will likely be historically high-probability signals and positions held throughout the month. This will likely occur every month for many years, and unless human behavior changes (it won't), buying fear and selling greed trading opportunities will be there for you for the rest of your life. There are limitless opportunities here for you.

By combining behavioral finance with quantitative analysis, you now have a blueprint to guide your trading. Buy the fear, sell the greed. The underpinnings have been there for centuries. It's up to you now as to how best to execute this in your trading.

I hope you've gained new knowledge from this book. If you have any questions or additional insights, please feel free to email me at QuestionsforLarry@cg3.com.

APPENDIX

Connors Research Traders Journal (Volume 7): Superior Trade Construction for Asymmetrical Gains

Reprinted from TradingMarkets.com, 2018.

APRIL 25, 2018 BY LARRY CONNORS

In this issue of the Connors Research Traders Journal we're going to learn one way to construct a fixed-risk trade with the potential to bring you asymmetrical returns.

Let's assume you are highly confident that a stock (or any market) is going to have a large move to the upside. You've done all your work, and you are ready to pounce on a high-probability trade which, if it's correct, has the potential to make you good money.

What would you do?

Most traders (especially non-professional traders) would buy the stock. Right? It only makes sense.

But does it? Could you do something better?

Could you put on a position in such a way where you fully contain the risk (to the penny), put up less money, and have the opportunity to make, on a risk/reward basis, far more money?

The answer is often yes, especially if you are trading setups which have historically had high-probability, large outsized moves. The key to this is known as "Professional Trade Construction".

Trade Construction, as described above is an art. It's looking at a trade signal and then asking, **"how can I risk the least amount of money in order to make the most amount of money from this trade?"**

Most traders spend most, if not all, their time "perfecting their directional strategy". Thousands of books have been written on how to do this (I've written a few of those) and

literally millions of “trading and investing ideas” are flooded upon us each year on CNBC, dozens of websites, blogs, investments services, social media, and more. On television they tell us to simply buy this stock and you’re good (what’s worse is only a few bother mentioning when to sell).

Stock picking (and because in most cases it’s not backed by any statistical evidence it’s more like “stock guessing”) predominates the world. And it has since the markets have come into play. Can you do better than this? The answer is yes – through the art of “trade construction”.

Trade construction is exactly as it sounds. It’s the ability to construct a trade in order to maximize your gains and minimize your risk. In fact, many professional trading firms will spend as much time if not more discussing how to best construct a trade as they do on the analysis of whether or not to take the trade. And the ones who often make the most money, did so partially because they got their analysis right and, more importantly, constructed their trade impeccably.

You only have to look at the tremendous amount of wealth that was made on the short side in 2008 when the housing bubble popped as an example. Many trading firms had the analysis correct. And many made good money. But then there were a handful of firms who did the exact same analysis and literally made billions! The firms who had it right had the analysis portion correct. **The firms though who made billions did so because they had the trade construction correct. They took on what amounted to be limited risk and when they were correct they were rewarded with life-changing profits.**

Today, we’ll look at an example of how to construct fixed risk trades with the potential for asymmetrical returns. This example is simply one of many ways to construct trades. The goal is to get you thinking about this because if you’re like the majority, you spend most of your time looking for directional strategies. Today, the goal is to take that knowledge and begin learning ways to potentially take those edges and construct trades to amplify the gains the times you are very correct.

Introducing Risk Reversal Credit (and Debit) Spreads

Let’s go with the following scenario:

1. For whatever reason, you believe a \$50 stock is going to have a substantial move to the upside over the next 1-2 weeks. Possibly it’s very oversold, possibly it’s having a parabolic move to the upside and you believe this move is going to continue, or for whatever reasons, you’re confident the stock is going higher – much higher.

2. Most traders go out and buy the stock. End of story. They risk the entire amount (100%), meaning \$5000 for every 100 shares they buy, needed to get long the stock. If they anticipate a 10% upward move or even a 20% upward move, they're essentially risking 10 to make 1 (or with a 20% move 5 to make 1). In this scenario the risk is always significantly greater than the reward. This is the way most of the world trades.
3. You, in the meantime, are going to be smarter. Instead of knee-jerking and impulsively buying the stock you first say to yourself: a) "how can I reduce my dollar risk?" and b) "how can I increase my risk/reward on this position?" By simply asking this question you're now ahead of 99% of the hundreds of thousands of traders out there who trade stocks – you've joined the ranks of the professionals.
4. Reminder: The stock is hypothetically at \$50 and you believe it can potentially move to \$55 or even \$60 in the near term. Instead of buying the stock, there are many things you can do, especially with options. For today though, with the scenario described above, one of the better ways to construct this is with a combination known as a **Risk Reversal Credit (or Debit) Spread**. For a fixed amount of risk, you can potentially make asymmetrical returns if your stock moves as anticipated.
5. There are many ways to construct this combination and we'll keep it simple to start.
6. Step 1: Sell an at-the-money (ATM) put credit spread. This means selling a 50 put and buying a 49 put. Simple. You will be taking in a credit of hypothetically .40 cents (\$40).
7. Step 2. You will now take that credit and buy an out-of-the-money (OTM) call with the proceeds. Let's hypothetically say it's the \$54 calls which expire next week.
8. So what have you done? In exchange for risking the difference between 50 – 49 (1 point or \$100 dollars per contract plus commissions) you have the opportunity to make unlimited returns if the stock goes above \$54 a share. Therefore, for every point above \$54 the stock travels you make \$100. At \$55 you had a 1-for-1 risk/reward. At \$56 it becomes 2-1. At 60 it's 6-1. And if you happen to get fortunate and the stock goes above \$64 you've made over 10 times the amount that you risked.

Had you instead bought the stock, your gain at \$65 was 30% of your risk. With the Risk Reversal Spread your gain was over 1000% on a risk/reward basis. You can now see why a handful of firms made billions in 2008 – they constructed trades with tremendous risk/reward characteristics such as the one we're discussing here.
9. So what's the catch? Well there is a catch. It's this – let's say the stock didn't go up as much as anticipated. Let's say it went to \$53 in a week. Long stock trader made \$3 or 6%

on his risk. Risk Reversal Combination trader made nothing. Why? Because the stock closed between 50-54 and that means the credit spread expired worthless and so did the long 54 call. The trade was a scratch plus commissions paid. And this is the risk of trading this strategy versus going outright long the stock. Small moves in your direction lead to no gains (unless your credit spread brought in more money than what you paid for the long calls). In my opinion, based on the fact that you anticipate a large upside move, this is a small price to pay in order to both fully contain your risk plus fully participate if the stock move substantially higher as you anticipated. **On a risk/reward basis, this is one of the better ways to construct a long trade.**

I recently had lunch with a friend who was an options market maker for over three decades. He told me the above trading strategy was one of his favorites. He referred to the strategy as a positive long-tail strategy with lots of trades breaking even and/or making a little and losing a little over a period of time. But, as with all long-tail strategies, tremendous gains can occur the handful of times the trade moves substantially in your favor. Basically, if you have the correct directional strategy in place combined with the right market conditions, this strategy puts you in the position to potentially make a substantial amount of money.

The above is a way to get you to start thinking more about constructing trades based on risk/reward characteristics. As you move ahead you will see there are many factors that will affect the pricing of the position including implied volatility, time to expiration, strike price selection, liquidity of the options (bid-ask spread), the skew, along with the availability of the options. Today is meant as an introduction to better construct your trades, especially if you believe a large, short-term move is imminent.

This same type of trade can be constructed on the short side; meaning you believe the stock or market is going to have a large move down. It's done with putting on a short call spread and using the proceeds to buy a put. Imagine having the foresight to have done this in any scenario where a stock or the market had a short-term sell-off. And now imagine having the foresight to have done this ahead of the sell-off in 2008. As you now understand, some traders were savvy enough to have constructed positions that had the characteristics of the Risk Reversal Credit (Debit) Spreads you learned about today.

Larry Connors

Connors Research LLC

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Connors Research Traders Journal (Volume 3): Stops (Still) Hurt

Reprinted from TradingMarkets.com, 2018.

APRIL 25, 2018 BY LARRY CONNORS

In my book *Short-Term Trading Strategies That Work*, we created a small firestorm with a chapter that was titled “Stops Hurt”. The chapter showed statistically that stops on pullbacks hurt performance (since publishing this test in the book many others have replicated it). Even stops placed 50% away (which few would even consider a stop) hurt performance.

Short-Term Trading Strategies That Work has been one of our best-selling books, surpassing even our high expectations. In no way though did we think the chapter on stops would become one of the chapters so widely discussed.

First, let me state right here that as of this writing (2018), I trade strategies that use stops. They usually also involve profit targets which means the position is held until a profit target is hit or the stop is hit. A strategy for example that has a 10% profit target and a 10% stop and is correct over 50% of the time will make money (commissions not included). Get up to 60% correct and you have a healthy trading strategy.

Therefore, I’m not “the guy who doesn’t use stops” as many mentioned in their discussions on the book. But, I am the guy who strongly feels (and more importantly sees statistically) that in many, many strategies, traders assume that as long as they have stops in place they are fully protected. The reality is they are not.

What are stops? Stops are partial insurance. Non-professionals often believe they provide full protection. In reality though, the equity markets are closed more hours than they are open. If something major occurs when the markets are closed, stops are often useless.

For example, imagine a trader being short a downtrending, poor-fundamentals stock at \$30. They did all their homework. Bad company, minimal earnings, cash-burning, terrible technicals, and on and on and on. They’re really onto something here on the short side, right? They tell themselves they’ll risk 10% shorting the stock at \$30 putting in a “protective stop” at \$33. At best though, possibly they’re “protected” for the 32 1/2 hours a week the market is open (assuming nothing extreme occurs intraday).

Now let’s imagine this same company is in an industry and the leading firm in the industry is outside the US and is hypothetically looking to enter the US as it does its global roll-up. Poor-technicals & poor-fundamentals company is approached, and

management's headache is now a bonanza. They get offered \$60 a share over a summer weekend on a buyout (and yes, this similar scenario played itself out in 2016).

The stock opens on Monday at \$58 and eventually rises to near \$60. How did that "protective" stop at \$33 work out? It's called losing nearly 100% overnight on a position (and sadly this happens to traders all the time both on the long and the short who rely on stops as their only protection). And in this case they got lucky. The buyout could have been at \$90 (or higher!) and they would have lost 200% or more overnight.

Here's a recent high-profile example on the long side. Look at the price history of the inverse long (short) volatility ETN – XIV. Thousands of traders owned that ETN on Feb 2, 2018 when it closed at \$99. Many of these traders had their "protective stops" in place just under that price. The problem is the stock opened at \$8 the next day. A 90% loss that was not supposed to happen because a "protective" stop was in place.

Here's the bottom line. Stops are band-aids that protect your skin when you scrape your elbow. Stops are useless though when a major car wreck occurs.

In *Short-Term Trading Strategies That Work*, we showed definitively with data-driven test results that stops hurt. Yes, they potentially have a place in one's portfolio, especially if other risk management tools are associated with it. As a stand-alone risk tool though, they're at best band-aids that cover the scrapes much of the time but leave one exposed to the more serious hits a position can take.

Larry Connors
Connors Research LLC
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AmiBroker Add-on Code for the Trading Strategies in *Buy the Fear, Sell the Greed*

If you are an AmiBroker user, we have programmed the add-on code for the following strategies from the *Buy the Fear, Sell the Greed* book including:

- RSI PowerZones
- CRASH
- Vol Panics
- VXX Trend Strategy
- Trading New Highs
- TPS; Fear and Greed Rising
- Terror Gaps

The AmiBroker Add-on Code will allow you to use the Explore button in the Analysis window to scan a universe of stocks each evening for trading signals for the next trading day. The signals will include entry, exit, and entry limit price information, if appropriate for the strategy, which you can export for further review.

To run the AmiBroker explorations requires AmiBroker version 6.2 or later and a data source that provides daily data. Our research team uses data from Norgate Data that is adjusted for splits and dividends, but you can use any data source that you consider reliable.

The price is \$100 per strategy or \$250 for all 7 strategies.

To order go to <http://bit.ly/amibrokeraddon> or call 888-484-8220 ext. 1 (international call +1-973-494-7311 ext. 1).

Please note: This code is not designed to allow you to backtest the *Buy the Fear, Sell the Greed* trading strategies on your computer.

More Research from Larry Connors

Short Term Trading Strategies That Work by Larry Connors and Cesar Alvarez (<https://bit.ly/2MkoIbB>)

Street Smarts by Laurence A. Connors and Linda Bradford Raschke (<https://bit.ly/2xZeURy>)

High Probability ETF Trading Strategies by Larry Connors, Cesar Alvarez, and Connors Research, LLC (<https://bit.ly/2LJRbqz>)

How Markets Really Work, 2nd Ed. by Larry Connors, Cesar Alvarez, and Connors Research, LLC (<https://amzn.to/2JlcE5z>)

Free Weekly Trading Research from Larry Connors

Receive weekly trading research from Larry Connors directly to your email inbox. Click here to subscribe: <http://tradingmarkets.com/newsletter>

Private Mentoring

If you are looking to learn directly from Larry Connors on a one-on-one basis, private mentoring is available on a limited basis.

For 15 years, Larry has been mentoring professional traders, including floor traders and hedge fund managers, along with those aspiring to become professional traders.

Larry's Private Mentoring Plan is a comprehensive 10-session program structured around where you are currently with your trading and where you ultimately want to get to.

Each session, done by phone or online, builds upon the previous session and you will learn how to build a business around your trading. Strategies and research are also provided for you as you work together over the 10 sessions.

Please note: Due to Larry's schedule space is limited.

If you would like to learn more about working one-on-one with Larry please call 1-888-484-8220 ext. 1 or email Mentoring@cg3.com.

Private Consulting for Asset Managers and Family Offices

If you run an asset management firm or a family office, we offer customized proprietary trading and investment strategies and research for your firm.

Consulting includes proprietary Equity, ETF, and Volatility Strategies and Research customized to your needs. The cost begins at \$100,000 per year and services are delivered quarterly.

For additional details please call 1-888-484-8220 ext. 1 or email Consulting@connorsresearch.com.

Introduction to AmiBroker Programming

Learn How To Backtest Your Best Trading Ideas In One Day - Guaranteed!

How would you like to have the ability to backtest your best trading ideas? And learn how to do this from one of the best in the industry?

Here is What You Will Receive

You will start the course with minimal to no knowledge on how to backtest in AmiBroker. Six hours later you will...

- 1) Know how to program a strategy.
- 2) Know how to backtest and validate that strategy.
- 3) Learn how to improve upon that strategy.
- 4) And be able to receive the daily signals for that strategy.

This is knowledge which you will be able to apply to your trading for the rest of your life.

In this class you will start from the beginning and within hours have the skill to take one of our Strategy Guidebooks and program it in AmiBroker yourself. Plus, your personal code will generate the signals for the upcoming day!

Course Objectives

This course is designed for traders who want to learn how to use AmiBroker to create backtests and/or generate trading signals, but who have little or no familiarity with the AmiBroker language.

At the completion of this course, you will be able to:

1. Create your own custom indicators, and add them to an AmiBroker chart.
2. Backtest basic trading strategies in order to see which ones have edges, which ones do not, and take the strategies that have edges and make them better.
3. Verify that your backtest results are correct.
4. Generate trading signals for the upcoming trading day.

What's Included

- Six hours of online instruction. The course was recorded live and you will be able to download it to your computer to view at your convenience.
- Several break-out sessions where you will spend hands-on time with AmiBroker.
- AFL code templates that you can easily modify for your own needs.
- A free copy of the Strategy Guidebook "ConnorsRSI Selective Strategy for ETFs and

Stocks”, which we will use as the basis for our backtest. By the end of the class, you will be able to take this strategy, program it in AmiBroker yourself, and get the signals for the upcoming day.

Your Instructor

You will be taught by Matt Radtke, one of the lead researchers for *Buy the Fear, Sell the Greed*. Matt has previously managed teams of professional programmers and has the gift of making programming simple-to-learn. Matt will walk you through step-by-step on how to properly backtest in AmiBroker so you too can begin testing your best trading strategies immediately.

Imagine having the ability to have a trading idea and to be able to backtest it on your own. You will learn exactly how to do this in this course.

Total Time Estimate: 6 hours

By the end of this course, you will be in the position to test your strategies, improve upon your strategies, and scan for the set-ups for your strategies.

Learning how to program in AmiBroker can save you hundreds of hours and make you a more profitable trader. When you get those great trading ideas, you will now be able to test them immediately on your own!

Order Today

The cost of “Programming in AmiBroker - Learn How to Backtest Your Best Trading Ideas in One Day” is \$1000. You will receive a full day of instruction, a copy of one of our Strategy Guidebooks, knowledge on how to backtest your strategies, along with being able to run your scans to get the trades that are signaling. Use this link to purchase the course today! <http://bit.ly/AmiBrokerCourse>

Correlation Risk

As I mentioned, an entire book can be written on correlation risk and it's impossible to fully cover this subject here. On a top-level though, there are five thoughts related to correlation risk I'd like to point out.

1. At times of extreme fear, many of the strategies in the book will become correlated. This is a book that identifies fear a number of different ways. At extremes, when fear is everywhere, many of the long strategies will be firing at the same time. Prepare ahead of time for this fact in deciding how you feel it's best for you to handle this based upon the risk you're willing to assume.
2. Global Correlations - Fear is a global contagion. A panic can trigger in any part of the world and move across the globe. For example, you'll see this on the long side of TPS in bull markets. You'll potentially see many country funds triggering buy signals within a few days of each other as fear spreads.
3. U.S Market Correlations - If an RSI PowerZones signal triggers on SPY, a number of other strategy signals could occur at the same time. Taking on too many strategies at the same time could increase your risk during these times because if the S&P 500 has pulled back because fear has increased, many of its underlying stocks are pulling back at the same time.
4. Sector/ETFs Correlations - For example, if oil prices sell-off and panic ensues, there could be signals both in the oil and oil-related ETFs along with stocks in those sectors. This can be partially mitigated by stating ahead of time you will only allow your portfolio to hold X% of a sector at any given time.
 - a. Thanks to open ended competition, over the past few years many once popular ETFs now have copy-cat ETFs (same holdings, different costs, or same concept, slightly different holdings). Therefore, if you see multiple signals from the same sector, select the ETF that's most liquid. Yes, EWG (Germany) and HEWG (Currency Hedged Germany) have different symbols and technically have different holdings. In reality though, on a short-term basis (3-7 days) they're closely aligned. Trading both is essentially doubling your position size in Germany. You can potentially avoid the additional risk by taking one signal, not both.
5. Correlations are dynamic - I've seen this for years and you may have too. This means correlations, especially short-term, constantly change. More importantly they tend to

change faster than most traders measure them. Correlations from a year ago are less relevant to a trader than correlations over the past month. This is a subject that can be extensively written on and I'm hopeful someone in the future writes a book on this subject. It would go a long way in assisting traders. For now, please just remember that correlations are dynamic and focusing on shorter-term correlations is better for trading.

Trading Resources

There is an abundance of great resources available for you to learn from.

The following is a list of authors who have written books, along with a handful of sites and individuals which I feel complement the knowledge in this book.

Please note, if an author (or site) is not listed here it does not mean I don't like their work. It could mean I either haven't read their book, or their material isn't an exact fit for the topic of this book.

In alphabetical order by author.

Options

Michael Benklifa - *Think Like an Option Trader: How to Profit by Moving from Stocks to Options* (You can order the book using this link: <https://amzn.to/2sMqmLk>). An under-the-radar gem. Devour this book.

Larry McMillan - *Options as a Strategic Investment: Fifth Edition* (<https://amzn.to/2Mgp0AB>). I believe Larry's options book is the most read options book in history. Well deserved!

Jon and Pete Najarian - I've known Jon for two decades and I'm sure you've seen the Najarian Brothers on CNBC during market hours and the show *Fast Money* that airs after the close. Jon and Pete walk the talk and I enjoy learning from their materials and their trading ideas. When you're asking "Who's on the other side of this trade?," it's probably best it's not the Najarian Brothers (<https://amzn.to/2HIyf8W>).

Martin O'Connell - *The Business of Options: Time-Tested Principles and Practices* (<https://amzn.to/2sYcaxT>). One of the leading professional options trainers in the world, this book is for intermediate/professional traders. Highly recommended.

Anthony (Tony) Saliba - I strongly recommend all of Tony's options books - they start with must-know entry level knowledge and build further from there (<https://amzn.to/2MiBnfe>). He was also the only options trader that made it into Jack Schwager's first book *Market Wizards*. One of the firms Tony founded was a leading education trading firm for professional trading desks globally. You'll be learning what many of the professional options traders over the years learned from Tony and his team.

Volatility Trading

I mentioned a number of volatility trading sites in Chapter 6. Here are books and authors that I recommend.

Sheldon Natenberg - *Option Volatility and Pricing: Advanced Trading Strategies and Techniques, 2nd Edition* (<https://amzn.to/2sZ6Udy>). This book is one of the bibles for options volatility trading.

Russell Rhoads - One of my favorite authors. I mentioned Russell in Chapter 6. I also recommend all his books (<https://amzn.to/2JMyHrK>).

Euan Sinclair - *Volatility Trading* (<https://amzn.to/2Js4SO1>). An excellent book - packed with knowledge.

General Trading

John Bollinger - *Bollinger on Bollinger Bands* (<https://amzn.to/2MmF57H>). John's "Bollinger Bands" are the staple of all charting packages today and have stood the test of time for a reason. There's multiple sources on the internet where you can find the many ways to applying John's Bands. Start first with anything John has written because he knows more about this topic than anyone in the world.

Robert Carver - *Systematic Trading: A Unique New Method For Designing Trading And Investing Systems* (<https://amzn.to/2Jt0Pkt>). Carver was responsible for the creation of AHL Partners' fundamental global macro strategy and then managed the fund's multi-billion dollar fixed income portfolio. I consider his book to be one of the authoritative sources on developing and executing trading systems.

Ernie Chan Ph.D. - I learned a great amount from Ernie's two books on quant trading: *Quantitative Trading: How to Build Your Own Algorithmic Trading Business* (<https://amzn.to/2Jui5pH>) and *Algorithmic Trading: Winning Strategies and Their Rationale* (<https://amzn.to/2t0A8IG>). Highly recommended.

Michael Covel - As I mentioned in the VXX Trend Strategy chapter, Michel Covel's trend-following information is about the best out there. Start with his book *Trend Following, 5th Edition: How to Make a Fortune in Bull, Bear and Black Swan Markets* (<https://amzn.to/2Jvleo0>) and build out from there.

Tom DeMark - *The New Science of Technical Analysis* (<https://amzn.to/2JlgJGV>). Tom has been an advisor to a number of the largest hedge funds in the world. His book *The New Science of Technical Analysis* was the start of him publishing his ideas and many trading strategies have been created over the years from the concepts in this book.

Kevin Haggerty - Kevin was the head of trading for Fidelity Capital Markets and arguably ran one of the most powerful trading desks in the 1990's. When he retired he took his knowledge to the TradingMarkets website influencing an entire generation of traders. Spending a day trading with Kevin is like spending a day with a maestro. Some of his courses are occasionally available for resale on the Internet. A great man and a market genius.

Chiente Hsu - *Rule Based Investing: Designing Effective Quantitative Strategies for Foreign Exchange, Interest Rates, Emerging Markets, Equity Indices, and Volatility* (<https://amzn.to/2JAxsss>). This book is on my top 5 book list of all time. Her chapter on using volatility as an on/off switch is worth 100 times more than you'll pay for the book.

Mark Minervini - *Trade Like a Stock Market Wizard: How to Achieve Super Performance in Stocks in Any Market* (<https://amzn.to/2sUXa12>). Mark is one of the leaders in teaching momentum trading. His books are well written and go into professional detail in explaining the how's and why's of trading momentum stocks.

George Pruitt - *The Ultimate Algorithmic Trading System Toolbox + Website: Using Today's Technology To Help You Become A Better Trader* (<https://amzn.to/2JDikFY>). George did a great job here in guiding you with the tools needed to build successful trading systems.

Linda Raschke - Linda and I co-write *Street Smarts: High Probability Short-Term Trading Strategies* (<https://amzn.to/2JMjXsU>). Anything written by Linda is gem-like. I still keep one of her writings from 25 years ago next to my desk because of how it still pertains to the markets today. Brilliant trader.

Igor Tulchinsky - *Finding Alphas: A Quantitative Approach to Building Trading Strategies* (<https://amzn.to/2MfsOSp>). Igor is the CEO of WorldQuant, a 600-employee, multi-billion dollar quant hedge fund. His bio alone should encourage you to own this book.

Larry Williams - (<https://amzn.to/2JGGh3U>). Larry is one of the true pioneers in the short-term trading industry. Many successful mean reversion strategies can find their roots in Larry's research over the years.

Risk & Money Management

Ken Grant - *Trading Risk: Enhanced Profitability through Risk Control* (<https://amzn.to/2MnayGY>). I love this book. Ken oversaw the Risk Management for three of the largest hedge funds in the world. The knowledge in this book is timeless.

William Poundstone - *Fortune's Formula: The Untold Story of the Scientific Betting System That Beat the Casinos and Wall Street* (<https://amzn.to/2l1B6B2>). A great book on Ed Thorp and

many others and how they applied Kelly Criterion to beat the market.

Nassim Taleb - (<https://amzn.to/2MnVR6k>). Nassim was a guest speaker at a conference I held for 500 traders in 2001. This was before he achieved his fame and everyone in the audience could see his brilliance. Agree or disagree with him - his books make you a smarter trader, especially as they pertain to risk.

History/Biography

Maneet Ahuja - *The Alpha Masters: Unlocking the Genius of the World's Top Hedge Funds* (<https://amzn.to/2y1W63T>). Many good trading ideas can be gained by reading this book.

Aaron Brown - *Red-Blooded Risk: The Secret History of Wall Street* (<https://amzn.to/2y2INQL>), and his answers to trading and investing questions on Quora.com. Aaron is one of the leading minds in the industry. Also, if you are on Quora.com, do a search for him. Aaron has been kind enough to answer hundreds of questions from traders around the globe providing them with world-class answers.

Jack Schwager - Any and all of Jack's *Market Wizards* books (<https://amzn.to/2sNR85Q>). Jack's books did as much to positively influence our industry and two generations of traders than any other trading books ever written.

Ed Thorp - *A Man for All Markets: From Las Vegas to Wall Street, How I Beat the Dealer and the Market* (<https://amzn.to/2sOmRnA>). There are no strategies in this book. It's a great story.

Podcasts

The Options Insider Radio Network (<http://www.theoptionsinsider.com/options-insider-radio-network/>). They generate a lot of timely trading information, along with ongoing education.

Option Alpha Podcast (<https://optionalpha.com/podcast>).

Extreme Self Improvement

There are many - here are my top three.

Ben Bergeron - *Chasing Excellence: A Story About Building the World's Fittest Athletes* (<https://amzn.to/2sPOTPn>). The CrossFit Games are a test to determine who are the fittest man and woman on Earth. Bergeron has coached more winners than any coach in CrossFit Games history. This book may use CrossFit as it's background but it's bigger. It's about applying the extreme mind-frame required to be the best at anything in life. I've recommended the book to a number of very successful friends. Each one has responded the same way after reading this book by basically saying "I thought I worked hard and was

focused until I read this book.” This book goes way beyond anything you’ve ever read before and there’s a reason it remains a best seller on Amazon. You’ll likely read this multiple times because there’s so much here.

Richard Machowicz - *Unleash the Warrior Within: Develop the Focus, Discipline, Confidence, and Courage You Need to Achieve Unlimited Goals* (<https://amzn.to/2l472VC>). Retired Navy Seal Richard Machowicz was a friend of mine. Mack impacted the lives of many people with this book, his friendship, and his television shows and his radio show. Sadly, he passed away from brain cancer in 2017. Mack left the world this book - a book that has changed many lives. My highest recommendation.

Jocko Willink - *Extreme Ownership: How U.S. Navy SEALs Lead and Win (New Edition)* (<https://amzn.to/2l4iXTi>). Former Navy Seal Commander Jocko Willink has written a book that became a #1 Best Seller on Amazon. Jocko’s philosophy is extreme and it’s spelled out in this book. Applying his concepts makes better traders and more importantly, better people. I also recommend listening to the “Jocko Podcast.”

Additional Books

Robert B. Cialdini - *Influence: The Psychology of Persuasion, Revised Edition* (<https://amzn.to/2sTqvFH>).

Marcos Lopez de Prado - *Advances in Financial Machine Learning* (<https://amzn.to/2HLtpYz>).

J. Welles Wilder - *New Concepts in Technical Trading Systems* (<https://amzn.to/2JxcccD>).

AmiBroker Coding

The two programmers I most rely on are:

Cesar Alvarez (<http://alvarezquanttrading.com/services/>). Cesar has been programming and testing my strategies since 2003 and may be the best AmiBroker programmer in the world. In the 1990’s he was a senior team member of the software development team that built Excel. You can email Cesar at Cesar@AlvarezQuantTrading.com.

Matt Radtke (www.quantforhire.com). Matt has programmed for me since 2012 and is also the instructor of our highly popular course on AmiBroker programming, *Introduction to AmiBroker Programming* (<https://bit.ly/2LK1KKe>). You can reach Matt by email at mradtkeboulder@gmail.com.

TradeStation Coding

Stuart Okorofsky (www.okor.com). Stuart has done my TradeStation coding since 1995 and many professionals consider him one of the top in the world. Stuart only takes on larger

projects now but if you have a serious project, he's the person to go to first. Stuart can be reached at krofs@aol.com.

Data, Charting and Quant Sites

Finviz (www.finviz.com). If you like data, you'll love Finviz. One of the best live data sites.

Quantopian and Quantcon (<https://www.quantopian.com/>). Quantopian is a large community of quant traders, investors and researchers (it has hundreds of thousands of members). It provides you with a platform to do your testing along with many other resources. Also, they hold an annual conference (Quantcon) in New York City and other cities around the globe which attracts some of the best speakers and teachers in the industry. Quantopian is highly recommended if you are serious about quantitative trading.

Stockcharts (www.stockcharts.com). There are so many solid charting tools available for you on this site. And they're continuously adding new features and improvements.

Relative Strength Indicator Formula and ConnorsRSI Calculation

Excerpt reprinted from An Introduction to ConnorsRSI from Connors Research, LLC, 2012.

Connors Research has been developing, testing, and publishing quantified trading strategies since the mid-1990s. During that time, we have had the opportunity to evaluate a great number of different technical indicators and to assess their effectiveness in predicting future price action. Now we've taken the next step and created an indicator of our own: *ConnorsRSI*. The purpose of this guidebook is to describe the indicator itself and also to provide a well-defined, quantified trading strategy that utilizes this new indicator.

ConnorsRSI is a composite indicator consisting of three components. Two of the three components utilize the Relative Strength Index (RSI) calculations developed by Welles Wilder in the 1970s, and the third component ranks the most recent price change on a scale of 0 to 100. Taken together, these three factors form a *momentum oscillator*, i.e. an indicator that fluctuates between 0 and 100 to indicate the level to which a security is overbought (high values) or oversold (low values).

Before we discuss how to calculate ConnorsRSI, let's review Wilder's RSI. RSI is a very useful and popular momentum oscillator that compares the magnitude of a stock's gains to the magnitude of its losses over some look-back period. Wilder himself believed that 14 periods was the ideal look-back. We often use the shorthand notation RSI(14) for the 14-period RSI. The formula below computes RSI(14) for a series of price changes:

Example A.1. RSI Calculation

$$RSI = 100 - \frac{100}{1 + RS}$$

$RS = \text{Average Gain} / \text{Average Loss}$

$\text{Average Gain} = [(\text{previous Average Gain}) \times 13 + \text{current Gain}] / 14$

$\text{First Average Gain} = \text{Total of Gains during past 14 periods} / 14$

$\text{Average Loss} = [(\text{previous Average Loss}) \times 13 + \text{current Loss}] / 14$

$\text{First Average Loss} = \text{Total of Losses during past 14 periods} / 14$

Note: "Losses" are noted as positive values.

$RS = \text{Average of } x \text{ days up closes} / \text{Average of } x \text{ days down closes}$

If we wanted to compute RSI for a different number of periods (N), then we would replace 14 in the formula above with N, and replace 13 with N-1. Regardless of the number of periods used in the calculation, the result will always be a number between 0 and 100. Traders who use RSI(14) typically look for values greater than 70 to identify overbought conditions, and values less than 30 to indicate oversold conditions.

Our previous research has shown that using shorter look-back periods makes RSI more effective in predicting short-term price movements. We have published many strategies that utilize RSI(2), as well as several that use RSI(3) and RSI(4). Changing the number of periods also has an effect on the RSI levels that best identify overbought and oversold conditions. For example, an RSI(2) value of less than 10 is usually a reliable indicator of an oversold condition, while an RSI(2) value over 90 is a good benchmark for an overbought condition.

Now let's turn our attention back to ConnorsRSI. As mentioned previously, ConnorsRSI combines three components, and as you might guess, they are all elements that our research has repeatedly shown to have significant predictive ability:

Price Momentum: As we just discussed, RSI is an excellent way to measure price momentum, i.e. overbought and oversold conditions. By default, ConnorsRSI applies a 3-

period RSI calculation to the daily closing prices of a security. We will refer to this value as $RSI(Close,3)$.

Duration of Up/Down Trend: When the closing price of a security is lower today than it was yesterday, we say that it has “closed down”. If yesterday’s closing price was lower than the previous day’s close, then we have a “streak” of two down close days. Our research has shown that the longer the duration of a down streak, the more the stock price is likely to bounce when it reverts to the mean. Likewise, longer duration up streaks result in larger moves down when the stock mean reverts. In effect, the streak duration is another type of overbought/oversold indicator.

The problem is, the number of days in a streak is theoretically unbounded, though we could probably place some practical limits on it based on past experience. For example, we might observe that there have been very few instances of either an up streak or a down streak lasting for more than 20 days, but that still doesn’t get us to a typical oscillator-type value that varies between 0 and 100.

The solution is two-fold. First, when we count the number of days in a streak, we will use positive numbers for an up streak, and negative numbers for a down streak. A quick example will help to illustrate this:

Example A.2. Sample Streak Calculation

Day	Closing Price	Streak Duration
1	\$20.00	
2	\$20.50	1
3	\$20.75	2
4	\$19.75	-1
5	\$19.50	-2
6	\$19.35	-3
7	\$19.35	0
8	\$19.40	1

The closing price on Day 2 is higher than on Day 1, so we have a one-day up streak. On Day 3, the price closes higher again, so we have a two-day up streak, i.e. the Streak Duration value is 2. On Day 4, the closing price falls, giving us a one-day down streak. The Streak Duration value is negative (-1) because the price movement is down, not up. The downward trend continues on Days 5 and 6, which our Streak Duration reflects with values of -2 and -3. On Day 7 the closing price is unchanged, so the Streak Duration is set to 0 indicating neither an up close nor a down close. Finally, on Day 8 the closing price rises again, bringing the Streak Duration value back to 1.

The second aspect of the solution is to apply the RSI calculation to the set of Streak Duration values. By default, ConnorsRSI uses a 2-period RSI for this part of the calculation, which we denote as $RSI(Streak,2)$. The result is that the longer an up streak continues, the closer the $RSI(Streak,2)$ value will be to 100. Conversely, the longer that a down streak continues, the closer the $RSI(Streak,2)$ value will be to 0. Thus, we now have two components -- $RSI(Close,3)$ and $RSI(Streak,2)$ -- that both use the same 0-100 scale to provide a perspective on the overbought/oversold status of the security we’re evaluating.

Relative Magnitude of Price Change: The final component of ConnorsRSI looks at the size of today’s price change in relation to previous price changes. We do this by using a Percent Rank calculation, which may also be referred to as a “percentile”. Basically, the Percent Rank value tells us the percentage of values in the look-back period that are less than the current value.

For this calculation, we measure price change not in dollars and cents, but as a percentage of the previous day’s price. This percentage gain or loss is typically referred to as the one-

day return. So if yesterday's closing price was \$80.00, and today's price is \$81.60, the one-day return is $(\$81.60 - \$80.00) / \$80.00 = 0.02 = 2.0\%$.

To determine the Percent Rank, we need to establish a look-back period. The Percent Rank value is then the number of values in the look-back period that are less than the current value, divided by the total number of values. For example, if the look-back period is 20 days, then we would compare today's 2.0% return to the one-day returns from each of the previous 20 days. Let's assume that three of those values are less than 2.0%. We would calculate Percent Rank as:

$$\text{Percent Rank} = 3 / 20 = 0.15 = 15\%$$

The default Percent Rank look-back period used for ConnorsRSI is 100, or PercentRank(100). We are comparing today's return to the previous 100 returns, or about 5 months of price history. To reiterate, large positive returns will have a Percent Rank closer to 100. Large negative returns will have a Percent Rank closer to 0.

The final ConnorsRSI calculation simply determines the average of the three component values. Thus, using the default input parameters would give us the equation:

$$\text{ConnorsRSI}(3,2,100) = [\text{RSI}(\text{Close},3) + \text{RSI}(\text{Streak},2) + \text{PercentRank}(100)] / 3$$

The result is a very robust indicator that is more effective than any of the three components used individually. In fact, ConnorsRSI also offers some advantages over using all three components together. When we use multiple indicators to generate an entry or exit signal, we typically set a target value for each one. The signal will only be considered valid when all the indicators exceed the target value. However, by using an average of the three component indicators, ConnorsRSI produces a blending effect that allows a strong value from one indicator to compensate for a slightly weaker value from another component. A simple example will help to clarify this.

Let's assume that Trader A and Trader B have agreed that each of the following indicator values identify an oversold condition:

- $\text{RSI}(\text{Close},3) < 15$
- $\text{RSI}(\text{Streak},2) < 10$
- $\text{PercentRank}(100) < 20$

Trader A decides to take trades only when all three conditions are true. Trader B decides to use ConnorsRSI to generate her entry signal, and uses a value of $(15 + 10 + 20) / 3 = 15$ as the limit. Now assume we have a stock that displays the following values today:

- $RSI(Close,3) = 10$
- $RSI(Streak,2) = 8$
- $PercentRank(100) = 21$
- $ConnorsRSI = (10 + 8 + 21) / 3 = 13$

Trader A will not take the trade, because one of the indicators does not meet his entry criteria. However, Trader B will take this trade, because the two low RSI values make up for the slightly high PercentRank value. Since all three indicators are attempting to measure the same thing (overbought/oversold condition of the stock) by different mechanisms, it makes intuitive sense to take this “majority rules” approach. More importantly, our research has shown ConnorsRSI to be superior to any other momentum indicator that we’ve tested.

To receive a free copy of the full report on ConnorsRSI, go to www.tradingmarkets.com and click on the ConnorsRSI link in the menu.